Preface

The outcome-based curriculum is developed by the department's teachers with the assistance, cooperation, and feedback of our graduates and some prestigious faculty members. Many universities' course curriculums were used in the development of the syllabus. In many ways, it differs from our previous syllabus. Students can now specialize in any of the many optional courses available, such as Pattern Recognition, Digital Signal Processing, Robotics Technology, Cloud Computing, Big Data Analysis, and so on, without sacrificing the fundamental and fundamental study of core Computer Science and Engineering courses.

The laboratory materials have also been updated as a result of this and other major modifications in course content, with more design-oriented classes emphasizing both practical and simulation components. The department has created a number of facilities to help such changes be implemented successfully, and efforts are being made to enhance the situation even further.

The syllabus had previously been set for 153.0 credit hours. A total of 152.0 credit hours are required to get a bachelor's degree from the department under this new updated syllabus. All of the department's faculties have contributed to updating the material of various disciplines. The editorial board put forth a lot of time and effort to rewrite and update the syllabus. The department owes them a particular thank you.

Students and those interested in learning about any changes made by the department in any course or in the university's rules and regulations may contact their mentors and the department office.

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Head,
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Table of Contents

SN	CONTENTS			PAGES
01.	**Title of the Academic Program**			004
02.	**Name of the University**			004
03.	**Vision of the University**			004
04.	**Mission of the University**			004
05.	**Name of the Program Offering Entity (Department/Faculty/Institute)**			004
06.	**Vision of the Program Offering Entity**			004
07.	**Mission of the Program Offering Entity**			004
08.	**Objectives of the Program Offering Entity**			005
09.	**Name of the Degree**			005
10.	**Description of the Program**			005
11.	**Graduate Attributes (based on need assessment)**			010
12.	**Program Educational Objectives (PEOs)**			010
13.	**Program Learning Outcomes (PLOs)**			011
14.	**Mapping of mission of the university with PEOs**			012
15.	**Mapping of PLOs with the PEOs**			012
16.	**Mapping of courses with the PLOs**			013
17.	**Structure of the Curriculum**			016
	17(a)	Duration of the program		016
	17(b)	Admission Requirements		016
	17(c)	Total minimum credit requirement to complete the program		016
	17(d)	Total class weeks in a semester		017
	17(e)	Minimum CGPA requirements for graduation		017
	17(f)	Maximum academic years of completion		017
	17(g)	Category of Courses		017
		g(i)	General Education Courses	017
		g(ii)	Mathematics and General Science Courses	017
		g(iii)	Core Courses	017
		g(iv)	Optional Courses	019
		G(v)	Capstone course/Internship/Thesis/Projects/Portfolio	019
18.	**Year/Level/Semester/Term wise distribution of courses**			020
19.	**Description of all courses of the program**			023
	19(a)	General Education Courses		023
	19(b)	Mathematics and General Science Courses		034
	19(c)	Core Courses		051
	19(d)	Optional Courses		117
20.	**Grading/Evaluation**			165

	20(a)	Grading Scale/Point System	165
	20(b)	Grades	165
	20(c)	Grade Point Average (GPA) and Cumulative Grade Point Average (CGPA)	165
	20(d)	Course Withdrawal	166
	20(e)	Incomplete (I) courses	166
	20(f)	Retake	166
	20(g)	Grade Improvement	166
	20(h)	Dropout	166
	20(i)	Grades Review Procedure	166
Appendix A : Template of Course Outline (Outcome-Based Curriculum)			167
Appendix B : Template of Course File (Outcome-Based Curriculum)			170

Outcome-Based Curriculum

Part A

1. **Title of the Academic Program:** Computer Science and Engineering

2. **Name of the University:** Hamdard University Bangladesh

3. **Vision of the University:**

Hamdard University Bangladesh envisions fostering national development by establishing an excellent higher education institution responsive to society's need, actively contributing to knowledge and being able to develop capable leaders to face the challenges of the Twenty-first Century.

4. **Mission of the University:**

Mission 1: Development of enlightened human beings with appropriate knowledge and skills capable of transcending them into professionally sound and enthusiastic entrepreneurs contributing their respective shares to national, regional and global development in human welfare.
Mission 2: Creation and advancement of knowledge through teaching and research; and innovation in modern science and natural medicine and disseminate results for the welfare of the nation, in particular and the world, at large.
Mission 3: Culmination of the cultural values and rich heritage of Sonargaon, Vikrampur and Bangladesh with other cultures to promote world harmony and peace.

5. **Name of the Program Offering Entity (Department/Faculty/Institute):**
   Department of Computer Science and Engineering

6. **Vision of the Program Offering Entity:**
   The Department of Computer Science and Engineering (CSE), Hamdard University Bangladesh (HUB) aspires to be a nationally recognized model for producing CSE graduates who can compete in and contribute to the demands of today's fast-paced, technology-driven world.

7. **Mission of the Program Offering Entity:**

   M1. To provide a high-quality education that will enable graduates to be active, productive, and engaged members of society on a local, national, and worldwide level by achieving the highest possible standards of teaching and research in Computer Science and Engineering and related areas.

M2. Prepare students as successful professionals for employment in any of the vast array of CSE-related fields including: Computer Programming, Networking, Software and Hardware Development, Database Administration, System Administration, Computer Graphics & Image Processing, Data Mining, Big Data Analysis, Machine Learning, IOT, Artificial Intelligence & Robotics, Cloud Computing, System Analysis Design & Development, and Information Systems Security etc.

M3. Develop students as excellent researchers on basic and applied challenges, allowing them to participate in transdisciplinary research and industries while also advancing the Scientific and Industrial Revolutions.

M4. Attain Personal academic success by the application of modern educational technology in secondary and tertiary education, independent thought, collegial interchange of ideas, and high ethical standards.

M5. Develop the abilities needed to find work in academia, industry, or government agencies.

8. Objectives of the Program Offering Entity:

The B.Sc. Program in Computer Science and Engineering (CSE) is designed to produce skilled graduates in the field to satisfy the growing demands of computer engineers in home and abroad. It provides the students an opportunity to obtain broad knowledge of Computer Science, Computer Engineering with some freedom to tailor the program according to the student's individual needs.

9. Name of the Degree: B.Sc. in CSE

10. Description of the Program:

A student must complete a minimum of 152 credits of course work to earn a B.Sc. in Computer Science & Engineering, which normally takes four years.

The CSE undergraduate degree courses are created with two goals in mind. First, students are introduced to computer science through a series of cleverly designed courses, such as Basic Computer Skills, which primarily introduces students to a computer programming environment using Python as a programming language; then, a more solid foundation on programming is provided through Structured Programming Language and Data Structures courses, which use C/C++ as a programming language; and finally, Object Oriented Programming is introduced.

To complement the software side, the Computer Science & Engineering degree also provides students with an in-depth study of hardware. Electrical Engineering and Electronic Devices & Circuits courses educate students to basic electrical circuit ideas. Following that, advanced hardware topics such as digital logic design and its integration with digital electronics and pulse technique, computer architecture, a close look at the microprocessor, and computer networks are covered. These classes will help students develop a strong foundation in computer hardware.

The degree also includes basic courses like System Analysis and Design, Operating Systems, Database Systems, Computer Graphics, Computer Peripherals and Interfacings, Communication Engineering, Compiler Design, Software Engineering, as well as advanced courses like Artificial Intelligence, Digital Image Processing, Simulation and Modeling, and others, all with their own labs.

Academic System

Hamdard University Bangladesh is a model of higher education consisting of semester, courses, credit hours, continuous evaluation and letter grading. In keeping its mission and goals, the University strives to ensure high academic standards by implementing well-designed curricula, carefully selecting quality students and faculty, utilizing modern and effective instructional methods and aids and by continuous monitoring with rigorous evaluation of all the pertinent activities.

Semesters

There are 2 (two) equal semesters named 'Spring' and 'Fall'. The duration of each Semester are 22 working weeks including Final Examination. Usually the Spring semester commences in March and the Fall semester in September. The Academic Calendar provides deadlines for registration; add/drop courses; withdrawals; examinations; publication of results; holidays etc. this has financial and academic implications too.

Course Numbering System

Each course is identified by a two-part numbering system. The first part with three alphabets represents the subject area and the second part refers to the level of the course. The series 100, 200, 300 and 400 numbers are intended to indicate progressively more demanding content of the course and correspondingly increasing competence on the part of the students enrolled in the course. For example CSE 327 Computer Peripherals and Interfacing is a third year course and it is assumed that a student registering in this course has already attended one or more second year (300 level) courses in Computer Science and Engineering. A student may register in this course before third year if he has already fulfilled the prerequisites for the course.

Course Duration

A student will be able to complete the program within 4 (four) years' time. The Maximum period to complete the undergraduate program is 6 (six) years from the date of first admission to the University.

Course Load

(a) **For regular students**

A student can register for Maximum 21.0 credits in a Semester. Students will have to take required number of credits as defined by the Department.

(b) **For Diploma students**

For 04 (Four) years Diploma students in the field of Computers / Electrical / Electronics / Telecommunications / Power / Mechanical and equivalent having Minimum CGPA of 2.5 (Average Grade C+), student can registers for Maximum 21.0 credits in a Semester for Evening section.

Admission Requirements

(a) **For regular students**

Every applicant, without any exception, must fulfill the admission requirements as laid by the University. A higher secondary certificate or its equivalent in Science group with the basic educational requirements. Minimum qualifying GPA to be eligible to apply is 2.5 individually in SSC/'O' level or equivalent and HSC/'A' level or equivalent examinations. For 'O' and 'A' level students, their Average B grade will be consider for admission. For all other foreign certificates, the University as per rules of the Bangladesh Government / University Grants Commission (UGC) of Bangladesh will determine equivalent.

(b) **For Diploma students**

The 04 (four) years Diploma student in the field of Computers / Electrical / Electronics / Mechanical / Telecommunications / Power and equivalent having Minimum CGPA of 2.5 (Out of 4) may also get admission at the Evening section. They will get Maximum 15 credits waiver from the GEC courses. Students must have minimum B grade in these waived courses.

Graduation / Degree requirements

The **B.Sc.** degree requirements are as follows:

(a) **For regular students**

(i) Completion of 152 credit hours courses (Theory + Laboratory)
(ii) Completion of 6 credits Dissertation with at least a 'C+' grade (GPA 2.5)
(iii) Passing of all the courses individually and maintaining a minimum CGPA of 2.5.

(b) **For Diploma students**

(i) Completion of 137 credit hours courses (Theory + Laboratory)
(ii) Completion of 6 credits Dissertation with at least a 'C+' grade (GPA 2.5)
(iii) Passing of all the courses individually and maintaining a minimum CGPA of 2.5.

Credit Hours

Students take courses in each semester and these courses have credits assigned to them and the credits are counted towards the degree. Credit Hours for a course are assigned on the basis of a 22 weeks Semester and 2 classes of 50 minutes duration each week. Laboratory classes will be of 2 hours duration in a week for each course.

Student Counseling

Each department will have a student counselor from among the teachers of the department. The counselor will help students to select his/her courses at the beginning of each semester. However, students and counselor share the responsibility for successful academic activities.

Performance Evaluation

The performance of the students will be evaluated throughout the semester by class tests, quizzes, assignments and midterm exams. End of semester evaluation includes comprehensive final exams, term papers, project reports, , presentation, viva voce etc. Numerical scores earned by students are cumulated and converted to letter grades at the end of each semester.

Student classification

The following classification applies to the student:

Freshman Year (First Year)
Sophomore Year (Second Year)
Junior Year (Third Year)
Senior Year (Fourth Year)

The Evaluation Procedures

Evaluation of students' performance will be based on final examination, terminal examinations, assignment & presentation / term paper and class-test / in course. The distribution of marks is as follows:

Theory:

Class Attendance	10%
Class Test	10%
Assignment & Presentation	10%
Mid-Term Examination	30%
Final Examination	40%
Total	**100%**

Laboratory:

Class Attendance	10%
Lab Report Assessment	20%
Project/Presentation	30%
Final Examination	40%
Total	**100%**

Grading and Point System

Numerical Grades	Letter Grades		Grade Point
80%-100%	A+	(A Plus)	4.0
75%-79.99%	A	(A Regular)	3.75
70%-74.99%	A-	(A Minus)	3.50
65%-69.99%	B+	(B Plus)	3.25
60%-64.99%	B	(B Regular)	3.0

55%-59.99%	B-	(B Minus)	2.75
50%-54.99%	C+	(C Plus)	2.50
45%-49.99%	C	(C Regular)	2.25
40%-44.99%	D	-	2.0
Less than 40%	F	Fail	0.0
	W	Withdrawn	
	I	Incomplete	

Note: Grades without numerical value
I: Incomplete
W: Withdrawal

Incomplete Grade
An incomplete (I) grade is assigned only when a student has failed to complete one or more requirements of the course for unavoidable reason / accidental circumstance and has applied for (I) grade. The students who are permitted to appear in make-up examination(s) will be assigned an (I) grade for that course and this grade will stay until the student appears in the make-up examination at the first available opportunity; if she/he fails to appear in the make-up examination the (I) grade will automatically be converted to (I) grade.

Withdrawal (W)
Withdrawal is assigned to a student who withdraws from the course within the deadline for withdrawal with 'W' grade. A student who withdraws after this date will earn the grade based on his performance before his withdrawal. Exception to this rule may be made on medical ground and on terms and condition of the University.

Grades Review Procedure
The Academic Council of the University administers the Grading Regulations and Reviews course grades submitted by the Departments.

Academic Standing
Students should take the requisite number of courses and maintain satisfactory grades in the courses. They are expected to maintain a GPA of 2.00 (both semester and cumulative) otherwise the student will be put on probation for the following semester. Course grade can be improved in the next available opportunity by paying the applicable course fee, in case of 'F' grade the students has to take the class in full with all the examinations. Grade 'C' and below may be improved by the students;

Credit Transfer
Credit transfer from an educational institution with a system similar to Hamdard University Bangladesh may be considered. Such candidates will have to apply with required documents and

are subject to credit transfer rules of this University. The total credits transferred by a student from other Universities will not exceed 50 credits for undergraduate program.

Methods and Medium of Instructions

The University teaching meets the demand of the time adopting present-day teaching methods including interactive internet, simulation, lab work, case analysis and field study. A special feature of Hamdard University Bangladesh teaching is the workshop/lab sessions designed to assist students in learning applications of concepts (mostly activity-based education) and theories. The medium of instruction in this University is English.

Credit Waiver Policy for Diploma Students

To apply for B.Sc. in CSE credit waiver, a student must have an average on C+ or better and courses must have been taken within the past 6 years. Student having four years Diploma in Computer Science & Engineering will get maximum 15 credits waiver from General Education Courses (GEC) under this waiver policy.

11. Graduate Attributes (based on need assessment):

Graduate qualities are more specific assertions that outline what students should learn during their time at university. These characteristics are linked to the information, abilities, and attitudes that students develop during their academic careers. During their time in the B.Sc. in CSE degree, students are expected to develop the following graduate qualities:

i. Solve computer science and engineering problems using their knowledge of mathematics, natural science, engineering basics, and an engineering specialization.
ii. Design computer science and engineering systems and components that satisfy specific needs using current engineering and IT techniques.
iii. Work in a multi-culturally varied multidisciplinary team and communicate effectively with key stakeholders.
iv. Participate in research, innovation, and entrepreneurship through life-long learning.

12. Program Educational Objectives (PEOs):

PEO1. Graduates will be able to formulate and solve Computer engineering problems using analytical talents and critical thinking using mathematical, scientific, and engineering principles

PEO2. Improve the ability to effectively communicate scientific discoveries in written, oral, and interactive presentations

PEO3. Help the Graduates to become top computational professionals and/or entrepreneurs, as well as continue to learn and take up new challenges in the disciplines of computer science and engineering.

PEO4. Assist Graduates pursuing knowledge and interdisciplinary learning throughout their lives in preparation for employment in industry, research, and academia, as well as leadership and entrepreneurship development.
PEO5. Facilitate the Graduates applying Computer Science and Engineering principles, techniques, and technologies in an ethical and responsible manner to contribute to society's long-term progress and well-being.

13. Program Learning Outcomes (PLOs):

PLOs	Learning Attribute	PLO statement
PLO1	Engineering Knowledge	Apply the knowledge of mathematics, sciences, engineering fundamentals, and manufacturing engineering to the solution of complex engineering problems.
PLO2	Problem Analysis	Identify, formulate, research relevant literature and analyze complex engineering problems, and reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences
PLO3	Design/Development of Solutions	Design solutions, exhibiting innovativeness, for complex engineering problems and design systems, components, or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, economic, ethical, environmental, and sustainability issues
PLO4	Investigation	Conduct investigation into complex problems, displaying creativeness, using research-based knowledge, and research methods including design of experiments, analysis, and interpretation of data, and synthesis of the information to provide valid conclusions
PLO5	Modern Tool Usage	Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modeling, to complex engineering activities, with an understanding of the limitations.
PLO6	The Engineer and Society	Apply reasoning based on contextual knowledge to assess societal, health, safety, legal, cultural, contemporary issues, and the consequent responsibilities relevant to professional engineering practices.
PLO7	Environment and	Understand the impact of professional engineering solutions in societal, global, and environmental contexts and demonstrate

	Sustainability	knowledge of and need for sustainable development.
PLO8	Ethics	Apply professional ethics with moral values and commit to the responsibilities and norms of the professional engineering code of practices
PLO9	Communication	Communicate effectively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions
PLO10	Individual and Team Work	Function effectively as an individual, and as a member or leader in diverse teams and in multidisciplinary settings.
PLO11	Project Management and Finance	Demonstrate knowledge and understanding of engineering management and financial principles and apply these to one's own work, as a member and/or leader in a team, to manage projects in multidisciplinary settings, and identify opportunities for entrepreneurship.
PLO12	Life-Long Learning	Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

14. Mapping of mission of the university with PEOs:

Mission/ PEO	PEO1	PEO2	PEO3	PEO4	PEO5
M1			√	√	
M2	√			√	√
M3		√		√	
M4		√			√
M5	√		√	√	

15. Mapping of PLOs with the PEOs:

PLOs	PEO 1	PEO 2	PEO 3	PEO 4	PEO 5
PLO 1	√				
PLO 2	√				
PLO 3			√	√	

PLO 4				√	
PLO 5			√	√	
PLO 6			√		
PLO 7					
PLO 8					√
PLO 9		√			
PLO 10				√	
PLO 11					
PLO 12	√	√			√

16. Mapping of courses with the PLOs:

Courses	**PLO 01**	**PLO 02**	**PLO 03**	**PLO 04**	**PLO 05**	**PLO 06**	**PLO 07**	**PLO 08**	**PLO 09**	**PLO 10**	**PLO 11**	**PLO 12**
General Education Courses												
ENG 111								√		√		√
ENG 127								√		√		√
BAN 211						√						
ENS 221						√	√					√
ACT 215										√	√	
MGT 329										√	√	
Mathematics and General Science Courses												
PHY 111	√											
MAT 115	√	√										
CSE 117	√	√				√		√				
MAT 125	√	√										
PHY 121	√	√										
PHY 122	√									√		
MAT 213	√											
MAT 217	√											
MAT 317	√	√										
MAT323	√	√										
Core Courses												
CSE 112	√		√		√							
CSE 113	√	√										
EEE 117	√	√	√	√								√
CSE 121	√	√	√	√								
CSE 122		√		√	√					√		√
EEE 127	√	√	√	√						√		√
EEE 128	√	√			√					√		√
CSE 213	√	√	√									

CSE 214		√	√	√	√					√		√
CSE 215		√	√	√								
CSE 216		√		√	√					√		√
CSE 221		√	√	√								
CSE 222		√		√	√					√		√
CSE 223	√	√	√									
CSE 224		√	√		√							√
CSE 225		√	√									
CSE 226			√		√					√		√
CSE 227		√		√					√			
CSE 229	√	√		√								
CSE 311	√	√	√	√								
CSE 313	√	√										
CSE 314		√	√	√	√							√
CSE 315	√		√									
CSE 316		√	√		√							√
CSE 317	√	√							√	√		
CSE 318		√		√	√					√		√
CSE 321	√	√	√									√
CSE 322		√	√	√	√							√
CSE 323	√	√	√	√								√
CSE 324			√	√								√
CSE 325	√			√								
CSE 326		√			√					√		√
CSE 411	√	√				√						
CSE 412		√	√		√							√
CSE 413		√		√								
CSE 414		√	√	√	√							√
CSE 415	√	√	√	√					√	√		
CSE 416			√		√				√	√		
CSE 417		√	√	√								
CSE 418		√	√		√							√
CSE 419		√		√								
CSE 421		√	√	√					√	√		√
CSE 423		√		√								√
CSE 424		√	√	√	√							√
Optional Courses												
CSE 431	√											
CSE 433	√		√									√
CSE 435	√	√	√	√	√							
CSE 436	√		√	√	√	√			√			√
CSE 437	√	√	√		√							√
CSE 439	√									√		√
CSE 441	√											√

CSE 443	√		√		√					√		√
CSE 445	√		√		√					√		
CSE 447	√		√	√	√							√
CSE 449	√	√	√					√				
CSE 451	√		√							√		
CSE 453		√	√		√					√		
CSE 455	√		√	√	√					√		
CSE 457	√	√			√					√		
CSE 459	√	√										√
CSE 461	√		√	√								
CSE 463	√		√								√	
CSE 465	√			√								√
CSE 467	√	√	√	√								
CSE 468		√	√	√	√							√
CSE 469		√						√		√		
CSE 471	√		√									
CSE 473	√		√		√							√
CSE 475	√		√							√		
CSE 477	√	√	√		√							
CSE 479	√		√		√							√
CSE 481	√	√	√	√								
CSE 482		√	√	√	√					√		√
CSE 483	√		√	√	√							√

Part B:

17. Structure of the Curriculum

a) Duration of the Program: Years: 4 Semesters: 8

A student will be able to complete the program within 4 (four) years' time. The Maximum period to complete the undergraduate program is 6 (six) years from the date of first admission to the University.

b) Admission Requirements

i) For regular students

Every applicant, without any exception, must fulfill the admission requirements as laid by the University. A higher secondary certificate or its equivalent in Science group with the basic educational requirements. Minimum qualifying GPA to be eligible to apply is 2.5 individually in SSC/'O' level or equivalent and HSC/'A' level or equivalent examinations. For 'O' and 'A' level students, their Average B grade will be consider for admission. For all other foreign certificates, the University as per rules of the Bangladesh Government / University Grants Commission (UGC) of Bangladesh will determine equivalent.

ii) For Diploma students

The 04 (four) years Diploma student in the field of Computers / Electrical / Electronics / Mechanical / Telecommunications / Power and equivalent having Minimum CGPA of 2.5 (Out of 4) may also get admission at the Evening section. They will get Maximum 15 credits waiver from the GEC courses. Students must have minimum B grade in these waived courses.

iii) Credit Transfer

Credit transfer from an educational institution with a system similar to Hamdard University Bangladesh may be considered. Such candidates will have to apply with required documents and are subject to credit transfer rules of this University. The total credits transferred by a student from other Universities will not exceed 50 credits for undergraduate program.

c) Total minimum credit requirement to complete the program: 152.0 Credits

For regular students

(i) Completion of 152.0 credit hours courses (Theory + Laboratory)
(ii) Completion of 6 credits Dissertation with at least a 'C+' grade (GPA 2.5)
(iii) Passing of all the courses individually and maintaining a minimum CGPA of 2.5.

(b) For Diploma students

(i) Completion of 137 credit hours courses (Theory + Laboratory)
(ii) Completion of 6 credits Dissertation with at least a 'C+' grade (GPA 2.5)

(iii) Passing of all the courses individually and maintaining a minimum CGPA of 2.5.

d) Total class weeks in a semester: 18 weeks

Students take courses in each semester and these courses have credits assigned to them and the credits are counted towards the degree. Credit Hours for a course are assigned on the basis of a 22 working weeks Semester and 2 classes of 50 minutes duration each week. Laboratory classes will be of 2 hours duration in a week for each course.

e) Minimum CGPA requirements for graduation: CGPA 2.5

f) Maximum academic years of completion: 6 years

g) Category of Courses:

i. General Education Courses:

Code	Course Title	Credits
ENG 111	English Basics	3.0
ENG 127	English Skills	3.0
BAN 211	History of Emergence of Bangladesh (Bangladesh Studies)	3.0
ENS 221	Environmental Science	3.0
ACT 215	Financial and Managerial Accounting	3.0
MGT 329	Industrial and Operational Management	3.0

ii. Mathematics and General Science Courses

Code	Course Title	Credits
PHY 111	Physics I	3.0
MAT 115	Differential & Integral Calculus	3.0
CSE 117	Engineering Ethics and Cyber Law	3.0
MAT 125	Linear Algebra and Complex Variables	3.0
PHY 121	Physics II	3.0
PHY 122	Physics II Lab	1.0
MAT 213	Statistics & Probability	3.0
MAT 217	Ordinary & Partial Differential Equation	3.0
MAT 317	Mathematical Methods	3.0
MAT 323	Numerical Analysis	3.0

iii. Core Courses

Code	Course Title	Credits
CSE 112	Basic Computer Skills	1.0
CSE 113	Discrete Mathematics	3.0

EEE 117	Electrical Engineering	3.0
CSE 121	Structured Programming Language	3.0
CSE 122	Structured Programming Language Lab	1.0
EEE 127	Electronic Devices & Circuits	3.0
EEE 128	Electrical and Electronic Circuits Lab	1.0
CSE 213	Digital Electronics	3.0
CSE 214	Digital Electronics Lab	1.0
CSE 215	Data Structures	3.0
CSE 216	Data Structures Lab	1.0
CSE 221	Computer Algorithms	3.0
CSE 222	Computer Algorithms Lab	1.0
CSE 223	Object Oriented Programming	3.0
CSE 224	Object Oriented Programming Lab	1.0
CSE 225	Digital Logic Design	3.0
CSE 226	Digital Logic Design Lab	1.0
CSE 227	Data Communication	3.0
CSE 229	Computer Architecture	3.0
CSE 311	Theory of Computing	3.0
CSE 313	Database Management System	3.0
CSE 314	Database Management System Lab	1.0
CSE 315	Microprocessors & Embedded System	3.0
CSE 316	Microprocessors & Embedded System Lab	1.0
CSE 317	Computer Networks	3.0
CSE 318	Computer Networks Lab	1.0
CSE 321	Operating System	3.0
CSE 322	Operating System Lab	1.0
CSE 323	System Analysis & Design	3.0
CSE 324	System Analysis & Design Lab	1.0
CSE 325	Computer Peripherals & Interfacing	3.0
CSE 326	Computer Peripherals & Interfacing Lab	1.0
CSE 411	Artificial Intelligence	3.0
CSE 412	Artificial Intelligence Lab	1.0
CSE 413	Computer Graphics	3.0
CSE 414	Computer Graphics Lab	1.0
CSE 415	Digital Signal Processing	3.0
CSE 416	Digital Signal Processing Lab	1.0
CSE 417	Compiler Design	3.0
CSE 418	Compiler Design Lab	1.0
CSE 419	Communication Engineering	3.0
CSE 421	Software Engineering	3.0

CSE 423	Digital Image Processing	3.0
CSE 424	Digital Image Processing Lab	1.0

iv. Optional Courses

Code	***Course Title***	***Credits***
CSE 431	Multimedia System Design	3.0
CSE 433	Telecommunication Engineering	3.0
CSE 435	Web Programming	3.0
CSE 436	Web Programming Lab	1.0
CSE 437	Software Project Management	3.0
CSE 439	Distributed System	3.0
CSE 441	Network Programming	3.0
CSE 443	Pattern Recognition	3.0
CSE 445	VLSI Design	3.0
CSE 447	Cellular & Mobile Communication	3.0
CSE 449	Digital Communications	3.0
CSE 451	Wireless & Mobile Communication	3.0
CSE 453	Telecommunication Transmission and Switching	3.0
CSE 455	Internet of Things	3.0
CSE 457	Mobile Network Management and Security	3.0
CSE 459	Embedded Network Systems	3.0
CSE 461	Client Server Computing	3.0
CSE 463	Decision Support System	3.0
CSE 465	Wireless Network	3.0
CSE 467	Simulation and Modeling	3.0
CSE 468	Simulation and Modeling Lab	1.0
CSE 469	Computer and Network Security	3.0
CSE 471	Parallel Processing	3.0
CSE 473	Robotics Technology	3.0
CSE 475	E-commerce & E-business	3.0
CSE 477	Cloud Computing	3.0
CSE 479	Advanced Database Management System	3.0
CSE 481	Machine Learning	3.0
CSE 482	Machine Learning Lab	1.0
CSE 483	Big Data Analytics	3.0

v. Capstone course/Internship/Thesis/Projects/Portfolio

Code	***Course Title***	***Credits***
CSE 429	Thesis/ Project/Internship	6.0

18. Year/Level/Semester/Term wise distribution of courses:

First Year 1st Semester

Course Code	*Course Title*	*Credits*
CSE 112	Basic Computer Skills	1.0
CSE 113	Discrete Mathematics	3.0
ENG 111	English Basics	3.0
PHY 111	Physics I	3.0
MAT 115	Differential & Integral Calculus	3.0
EEE 117	Electrical Engineering	3.0
CSE 117	Engineering Ethics and Cyber Law	3.0
	Total:	***19.0***

First Year 2nd Semester

Course Code	*Course Title*	*Credits*
ENG 127	English Skills	3.0
CSE 121	Structured Programming Language	3.0
CSE 122	Structured Programming Language Lab	1.0
MAT 125	Linear Algebra & Complex Variables	3.0
PHY 121	Physics II	3.0
PHY 122	Physics II Lab	1.0
EEE 127	Electronic Devices & Circuits	3.0
EEE 128	Electrical and Electronic Circuits Lab	1.0
	Total:	***18.0***

Second Year 3rd Semester

Course Code	*Course Title*	*Credits*
BAN 211	History of Emergence of Bangladesh	3.0
CSE 213	Digital Electronics	3.0
CSE 214	Digital Electronics Lab	1.0
CSE 215	Data Structure	3.0
CSE 216	Data Structure Lab	1.0
ACT 215	Financial and Managerial Accounting	3.0
MAT 213	Statistics and Probability	3.0
MAT 217	Ordinary & Partial Differential Equation	3.0
	Total:	***20.0***

Second Year 4th Semester

Course Code	*Course Title*	*Credits*
ENS 221	Environmental Science	3.0

CSE 221	Computer Algorithms	3.0
CSE 222	Computer Algorithms Lab	1.0
CSE 223	Object Oriented Programming	3.0
CSE 224	Object Oriented Programming Lab	1.0
CSE 225	Digital Logic Design	3.0
CSE 226	Digital Logic Design Lab	1.0
CSE 227	Data Communication	3.0
CSE 229	Computer Architecture	3.0
	Total:	***21.0***

Third Year 5th Semester

Course Code	***Course Title***	***Credits***
CSE 311	Theory of Computing	3.0
CSE 313	Database Management System	3.0
CSE 314	Database Management System Lab	1.0
CSE 315	Microprocessors & Embedded System	3.0
CSE 316	Microprocessors & Embedded System Lab	1.0
CSE 317	Computer Networks	3.0
CSE 318	Computer Networks Lab	1.0
MAT 317	Mathematical Methods	3.0
	Total:	***18.0***

Third Year 6th Semester

Course Code	***Course Title***	***Credits***
CSE 321	Operating System	3.0
CSE 322	Operating System Lab	1.0
CSE 323	System Analysis & Design	3.0
CSE 324	System Analysis & Design Lab	1.0
CSE 325	Computer Peripherals & Interfacing	3.0
CSE 326	Computer Peripherals & Interfacing Lab	1.0
MGT 329	Industrial and Operational Management	3.0
MAT 323	Numerical Analysis	3.0
	Total:	***18.0***

Fourth Year 7th Semester

Course Code	***Course Title***	***Credits***
CSE 411	Artificial Intelligence	3.0
CSE 412	Artificial Intelligence Lab	1.0
CSE 413	Computer Graphics	3.0
CSE 414	Computer Graphics Lab	1.0
CSE 415	Digital Signal Processing	3.0

CSE 416	Digital Signal Processing Lab	1.0
CSE 417	Compiler Design	3.0
CSE 418	Compiler Design Lab	1.0
CSE 419	Communication Engineering	3.0
	Total :	***19.0***

Fourth Year 8th Semester

Course Code	***Course Title***	***Credits***
CSE 421	Software Engineering	3.0
CSE 423	Digital Image Processing	3.0
CSE 424	Digital Image Processing Lab	1.0
CSE 4**	**Option – (Any Two)**	6.0
CSE 429	Project/Thesis/Internship	6.0
	Total :	**19.0**
	Grand Total :	***152***

Part C

19. Description of all courses of the program:

19. a) General Education Courses:

Course Code: ENG 111	**Credits: 3.0**	**Year: First**	**Semester: First**
Course Title: English Basics		**Course Status: Theory**	

Rationale of the Course:
This is an upper-intermediate level course where you will learn about the skills that are required for your development in learning English. The course **English Basics** is offered to the students requiring a semi-mastery level skill in reading, writing, speaking, and listening. The aim of this course is to provide knowledge of English language in terms of grammar, composition and spoken English.

Course Content:
Introduction to Language and Communication, Parts of Speech: Noun; Count ability, number, Possessive case; Use of Articles; Parts of Speech: Adjective and Adverb, conversion of adjectives, Position of adjectives, Formation of Adverbs, Position of Adverbs, Degree of comparison: comparatives and superlatives; Parts of Speech: Verbs, Gerund, Infinitive, participle; Modal auxiliaries; Tenses; Active and Passive Voice; Parts of Speech: Preposition; Structures of sentences: simple, compound and complex; Asking questions; Direct & Indirect Speech; Punctuation and Capitalization; Reading comprehension; Paragraph writing; Writing formal letters.

Recommended Books:

1. Murphy, R. Intermediate English Grammar.
2. Wren and Martin. High School English Grammar and Composition.
3. Das, P.C. Applied English Grammar and Composition.
4. Lott, H. Real English Grammar.
5. Islam, J. Advanced Reading and Writing.
6. Ahmed, S. Learning English The Easy Way.
7. Shahidullah, M. Writing Effective Paragraphs.

Course Learning Outcomes: At the end of the course the students will be able to

CLO1	**Define** different types of grammar; **Understand** the rules and uses of grammar, the audio clips, reading comprehension;

CLO2	**Explain** any reading passages, read critically, express and support opinions, summarize, and analyze texts using a variety of methods;
CLO3	**Discuss** different topics, introduce themselves accurately;
CLO4	**Create** paragraphs, picture description, letter and application with innovative ideas;
CLO5	**Identify** and **apply** common sentence elements such as spelling, punctuation, grammar, and vocabulary; and

Mapping of Course Learning Outcomes (CLOs) to Program Learning Outcomes PLOs:

CLOs	Program Learning Outcomes (PLOs)											
	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1										√		√
CLO2												√
CLO3								√				√
CLO4										√		√
CLO5												√

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lecture	Assignment / Midterm Exam
CLO2	Lecture	Assignment / Class Test /Midterm Exam
CLO3	Lecture / Group Discussion	Class Test / Midterm Exam/ Final Exam/Viva Voce/ Presentation
CLO4	Lecture / Group Discussion	Presentation/ Class Test / Final Exam
CLO5	Lecture	Assignment / Class Test/ Midterm Exam/ Final Exam

Course Code: ENG 127	**Credits: 3.0**	**Year: First**	**Semester: Second**
Course Title: English Skills		**Course Status: Theory**	

Rationale of the Course:

This course is designed to equip students with the necessary skills to perform effectively and competently in the social and academic contexts. This is done through the integration of the four language skills (listening, speaking, reading and writing). The course focuses on enhancing the

students' abilities to use the language by exploiting a variety of materials in varied situations. Appropriate consideration is given to the development of higher-level grammatical construction, vocabulary expansion and extensive reading activities which are intended to increase students' lexical density.

Course Content:

This course is designed to equip students with the necessary skills to perform effectively and competently in the social and academic contexts. This is done through the integration of the four language skills (listening, speaking, reading and writing). The course focuses on enhancing the students' abilities to use the language by exploiting a variety of materials in varied situations. Appropriate consideration is given to the development of higher-level grammatical construction, vocabulary expansion and extensive reading activities which are intended to increase students' lexical density.

Listening: Listening for main ideas/ key information, listening for specific details, listening and understanding both implicit and explicit messages, listening and responding to texts (i.e., following instructions, answering questions, reacting to texts, etc.), listening and note-taking.

Speaking: Understanding social conventions (i.e., formal/informal speech, turn taking etc.), guided conversations (involving different functions and situations, e.g., greetings, requesting, apologizing, etc.), impromptu talks, reporting news items, role-plays and simulations, preparing and presenting talks on a given theme, interviews, story telling, informal debates and group discussions, public speaking.

Reading: Reading strategies (skimming, scanning, predicting, inference, etc.), speed reading, active reading (highlighting, getting information from text quickly finding your way around texts noting key words, following main arguments, interacting with the text and summarizing), reading to improve linguistic skills and expand vocabulary; understanding overall discourse to relate structure and meaning to the analysis and comprehension of the text, extensive reading (reading outside class books selected by teachers), reading journal articles.

Writing: Paraphrasing, summarizing, organizing an essay: thesis statement, Supporting sentences, logical order and conclusions, writing essays following different modes of writing: argumentative essay and research essay, business Letters: Sales Letter, Order Letter, Complaint Letter, Letter of Recommendation, Letter of Resignation.

Course Learning Outcomes: At the end of the course the students will be able to

CLO1	**Define** different terms; **Understand** the rules and uses of sentences, the audio clips, reading comprehension
CLO2	**Explain** any reading passages, read critically, express and support opinions, summarize, and analyze texts using a variety of methods
CLO3	**Discuss** different topics, introduce themselves accurately;
CLO4	**Create** paragraphs, picture description, letter and application with innovative ideas

CLO5	**Identify** and **apply** common sentence elements such as spelling, punctuation, grammar, and vocabulary

Mapping of Course Learning Outcomes (CLOs) to Program Learning Outcomes PLOs:

CLOs	Program Learning Outcomes (PLOs)											
	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1										√		√
CLO2												√
CLO3								√				
CLO4										√		√
CLO5												√

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lecture	Assignment / Midterm Exam
CLO2	Lecture	Assignment / Class Test /Midterm Exam
CLO3	Lecture / Group Discussion	Class Test / Midterm Exam/ Final Exam/Viva Voce/ Presentation
CLO4	Lecture / Group Discussion	Presentation/ Class Test / Final Exam
CLO5	Lecture	Assignment / Class Test/ Midterm Exam/ Final Exam

Course Code: BAN 211	**Credits: 3.0**	**Year: Second**	**Semester: Third**
Course Title: History of Emergence of Bangladesh		**Course Status: Theory**	

Rationale of the Course:

History of Emergence of Bangladesh has been designed for the purpose of learning the entire history of Bangladesh, to understand present Bangladesh in the light of history and to provide them with basic knowledge of current politics and economy of the country in the focus of B.Sc. level students. This course will intensify students understanding of multifarious interconnection of historical events which lead to the formation of Bangladesh, current trend in political and economic development thereby improving critical thinking along with their written and oral communication skills, quantitative skills and technical literacy. It will also enhance their understanding of current phenomena in the light of history which will make them responsible global citizen.

Course Content:

Sources of History, History in Nation Building, Ancient Geography and trade links with other world, Pala and Sena dynasties, Muslim conquest of Bengal, socio-economic and cultural changes, Unification of Bengal, The development of Bengali Language and Literature. The independent Sultanate in Bengal, Bengal under the Mughals, The Nawabi Rule in Bengal (1700-1765). British Colonial Rule and Response, Introduction of Zamindari system and its decline, changes of socio-economic condition, Resistance movements, English education and its impact. Revival of statehood in Bengal, the growth of Indian National Congress, the creation of New Province of East Bengal and Assam, Muslim League (1906), Bengal Pact (1923). Foundation of Awami League, Language Movement of 1952, United Front and Fall of Muslim League, the Military Rule of Ayub Khan, Economic Disparity between the two regions, cultural suppression of West Pakistan, 6-point Movement, Mass Upsurge in 1969, Rule of Yahya Khan, Election of 1970, the War of Independence and the Emergence of Bangladesh.

Books Recommended:

1. Sirajul Islam - History of Bangladesh 1704-1971, Asiatic Society of Bangladesh.
2. Khan A.R. & M. Hassain -Development Strategy for Bangladesh.
3. Anu Mahamood - Inside of Bangladesh Economy.

Course Learning Outcomes (CLOs): At the end of the Course, the Student will be able to:

CLO1	**Remember** the history, geography, trade, socio-economic and cultural changes, Language and Literature, the independent Sultanate, Bengal under the Mughals, BritishCLOonial Rule and Response, Introduction of Zamindari system, Resistance movements.
CLO2	**Understand** the revival of statehood in Bengal, the growth of Indian National Congress, the creation of New Province of East Bengal and Assam, Muslim League, foundation of Awami League and Language Movement, Economic Disparity between the two regions, 6-point Movement, Mass Upsurge, Election of 1970, the War of Independence and the Emergence of Bangladesh

Mapping of Course Learning Outcomes (CLOs) to Program Learning Outcomes PLOs:

CLOs	Program Learning Outcomes (PLOs)											
	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1						√						
CLO2						√						

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lecture	Assignment / Midterm Exam
CLO2	Lecture / Group Discussion	Assignment/Class Test/Presentation/ Midterm Exam/ Final Exam

Course Code: ENS 221	**Credits: 3.0**	**Year: Second**	**Semester: Fourth**
Course Title: Environmental Science		**Course Status: Theory**	

Rationale of the Course:

The human civilization has been facing the climate change impacted phenomena for a couple of decades. Moreover, there have seen an extensive range of challenges like flood, cyclone, sea level rise and storm surge in Bangladesh. But, the root cause of global warming is the emission of greenhouse gas in a large scale. So, the renewable energy and efficient technology are quite required in the present context. In this regard, the course environmental science would be able to create a huge awareness among the students.

Course Content:

Introduction: Environmental science and sustainability of ecosystems and biodiversity. Atmospheric structure. Radiation balance of earth. Chemical species and particulates in the atmosphere. Weather and climate. The climate variables and its components. Human-induced climate variables. Climate change models. Global and local climate change predictability. Environmental pollution and its effect on human, animal and plant health. Population density and sanitation. Availability of safe water. Air pollution. Global warming. Ozone depletion. Indoor air quality. Air, water and soil pollution by contamination with hazardous chemicals. Use of plant growth regulators, fertilizers, fruit ripening agents and pesticides in agriculture. Industrial waste. Hospital waste. Waste management systems. Sewage management. Sound pollution. Radioactive hazard. Food and waterborne diseases. Infectious diseases. Protection against diseases caused by environmental effects. Depletion of biota by environmental effects. Threatened medicinal plant species. Preservation of such species. Regulatory acts and actions to protect environment and their proper and effective implementation.

Text Books :

1. S.S Dara: A Text Book of Environmental Chemistry and Pollution Control, Chand & Company Ltd.
2. Spengler, John D. and Sexton, Ken (1983) "Indoor Air Pollution: A Public Health Perspective" *Science* (New Series) 221(4605)

Reference Books:

1. Hong, Sungmin *et al.* (1996) "History of Ancient Copper Smelting Pollution During Roman and Medieval Times Recorded in Greenland Ice" *Science* (New Series) 272(5259)
2. L. Gari (2002), "Arabic Treatises on Environmental Pollution up to the End of the Thirteenth Century", *Environment and History* **8** (4).
3. State of the Environment, Issue: Air Quality (Australian Government website page) Pollution and Society Marisa Buchanan and Carl Horwitz, University of Michigan

Course Learning Outcomes (CLOs): At the end of the Course, the Students will be able to:

CLO1	**Remember** the basic components of the environment, ecology, atmosphere, hydrosphere, lithosphere, and biosphere, Green House gas
CLO2	**Understand** the relation between biotic and a-biotic components, climate, climate change, global warming, global environmental change, greenhouse mechanism, sedimentation process, sustainable development.
CLO3	**Analyze** the impacts of sea-level rise, storm surge, cyclone, floods, droughts etc.
CLO4	**Analyze** the present laws and regulations in the context of Bangladesh like Bangladesh Climate change Strategy Action Plan, National Adaptation Programme of Action

Mapping of Course Learning Outcomes (CLOs) to Program Learning Outcomes (PLOs):

CLOs	**Program Learning Outcomes (PLOs)**											
	PLO1	**PLO2**	**PLO3**	**PLO4**	**PLO5**	**PLO6**	**PLO7**	**PLO8**	**PLO9**	**PLO10**	**PLO11**	**PLO12**
CLO1							√					
CLO2							√					
CLO3						√	√					
CLO4						√	√					√

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	**Teaching-Learning Strategy**	**Assessment Strategy**
CLO1	Lecture	Assignment / Midterm Exam
CLO2	Lecture	Assignment / Class Test /Midterm Exam
CLO3	Lecture / Group Discussion	Mini Project/ Presentation/ Midterm Exam/ Final Exam
CLO4	Lecture / Group Discussion	Assignment/Final Exam

Course Code: ACT 215	Credits: 3.0	Year: Second	Semester: Third
Course Title: Financial and Managerial Accounting		Course Status: Theory	

Rationale of the Course:

Accounting is the process of organizing, analyzing, and communicating financial information that is used for decision-making. Financial information is typically prepared by accountants—those trained in the specific techniques and practices of the profession. This course explores many of the topics and techniques related to the accounting profession. While many students will directly apply the knowledge gained in this course to continue their education and become accountants and business professionals, others might pursue different career paths. However, a solid understanding of accounting can for many still serve as a useful resource. In fact, it is hard to think of a profession where a foundation in the principles of accounting would not be beneficial. Therefore, one of the goals of this course is to provide a solid understanding of how financial information is prepared and used in the workplace, regardless of your particular career path.

Course Content:

Introduction: Definition of Accounting and Its Functions-Users of Accounting Data-The Work of an Accountant-Ethics in Financial Reporting-Generally Accepted Accounting Principles (GAAP)-Understanding of Accounting Principles: Accounting Concepts, Assumptions and Conventions-The Basic Accounting Equation/Balance Sheet Equation-Financial Transactions and their Effect on Accounting Equation -Understanding of Different Financial Statements and Summary of the Accounting Cycle.

The Recording Process: The Account- Mechanics of Double-Entry Accounting-Transactions Illustrating the Rules of Debit and Credit-The Account and the Equation-Steps in the Recording Process-Meaning of Journal-Various Kinds of Journal-Journalizing Business Transactions-Meaning of Ledger–Importance of Ledger–Posting Mechanism–Accounts and their Balances-Definition of Trial Balance–Trial Balance and Accounting Accuracy-Preparation of Trial Balance.

Adjusting the Accounts and Preparing the Statements: Need for Adjustments at the End of Accounting Period-Accrual Vs. Cash-Basis Accounting-Recognizing Revenues and Expenses-Types of Adjusting Entries-Adjusting the Accounts-The Adjusted Trial Balance-Preparing Statements from the Adjusted Trial Balance-The Adjustment Process-Cash Flow Statement-The Classified Balance Sheet/Classification of Balance Sheet Items- Owner's Equity on the Balance Sheet.

Completing the Accounting Cycle: Definition of Worksheet-Steps in Preparing Worksheet-Preparing Financial Statements from Worksheet-Preparing Adjusting Entries from a Worksheet-Preparing Closing Entries-Posting Closing Entries-Preparing Post-Closing Trial Balance.

Accounting for Merchandising Operations: Merchandising Operations-Recording Purchases of Merchandise-Recording Sales of Merchandise-Completing the Accounting Cycle-Forms of Financial Statements: Multiple-Step and Single-Step Income Statements and Classified Balance Sheet-Determining Cost of Goods Sold Under a Periodic System-Periodic Inventory System-Classifying Inventories- Determining Inventory Quantities-Inventory Costing-Inventory Errors-Estimating Inventories-Worksheet for Merchandising Company.

Accounting Information Systems: Basic Concepts of Accounting Information Systems-Computerized Accounting Systems vs. Manual Accounting Systems.

Accounting for Receivables: Types of Receivables-Accounts Receivables-Notes Receivables-Statement Presentation and Analysis.

Accounting for Plant Assets, Natural Resources, and Intangible Assets: Determining the Cost of Plant Assets-Depreciation-Depreciation Methods-Expenditures for Useful Life-Plant Assets Disposals-Exchange of Plant Assets-Natural Resources-Accounting for Intangible Assets.

Accounting for Current Liabilities and Payroll Accounting: Accounting for Current Liabilities - Contingent Liabilities - Payroll Accounting.

Text Books:

1. Weygandt & Kieso , **Accounting Principles**, (8/e), Wiley Publishing.

Reference Book:

1. William W. Pyle and Kermit D. Larson, **Fundamental Accounting Principles**, (10/e). Richard D. IRWIN. ICC. ILLINOIS.
2. Wild, Larson, Chiappetta, **Financial and Managerial Accounting**, (Latest edition)

Course Learning Outcomes (CLOs): At the end of the Course, the Student will be able to:

CLO1	Understand the conceptual basis of accounting, the basic accounting rules for recording business operations, assets and liabilities, the methodology of preparing accounting reports, and the requirements for their contents.
CLO2	Explain the basic accounting rules for recording business operations.
CLO3	Analyze managerial accounting analytical tools for internal control, pricing and developing different budgets for financial planning purposes as well as to get essential skills in capital budgeting techniques.
CLO4	Compare between the profitability ratios and investment analysis

Mapping of Course Learning Outcomes (CLOs) to Program Learning Outcomes PLOs:

CLOs	**Program Learning Outcomes (PLOs)**											
	PLO1	**PLO2**	**PLO3**	**PLO4**	**PLO5**	**PLO6**	**PLO7**	**PLO8**	**PLO9**	**PLO10**	**PLO11**	**PLO12**
CLO1											√	

CLO2											√	
CLO3										√	√	
CLO4										√	√	

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lecture	Assignment / Midterm Exam
CLO2	Lecture	Assignment / Class Test /Midterm Exam
CLO3	Lecture / Group Discussion	Class Test / Midterm Exam/ Final Exam
CLO4	Lecture / Group Discussion	Presentation/ Class Test / Final Exam

Course Code: MGT 329	**Credits: 3.0**	**Year: Third**	**Semester: Fifth**
Course Title: Industrial and Operational Management		**Course Status: Theory**	

Rationale of the Course:

This course reviews the management of operations in manufacturing, service, and government organizations. Topics include a review of the activities and responsibilities of operations management, the tools and techniques available to assist in running the operation, and the factors considered in the design of the system.

Course Content:

Introduction, evolution, management function, organization and environment. Organization: Theory and structure; Coordination; Span of control; Authority delegation; Groups; Committee and task force; Manpower planning. Personnel Management: Scope; Importance; Need hierarchy; Motivation; job redesign; Leadership; Participative management; Training; Performance appraisal; Wages and incentives; Informal groups; Organizational change and conflict. Cost and Financial Management: Elements of costs of products depreciations; Break-even analysis; Investment analysis; Benefit cost analysis. Management Accounting: Cost Planning and control; Budget and budgetary control; Development planning process; Project management; Network models for project time estimation; Estimation of confidence level; Simplex method for minimization of project time; Project effort analysis methods; Marketing Management: Concepts; Strategy; Sales promotion; Patent laws. Technology Management: Management of innovation and changes; Technology life cycle; Case studies.

Recommended Readings:

1. Min-Jung Yo, Remy Glardon, *Manufacturing Operations Management*, World Scientific Europe Ltd (July 7, 2018)

Supplementary Readings:

1. Courtney Hoover, ***Industrial Engineering and Operations Management***, Willford Press (June 3, 2016)

Course Learning Outcomes (CLOs): At the end of the Course, the Student will be able to:

CLO1	Understand the fundamentals of operations management.
CLO2	Explain the cost and financial management
CLO3	Analyze management accounting and project management.
CLO4	Compare between quantitative and qualitative analysis techniques to improve understanding of operations.

Mapping of Course Learning Outcomes (CLOs) to Program Learning Outcomes PLOs:

CLOs	**Program Learning Outcomes (PLOs)**											
	PLO1	**PLO2**	**PLO3**	**PLO4**	**PLO5**	**PLO6**	**PLO7**	**PLO8**	**PLO9**	**PLO10**	**PLO11**	**PLO12**
CLO1											√	
CLO2											√	
CLO3											√	
CLO4										√	√	

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	**Teaching-Learning Strategy**	**Assessment Strategy**
CLO1	Lecture	Assignment / Midterm Exam
CLO2	Lecture	Assignment / Class Test /Midterm Exam
CLO3	Lecture / Group Discussion	Class Test / Midterm Exam/ Final Exam
CLO4	Lecture / Group Discussion	Presentation/ Class Test / Final Exam

19. b) Mathematics and General Science Courses

Course Code: PHY 111	Credits: 3.0	Year: First	Semester: First
Course Title: Physics I		Course Status: Theory	

Rationale of the course:

Studying physics strengthens quantitative reasoning and problem solving skills that are valuable in areas beyond physics. Students who study physics or engineering physics are prepared to work on forefront ideas in science and technology, in academia, the government, or the private sector.

Course Content:

Waves: Waves and particles, different types, Transverse & Longitudinal waves, Wavelength & Frequency, Principle of Superposition waves, Standing Waves and Resonance

Oscillations: Oscillations, SHM, The force law for SHM, Differential equation of simple harmonic oscillator, total energy and average energy, combination of simple harmonic oscillations, spring mass system, Pendulums, Torsional pendulum; two body oscillation, reduced mass, damped oscillation, forced oscillation, resonance, progressive wave, power and intensity of wave, stationary wave, group and phase velocities.

Optics: Light: Light theories, source and properties of light, Refractive index, Optical path, Dispersion, Light visible range. **Fermats Principle and its Applications:** Fermats principle of least time, Rectilinear propagation of light, Reversibility of light rays, Optical path stationary. **Reflection and Refraction:** Reflection at plane surfaces & Spherical mirrors, Refraction of light, Reflecting prisms, Dispersion, Dispersive prisms, Refraction at spherical surfaces, magnification, Smith-Helmholtz Equation and Langrange Law. **Defects of images**: spherical aberration, astigmatism, coma, distortion, curvature, chromatic aberration. Theories of light, interference of light, Young's double slit experiment, displacement of fringes and its uses, Fresnel bi-prism, interference in thin films, Newton's rings, interferometers, **Diffraction:** diffraction by single slit, diffraction from a circular aperture, resolving power of optical instruments, diffraction at double slit and N slits, diffraction grating; polarization; production and analysis of polarized lights, Brewster's law, Malus law, polarization by double refraction, Nicol prism, optical activity, Polari-meters.

Thermal Physics:

Heat: Heat, Measuring temperature, Different measuring scales, Thermal expansion, Heat and temperature, Absorption of heat by solids and liquids.

Thermodynamics: First Law of Thermodynamics and its applications, **The Kinetic Theory of Gases:** Avogardo's number, Ideal gases, Pressure, temperature and RMS speed, Translational K.E., Mean free path, Distribution of molecular speeds, Molecular specific heats of an ideal gas, Degree of freedom**,** Maxwell's distribution of molecular speeds, reversible and irreversible processes, Carnot's cycle, second law thermodynamics, Carnot's theorem, entropy, Thermodynamic functions, Maxwell relations, Clausius and Clapeyron equation.

Text Books:

1. David Halliday and Robert Resnick: Physics I & II
2. N Subrahmanyam & Brijit Lal: A Text Book of Optics
3. V P Vasandani & D S Kumar : Treatise on Heat Engine

Reference Books:

1. B. L. Theraja: Engineering Physics
2. A. Beiser: Modern Physics
3. D Halliday, R Resnick & J Walker : Fundamentals of Physics
4. Y A Cengel & M A Boles : Thermodynamics

Course Learning Outcomes (CLOs): At the end of the Course, the Student will be able to:

CLO1	**Understand** Simple Harmonics Motion of the particles, concept of Two body oscillation, damped oscillations, resonances, group and phase velocities.
CLO2	**Explain** different theories of light (Reflection, Refraction, Interference, Dispersion, Diffraction, Polarization).
CLO3	**Explain** Smith-Helmholtz Equation and Langrange Law, Defects of images, coma, distortion, curvature.
CLO4	**Explain** different Thermodynamics law and their applications, Maxwell's distribution of molecular speeds, Maxwell relations, Clausius and Clapeyron equation.

Mapping of Course Learning Outcomes (CLOs) to Program Learning Outcomes (PLOs):

CLOs	Program Learning Outcomes (PLOs)											
	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√											
CLO2	√											
CLO3	√											
CLO4	√											

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lecture / Group Discussion	Assignment / Midterm Exam
CLO2	Lecture / Group Discussion	Assignment / Class Test /Midterm Exam
CLO3	Lecture / Group Discussion	Presentation/ Midterm Exam/ Final Exam
CLO4	Lecture / Group Discussion	Mini Project/ Presentation/ Class Test / Final Exam

Course Code: MAT 115	Credits: 3.0	Year: First	Semester: First
Course Title: Differential and Integral Calculus		Course Status: Theory	

Rationale of the Course:

After completing this course, the Students will be able to know about differential and integral Calculus and be able to draw graphs of different curves. They will have the idea of differentiation and Integration. This will also lead the students to calculate area and volume of regular and irregular shape and enhance the student's ability to the entrance the deep course and knowledge of Mathematics. Mathematics is key to Engineering and Calculus is a key to Mathematics.

Course Content:

Differential Calculus: Function: Domain and Range, Plotting, Limit, Continuity, Differentiation and its applications, Successive differentiation and Leibnitz's theorem. General theorems: Roll's theorem, MVT in various versions, geometric significance. Expansion of Functions Indetermine forms, Extrema; Functions of several variables; Partial differentiation to differentiation, Lagrange multipliers, Integral Calculus: Integration: Methods of Substitutions ,By parts, Partial Fraction, Successive Reduction, Definite integral: properties, geometric interpretation, use in summing series. Improper integral::classification, beta function and gamma function. Applications: Volumes of solids of revolution, Area of surface of revolution. Multiple integrals and their applications.

Recommended Readings:

1. Anton, Howard, Bivens, Irl& Davis, Stephen (2013), *Calculus*, 10th ed. John Wiley & Sons, Singapore Pte. Ltd. Indian Reprint (2016) by Wiley India Pvt. Ltd. Delhi.

Supplementary Readings:

1. Strauss, Monty J., Bradlley, Gerald L., & Smith, Karl J. (2007). *Calculus*, 3rd ed. Dorling Kindersley (India) Pvt. Ltd. (Pearson Education), Delhi. India Reprint 2011
2. Prasad, Gorakh (2016). *Differential Calculus*, 19th ed. Pothishala Pvt. Ltd. Alahabad.
3. Thomas, Jr. George B., Weir, Maurice D., & Has, Joel (2014).*Thomas' Calculus*, 13 ed. Pearson Education, Delhi, Indian Reprint 2017.

Course Learning Outcomes (CLOs): At the end of the Course, the Student will be able to:

CLO1	**Understand** Function, basic rules of differentiation and integration, Beta Function and Gamma Function.
CLO2	**Apply** Derivatives applied in rate of change, geometrically and physically application field of derivatives.
CLO3	**Apply** Determination of area, volume of regular and irregular shape and of solid geometry.

CLO4	**Evaluate** Rolle's theorem, Mean value theorem, Leibnitz thorem, Taylor and Maclaurin's series and their real life application.

Mapping of Course Learning Outcomes (CLOs) to Program Learning Outcomes (PLOs):

CLOs	Program Learning Outcomes (PLOs)											
	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√											
CLO2		√										
CLO3		√										
CLO4		√										

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lecture	Assignment / Midterm Exam
CLO2	Lecture	Assignment / Class Test /Midterm Exam
CLO3	Lecture / Group Discussion	Presentation/ Midterm Exam/ Final Exam
CLO4	Lecture / Group Discussion	Presentation/ Class Test / Mid/Final Exam

Course Code: CSE 117	**Credits: 3.0**	**Year: First**	**Semester: First**
Course Title: Engineering Ethics and Cyber Law		**Course Status: Theory**	

Rationale of the course:

This course encourages engineers to conduct themselves in a professional manner that includes adhering to the highest ethical standards and effectively managing resources and choices. It enhances the profession, increases future standards, and impresses moral mindsets and behaviors on individuals.

Course Content:

Engineering Ethics: Introduction to Ethics. Theories of Ethics. Principles of Engineering Ethics. Ethical expectation: Employers and employees, inter-professional relationship, Standards and codes: Fundamental Canons, NSPE codes, IEEE codes of conduct, ACM codes; Institutionalization of ethical conduct. Ethical Dilemmas, Choices (Whistle Blowing), Ethical Challenges for CSE Engineers with the advancement of Technology; Case studies related to ethical issues in ICT and other Engineering disciplines.

Cyber Law: Cyber laws and rights in today's digital age, Computer Crime and Cyber Security, Privacy and Confidentiality issue in CSE, Legal Framework in CSE-Copyright laws, ICT Act, Right To Information (RTI), Patents, Royalty etc. Intellectual Property Issues connected with use

and management of Digital Data The similar Acts of other countries Information Warfare: Nature of information warfare, including computer crime and information terrorism; Threats to information resources, including military and economic espionage, communications eavesdropping, computer break-ins, denial-of-service, destruction and modification of data, distortion and fabrication of information, forgery, control and disruption of information How, electronic bombs, and sops and perception management. Countermeasures, including authentication, encryption, auditing, monitoring, intrusion election, and firewalls, and the limitations of those countermeasures. Cyberspace law and law enforcement, information warfare and the military, and intelligence in the information age. Information warfare policy and ethical Issues.

Recommended Books:

1. Engineering Ethics Concepts and Cases - Charles E. Harris. Jr. , Michael S. Pritchard , Michael J. Rabins; Wadsworth Cengage Learning
2. Introduction to Engineering Ethics - Schinzinger and Martin; McGraw Hill
3. Cyber law: The Law of the Internet and Information Technology - **Brian Craig**, Prentice Hall
4. Computer Ethics and Professional Responsibility - **Terrell Ward Bynum, Simon Rogerson**, Weily Blackwel

Course Learning Outcomes (CLOs):

Upon completion of the course, the students will be able to:

CLO 1: Solve and **apply** cognitive skills and ability to identify, analysis, and **articulate** the importance of team building, project risk, and time cost management.

CLO 2: Explain the theoretical aspects of 'ethics', 'social norms', 'virtues', 'values', 'legal bindings in professional fields' etc.

CLO 3: Make Learner Conversant With The Social And Intellectual Property Issues Emerging From 'Cyberspace.

CLO 4: Explore the Legal and Policy Developments in Various Countries to Regulate Cyberspace;

CLO 5: Give Learners in Depth Knowledge of Information Technology Act and Legal Frame Work of Right to Privacy, Data Security and Data Protection.

Mapping of Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs):

Program Learning Outcomes (PLOs)	CLO 1	CLO 2	CLO 3	CLO 4	CLO 5
PLO 1: Engineering Knowledge			√	√	
PLO 2: Problem Analysis		√			
PLO 3: Design/Development of Solutions					
PLO 4: Investigation					
PLO 5: Modern Tool Usage					
PLO 6: The Engineer and Society	√				
PLO 7: Environment and Sustainability					

PLO 8: Ethics		√	√		
PLO 9: Communication					
PLO 10: Individual and Team Work					
PLO 11: Project Management and Finance					
PLO 12: Life-Long Learning					

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lecture	Assignment / Midterm Exam
CLO2	Lecture / Case Study	Assignment / Class Test /Midterm Exam
CLO3	Lecture / Case Study	Assignment / Class Test /Midterm Exam
CLO4	Lecture / Group Discussion	Class Test / Midterm Exam/ Final Exam
CLO5	Lecture / Group Discussion	Presentation/ Class Test / Final Exam

Course Code: MAT 125	**Credits: 3.0**	**Year: First**	**Semester: Second**
Course Title: Linear Algebra and Complex Variables		**Course Status: Theory**	

Rationale of the Course:

Linear algebra is fundamental in modern presentations of geometry, including for defining basic objects. It is used in engineering, physics, mathematics, astrophysics, and many other fields. It provides powerful tools for doing mathematical analysis, and often yields pleasing and unanticipated answers.

Course Content:

Determinants: Evaluation, Cramer's Rule. Matrix: operations, different types, Echelon form – elementary row operations, Rank. Linear equation: systems, solution, Gaussian elimination. Vectors, liner dependency, bases, Euclidian space, inner product and inner product space, Gram-Schimidt Orthogonalization. Liner transformations: basic concepts, linear operators. Eigenvalue and Eigenvectors: diagonalization, Cayley-Hamilton theorem; Quadraric forms and Hermitian forms. Concept of complex number, general functions of a complex variable, Complex differentiation and derivatives, Analytic functions, Cauchy-Rieman equations, Line integral of a complex function, Cauchy theorem, Cauchy's integral theorem, Taylor's and Laurent's theorem, Singular points, Residue, Cauchy's residue theorem.

Recommended Books:

1. Elementary Linear Algebra by Howard Anton
2. Linear Algebra and Its Applications, 4th Edition by David C. Lay
3. Complex variable – Schaum's Out-line Series by Spiegel

4. Complex Analysis – Prof. Haridashalder & Dr. Md. Amirul Islam

Course Learning Outcomes (CLOs): At the end of the Course, the Student will be able to:

CLO1	**Define** basis, dimensions, image, kernel, nullity, rank of matrix, linear transformations, singularities, analytic function, harmonic function, conformal mapping, zeros, poles, essential singularity etc. Understand system of linear equations, how to diagonalize a matrix, represent linear transformations by matrices and vice- versa.
CLO2	**Apply** Cauchy's Theorem and the Cauchy Integral Formulas to evaluate complex integrals.
CLO3	**Apply** characteristic polynomials to compute eigenvalues and eigenvectors.
CLO4	**Evaluate** real definite Integrals as application of Residue Theorem.

Mapping of Course Learning Outcomes (CLOs) to Program Learning Outcomes PLOs:

CLOs	**Program Learning Outcomes (PLOs)**											
	PLO1	**PLO2**	**PLO3**	**PLO4**	**PLO5**	**PLO6**	**PLO7**	**PLO8**	**PLO9**	**PLO10**	**PLO11**	**PLO12**
CLO1	√											
CLO2	√											
CLO3	√											
CLO4	√	√										

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	**Teaching-Learning Strategy**	**Assessment Strategy**
CLO1	Lecture	Assignment / Midterm Exam
CLO2	Lecture	Assignment / Class Test /Midterm Exam
CLO3	Lecture / Group Discussion	Midterm Exam/ Final Exam
CLO4	Lecture / Group Discussion	Class Test / Final Exam

Course Code: PHY 121	**Credits: 3.0**	**Year: First**	**Semester: Second**
Course Title: Physics II		**Course Status: Theory**	

Rationale of the Course:

Physics is a fundamental science that endeavors to explain all the natural phenomena that occur in the universe. Its power lies in the use of a comparatively small number of assumptions, models, laws and theories to explain a wide range of phenomena, from the incredibly small to the

incredibly large. Physics has helped to unlock the mysteries of the universe and provides the foundation of understanding upon which modern technologies and all other sciences are based.

Course Content:

Motion along a straight line: Position, Displacement, Speed, Velocity, Acceleration General and free fall.

Motion in Two and Three Dimensions: Position, Displacement, Speed, Velocity, Acceleration and instantaneous acceleration, projectile motion. Linear momentum of a particle, linear momentum of a system of particles, conservation of linear momentum, some applications of the momentum principle; Angular momentum of a particle, angular momentum of a system of particles, Kepler's law of planetary motion, the law of universal Gravitation, the motion of planets and satellites

Electricity: Electric charge, conductors, insulators, Charge and matter, Coulomb's Law, Electric field, lines of force, field strength, concept of electric flux and the Gauss's law- some applications of Gauss's law, Gauss's law in vector form, Electric potential, relation between electric field and electric potential, capacitance and dielectrics, gradient, Laplace's and Poisson's equations. Current, Current density, resistivity, the magnetic field, Ampere's law, Biot-Savart law and their applications, Laws of electromagnetic induction- Maxwell's equation.

Magnetism: Current, Current density, resistivity, the magnetic field, The Hall Effect, Ampere's law, Biot-Savart law and their applications, The Faraday's Laws of electromagnetic induction- Maxwell's equation.

Modern Physics: Galilean relativity and Einstein' special theory of relativity; Lorentz transformation equations, Length contraction, Time dilation and mass-energy relation, photoelectric effect, Compton effect; De Broglie matter waves and its success in explaining Bohr's theory, Pauli's exclusion principle, Constituent of atomic nucleus, Nuclear binding energy, different types of radioactivity, radioactive decay law; Nuclear reactions, nuclear fission, nuclear fusion, atomic power plant.

Reference Books:

1. David Halliday and Robert Resnick: Physics I & II
2. N Subrahmanyam & Brijit Lal: A Text Book of Optics
3. V P Vasandani & D S Kumar : Treatise on Heat Engine
4. B. L. Theraja: Engineering Physics
5. A. Beiser: Modern Physics
6. D Halliday, R Resnick & J Walker : Fundamentals of Physics
7. Y A Cengel & M A Boles : Thermodynamics

Course Learning Outcomes (CLOs): At the end of the Course, the Student will be able to:

CLO1	**Define** electricity, charge, Capacitance, Position and **Understand** various laws of physics like Coulomb's Law Gauss's law Ampere's Law, Biot-Savart law, Faraday's Law, Lenz' Law.

CLO2	**Apply** knowledge of electricity and magnetism to explain natural physical processes and related technological advances.
CLO3	**Apply** the concepts of modern physics, Elementary quantum mechanics and nuclear physics to various fields of engineering.
CLO4	**Evaluate** typical simple and moderately complex problems of selected topics and of the linked topics of science/engineering courses.

Mapping of Course Learning Outcomes (CLOs) to Program Learning Outcomes (PLOs):

CLOs	Program Learning Outcomes (PLOs)											
	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√											
CLO2		√										
CLO3		√										
CLO4	√											

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lecture	Assignment / Midterm Exam
CLO2	Lecture	Assignment / Class Test /Midterm Exam
CLO3	Lecture / Group Discussion	Presentation/ Midterm Exam/ Final Exam
CLO4	Lecture / Group Discussion	Mini Project/ Presentation/ Class Test / Final Exam

Course Code: PHY 122	**Credits: 1.0**	**Year: First**	**Semester: Second**
Course Title: Physics II Lab		**Course Status: Lab**	

Rationale of the Course:

The introductory physics laboratories have the following purposes and goals such as To provide an experimental foundation for the theoretical concepts introduced in the lectures, To familiarize students with experimental apparatus, the scientific method, and methods of data analysis so that they will have some idea of the inductive process by which the ideas were originated, To introduce the methods used for estimating and dealing with experimental uncertainties, including simple ideas in probability theory and the distinctions between random (statistical) and systematic "errors." The laboratory is not a contest whose object is to get the "right answer." The purpose is to learn how to gain knowledge by looking at reality, not an attempt to make reality conform to preconceptions.

Course Content:
Use of Slide Caliper/ Vernier Calipers, Screw Gauge, simple pendulum, determining spring constant of a spring, Use of the Searle's Apparatus, Simple Balance, Sono-meter, Plane Mirror, Verification of Ohm's Law, Measurement of Unknown Resistance of a sample wire, Use of a Spherical/Simple Glass Sample.

Course Learning Outcomes (CLOs): At the end of the Course, the Student will be able to:

CLO1	**Verify** the theoretical laws of physics and determination of several physical parameters using hardware tools.
CLO2	**Write** comprehensive reports on the work done in laboratory.

Mapping of Course Learning Outcomes (CLOs) to Program Learning Outcomes (PLOs):

CLOs	Program Learning Outcomes (PLOs)											
	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√											
CLO2										√		

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lab demonstration/ Lecture	Lab report/Final exam
CLO2	Lecture	Lab report/Final exam

Course Code: MAT 213	**Credits: 3.0**	**Year: Second**	**Semester: Third**
Course Title: Statistics & Probability		**Course Status: Theory**	

Rationale of the Course:
This course is designed to provide fundamental concepts of statistics and basic concepts of probability, probability distribution. Also this course describes the sampling technique, Index number, Survey Methodology and statistical hypothesis testing to test the validity of data.

Course Content:
Introduction: Meaning and scope, variables and attributes, collection, source of data, methods of data collection, level of measurements; nominal, ordinal, interval & ratio level data. and the presentation of statistical data. **Data description :** Constructing a Frequency Distribution, class interval and midpoints, Relative frequency distribution, graphical presentation, Histogram, frequency polygon. **Statistical Measures:** Introduction, the population mean, the sample mean, arithmetic mean, weight mean, the median, the mode, the Geometric Mean, median and mode of

a grouped data. **Measures of Dispersion:** Introduction, measure of dispersion, range, mean deviation, variance and standard deviation. **Linear Regression & Correlation:** Introduction, coefficient of correlation**,** the coefficient of determination, testing, linear correlation, measures of correlation and significance of correlation coefficient, Regression Analysis, least Square principles drawing the regression line, the standard error of estimate. **Basic concepts of probability:** Introduction, birth of the theory, the approach to probability, sample space & events, joint marginal and conditional probability, addition and multiplication rules. **Probability Theorems:** Random variable, conditional probability, Baye's Theorem; Standard discrete Distribution: Binomial, Poisson and Geometric Distributions; Standard Continuous Distribution Normal Exponential Distributions, Addition Law of probability.
Analysis of variance: One way and two way classification correlation and Regression Analysis: Rank correlation co efficient and Index Numbers.

Text Book:

1. R E Walpole and R H Myers: Basic Statistics and Probability

Reference Books:

1. M R Spiegel: Saum's Outline Statistics
2. J. N. Merian: Statistics

Course Learning Outcomes (CLOs): At the end of the Course, the Student will be able to:

CLO1	**Define** primary and secondary data, Frequency distribution, mean, median and mode, probability, discrete and continuous random variables, Conditional and joint probability distribution and Understand various Figure of statistical data, basic statistical concepts and measures, the basic knowledge on fundamental probability concepts, including random variable, probability of an event, additive rules and conditional probability and perform hypotheses testing.
CLO2	**Apply** the concepts of mean, median and mode, quartile deviation, mean deviation, Standard deviation, Box plot, Karl Pearson's and Bowley's coefficient of Skewness.
CLO3	**Apply** the concepts of probability independents and depends events, Bayes' Theorem, Binomial, Poisson and normal distribution, Moment generating function, some sampling Distributions, Test of Significance such as Hypothesis testing, Coefficient of rank correlation, Forecasting Methods, Components of Time Series, trend equation, Types of Index Numbers such as Unweighted, weighted, Laspeyre's, Fisher's, etc.
CLO4	**Evaluate** typical simple and moderately complex problems of selected topics and of the linked topics of all science and engineering courses.

Mapping of Course Learning Outcomes (CLOs) to Program Learning Outcomes (PLOs):

CLOs	**Program Learning Outcomes (PLOs)**											
	PLO1	**PLO2**	**PLO3**	**PLO4**	**PLO5**	**PLO6**	**PLO7**	**PLO8**	**PLO9**	**PLO10**	**PLO11**	**PLO12**

CLO1	√											
CLO2	√											
CLO3	√											
CLO4	√											

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lecture/ Group Discussion	Class Test /Assignment / Midterm Exam
CLO2	Lecture	Assignment / Class Test /Midterm Exam
CLO3	Lecture / Group Discussion	Presentation/ Final Exam
CLO4	Lecture / Group Discussion	Assignment/ Class Test / Final Exam

Course Code: MAT 217	**Credits: 3.0**	**Year: Second**	**Semester: Third**
Course Title: Ordinary and Partial Differential Equations		**Course Status: Theory**	

Rationale of the Course:

Ordinary and partial differential equation is a branch of Applied Mathematics concerning mathematical methods and techniques that are typically used in engineering and industry. Along with fields like engineering physics and engineering geology, both of which may belong in the wider category engineering science, engineering mathematics is an interdisciplinary subject motivated by engineers' needs both for practical, theoretical and other considerations out with their specialization and to deal with constraints to be effective in their work. Historically, this course consisted mostly of applied analysis, most notably: differential equations; ordinary differential equation, and partial differential equations.

Course Content:

Introduction and formation; First order ODE: variable separation, Linear, application; Second and higher order ODE with constant coefficient: Homogeneous and non-homogeneous case, annihilation method, variation of parameters, Cauchy-Euler equation; Series Solution: about ordinary points, about singular points, special equations for applications. Systems of linear first order ODE. Partial differential Equation (PDE) : Basic Concepts ; PDE of order one linear and non-liner; PDE of higher order with constant co –efficient :Homogeneous and non-homogeneous; PDE of order two with variable co-efficient.

Course Learning Outcomes (CLOs): At the end of the Course, the Student will be able to:

CLO1	**Define** Differential equation, Ordinary differential equation, Partial differential equation, Formation of differential equation, Formation of partial differential equation.
CLO2	**Apply** Separation of Variables, Homogenous equation, Equation reducible to homogenous, Exact equation, Linear Equation, Reducible to Linear Equation, Linear Differential Equations with Constant Coefficients, Linear Differential Equation with right-hand side non zero, Variation of parameter, Method of Successive approximation to solve ordinary differential equation**.**
CLO3	**Apply** Linear and Non-Linear first-order equation, Standard forms, Linear Equation of higher order, Partial Differential Equations with Constant Coefficients, Equation of second order with variable coefficients, Particular solution with boundary and initial conditions to solve partial differential equations.
CLO4	**Evaluate** mathematical models of physical systems, CR circuit, Wave equations, Boundary value problems.

Mapping of Course Learning Outcomes (CLOs) to Program Learning Outcomes (PLOs):

CLOs	**Program Learning Outcomes (PLOs)**											
	PLO1	**PLO2**	**PLO3**	**PLO4**	**PLO5**	**PLO6**	**PLO7**	**PLO8**	**PLO9**	**PLO10**	**PLO11**	**PLO12**
CLO1	√											
CLO2	√											
CLO3	√											
CLO4	√											

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	**Teaching-Learning Strategy**	**Assessment Strategy**
CLO1	Lecture	Assignment / Midterm Exam
CLO2	Lecture	Assignment / Class Test /Midterm Exam
CLO3	Lecture / Group Discussion	Presentation/ Final Exam
CLO4	Lecture / Group Discussion	Presentation/ Class Test / Final Exam

Course Code: MAT 317	**Credits: 3.0**	**Year: Third**	**Semester: Fifth**
Course Title: Mathematical Methods		**Course Status: Theory**	

Rationale of the Course:

This course is to establish a relationship between mathematics and application to other discipline such as physical science and engineering students. Also, the course is not designed to be over-

technical in terms of theoretical mathematics; rather it features a range of highly useful concepts that are at the core of applied mathematics. It contains Power Series Solution, Legendre function, Bessel's function, Green's function, Eigen function method and Strum-Liouville problem, Laplace transforms, Fourier series and Fourier transforms.

Course Content:

Complex number system, complex functions, De Moiver's theorem, Euler's identity, Unitary and Hermitian matrices, Line integrals. Fourier Analysis: Real and complex form of Fourier series, finite transform, Fourier integral, Fourier transforms & their uses in solving BVP of wave equation, Properties of Fourier series and transforms. Discrete Fourier Transform (DFT) Laplace Transform: introduction, inverse transform and transforms of derivative, Translation theorems, Dirac Delta function, Systems of linear equations.

Recommended Readings:

1. Churchill, R.V: Fourier series and Boundary Value Problem
2. Md. Abdur Rahman: Mathematical Methods (Volume One& Two)

Supplementary Readings:

1. Jeffreys and Jeffreys: Methods of Mathematical Physics
2. Stephenson: Mathematical Methods.
3. Spiegel, M.R: Fourier and Laplace Transform
4. Raisinghania, M.D: Laplace & Fourier Transform
5. Dass: Engineering Mathematics
6. Kreyszig: Advanced Engineering Mathematics

Course Learning Outcomes (CLOs):

At the end of the Course, the Student will be able to:

CLO1	**Define** Fourier coefficients, sine and cosine series, Fourier sine and cosine transforms, complex Fourier transform, Differential equations, Generating functions, Orthogonality, Recurrence relations, Integral representations, Rodrigue's formula.
CLO2	**Understand** Existence and basic properties, Inverse Laplace Transform and convolution, singularity of a rational function, series solution of linear differential equations at non singular and regular singular points. Orthogonal and orthonormal set of functions.
CLO3	**Apply** conditions to solve boundary value problems related to PDE.
CLO4	**Evaluate** integrals, Solution of linear differential equations with constant coefficients and linear systems. Generating functions, Recurrence relations, Integral representations, Neumann function.

Mapping of Course Learning Outcomes (CLOs) to Program Learning Outcomes (PLOs):

CLOs	Program Learning Outcomes (PLOs)											
	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√											
CLO2	√											
CLO3	√	√										
CLO4	√	√										

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
COL1	Lecture	Assignment / Midterm Exam
COL2	Lecture	Assignment / Class Test /Midterm Exam
COL3	Lecture / Group Discussion	Midterm Exam/ Final Exam
COL4	Lecture / Group Discussion	Class Test / Final Exam

Course Code: MAT 323	Credits: 3.0	Year: Third	Semester: Sixth
Course Title: Numerical Analysis		Course Status: Theory	

Rationale of the Course:

Numerical analysis is a technique by which the mathematical problems are formulated, so that can be solved with arithmetic. Numerical analysis is the branch of mathematics that is used to find approximations to difficult problems such as finding the roots of non-linear equations, interpolation, numerical differentiation and integration and numerical solutions of the differential equations. Most of the Mathematical problems that arise in science and engineering are very hard and sometime impossible to solve exactly. Thus, an approximation to a difficult Mathematical problem is very important to make it easier to solve. This course is studying the use of Numerical Analysis in modeling and simulation of engineering process.

Course Content:

Polynomials: Numerical solution of polynomials. Computational methods for solving problems in linear algebra, linear programming, nonlinear equations,

Solution of Algebraic and Transcendental Equation: Solution of Algebraic and Transcendental Equation using Graphical method, Analytical method, Regular falsi method, Iteration & Newton's Raphson method. **Simultaneous Linear Algebraic Equation:** Solution of simultaneous linear algebraic equation using Matrix inverse method, Cramer's method, Gauss

elimination method, Crouts method, Factorization method and Gauss Seidel iterative method. **Interpolation with Equal Interval:** Different interpolation method, Graphical method, Method of curve fitting method. **Interpolation with Unequal Interval:** Divided difference method, Newton's formulae for unequal interval, Relation between divided difference and ordinary difference, Lagrange's interpolation formula unequal interval & Hermite's interpolation formula. **Errors:** Errors and their analysis; absolute, relative and percentage errors; remainder terms of various interpolation formulae. **Numerical Integration:** General quadrature formula, Trapezoidal rule, Simpson's one third-three-eight rule, Weddle's rule, Cote's method, Numerical double and Gaussian Integration. **Numerical Solution of Ordinary Diff. Equation:** Pichards method, Euler method, Taylor's method, Rungs method, Rungs-Kutta method. **Curve Fitting and Principle of least-squares**. Scatter diagram, Methods of curve fitting- Grapgic method, Method of Least Square, Change of origin and scales, Most plausible solution og a system of linear equations, Fitting of the curve of type $y = ab^x$, $y = ax^b$, $pv^r = k$. **Eigen Value and Eigen Vectors:** Iterative method for dominant latent root, Evaluation of all the eigen values, the eigen values of real symmetric matrix, Given's method.

Text Book:

1. S. Balachandra Rao and C.K. Shantha : Numerical Methods

Reference Books:

1. Ferziger and J.H : Numerical Methods for Engineering Application.
2. Gerald and C.F.: Applied Numerical Analysis.

Course Learning Outcomes (CLOs): At the end of the Course, the Student will be able to:

CLO1	**Define** Difference operator, Algebraic and transcendental equations, Interpolation, Extrapolation, Differential Equations, Fundamental theorem of definite integral, Spline, Error analysis.
CLO2	**Apply** the concepts of numerical methods to solve Algebraic and Transcendental equations, Differential Equations in various fields of Science and Engineering.
CLO3	**Apply** mathematical technique to solve interpolation and extrapolation from a data set, linear systems by direct method and iterative techniques, and Non-linear system
CLO4	**Work out** numerical differentiation and integration whenever and wherever routine methods are not applicable.
CLO5	**Evaluate** typical physical problems of science/engineering courses by analyzing them properly.

Mapping of Course Learning Outcomes (CLOs) to Program Learning Outcomes PLOs:

CLOs	Program Learning Outcomes (PLOs)											
	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√											

CLO2		√										
CLO3	√											
CLO4		√										
CLO5		√										

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lecture	Class Test / Midterm Exam
CLO2	Lecture	Assignment / Class Test /Midterm Exam
CLO3	Lecture/Group Discussion	Assignment/Presentation/Midterm Exam/Final Exam
CLO4	Lecture/Group Discussion	Assignment / Presentation/ Class Test / Final Exam
CLO5	Lecture/Group Discussion	Assignment/Mini Project/ Final Exam

19. c) Core Courses:

Course Code: CSE 112	Credits: 1.0	Year: First	Semester: First
Course Title: Basic Computer Skills		Course Status: Lab	

Rationale of the course:
To introduce with the most recent technology of the computer, computer equipment, computer problem solving, computer programming and memory.

Course Content:
Introduction to Computer fundamentals: types and generation of computers, basic organization and functional units. Number systems, codes and logic functions. Processors, input, output and memory devices, keyboard, mouse, OMR, OCR, MICR, CD-ROM, Printers, CRT, microfilm, floppy, Study of basic hardware (internal and external) components, BIOS, Windows, MS-DOS, UNIX, and LINUX Operating System, Details study of Windows operating systems, Word processing software, MS-WORD Commands, Practice on MS-WORD, Spreadsheet Analysis software MS-Excel Commands, Practice on MS-Excel, Study of MS-Power Point Commands, Practice on MS-Power Point, Use of Internet, Browsing through Internet, Introduction to Programming with Python: Introduction to programming basics (what it is and how it works), binary computation, problem-solving methods and algorithm development. Includes procedural and data abstractions, program design, debugging, testing, and documentation. Covers data types, control structures, functions, parameter passing, library functions, arrays, inheritance and object oriented design. Laboratory exercises in Python.

Text Books:
1. Sinha P K: Computer Fundamentals
2. The Complete PC Upgrade and Maintenance Guide (16th Edition) – Mark Minasi, Faithe Wempen, Quentin Docter.

Reference Books:
1. Norton: Introduction to Computers
2. C S Frence: Computer Science

Course Learning Outcomes (CLOs):
Upon completion of the course, the students will be able to:
CLO 1: Enhance Knowledge of computer equipment, including both hardware and software and their usage.
CLO 2: Identify the components of a computer system, their functions, and the basics of the microprocessor function.
CLO 3: Demonstrate the fundamentals of computers and computer nomenclature, particularly with respect to personal computer hardware and software, the Web, and enterprise computing.

CLO 4: Learn the usage of Internet and computing resources and maintenance of hardware and software components.
CLO 5: Understand the programming basics (operations, control structures, data types, etc.) and readily **use** the Python programming language

Mapping of Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs):

Program Learning Outcomes (PLOs)	CLO 1	CLO 2	CLO 3	CLO 4	CLO 5
PLO 1: Engineering Knowledge	√		√	√	√
PLO 2: Problem Analysis					
PLO 3: Design/Development of Solutions			√		√
PLO 4: Investigation					
PLO 5: Modern Tool Usage			√		√
PLO 6: The Engineer and Society					
PLO 7: Environment and Sustainability					
PLO 8: Ethics					
PLO 9: Communication					
PLO 10: Individual and Team Work					
PLO 11: Project Management and Finance					
PLO 12: Life-Long Learning					

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lecture	Assignment / Midterm Exam
CLO2	Lecture / Case Study	Assignment / Class Test /Midterm Exam
CLO3	Lecture / Case Study	Assignment / Class Test /Midterm Exam
CLO4	Lecture / Group Discussion	Class Test / Midterm Exam/ Final Exam
CLO5	Lecture / Group Discussion	Presentation/ Class Test / Final Exam

Course Code: CSE 113	**Credits: 3.0**	**Year: First**	**Semester: First**
Course Title: Discrete Mathematics		**Course Status: Theory**	

Rationale of the course:
To understand the propositional logics as well as other fundamental concepts of computer science like algorithms, data structures, and other mathematical concepts.

Course Content:

Prepositional and Periodic Calculus: Statements and compound statement, conjunction, negation, proposition and truth-tables, algebra of propositions and propositional calculus.

Set theory: Basic concepts of sets and set elements, universal set, subsets, set operations, algebra of sets, power sets, mathematical induction.

Relations: Basic concept, pictorial representation of relation, inverse relations and properties of relations, Equivalence relations, Generating functions, Algebraic structures

Functions: Basic concepts, graphs of function; one-one, onto and invertible functions.

Vectors and Matrices: Vectors, Matrices, Matrix addition and multiplication, Transpose, Invertible Matrices and Determinants.

Graph theory: Graphs and multigraphs, degree, connectivity, special graphs, labeled graphs. Paths and trees, Semi graph.

Planar graphs: Planar and Nonplanar graphs, Rooted trees, ordered rooted trees.

Posets and lattices: Partially ordered sets, supermen and infimum Lattices.

Proposition Calculus: Statements and compound statements, conjunction, Disjunction, negation, Truth tables, Tautologies and contradictions, conditional and bi-conditional statements.

Text Book:

B. Discrete Mathmetics and its application - Kcnncih H. Roscn.

Reference Books:

1. Stant : Discrete Structure
2. Bernard Kolman, Robert C. Busby, Sharon Ross : Discrete Mathematics

Course Learning Outcomes (CLOs):

After completing the course, the students will be able to:

CLO 1: Apply propositional logic, predicate logic, and theorem providing for mathematical reasoning.

CLO 2: Understand counting, permutations, and combinations for combinational analysis.

CLO 3: Derive the growth of functions, complexity analysis of algorithms, and integer algorithms for algorithmic thinking and demonstrate this knowledge and write reports for realistic problem solving.

CLO 4: Learn discrete structures such as sets, functions, relations, graphs, and trees for modeling discrete objects and demonstrate this knowledge and **write** reports for realistic problem solving.

Mapping of Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs):

Program Learning Outcomes (PLOs)	CLO 1	CLO 2	CLO 3	CLO 4
PLO 1: Engineering Knowledge	√			
PLO 2: Problem Analysis	√		√	√

PLO 3: Design/Development of Solutions				
PLO 4: Investigation				
PLO 5: Modern Tool Usage				
PLO 6: The Engineer and Society				
PLO 7: Environment and Sustainability				
PLO 8: Ethics				
PLO 9: Communication				
PLO 10: Individual and Team Work				
PLO 11: Project Management and Finance				
PLO 12: Life-Long Learning				

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lecture	Assignment / Midterm Exam
CLO2	Lecture	Assignment / Class Test /Midterm Exam
CLO3	Lecture / Group Discussion	Presentation/ Final Exam
CLO4	Lecture / Group Discussion	Presentation/ Class Test / Final Exam

Course Code: EEE 117	**Credits: 3.0**	**Year: First**	**Semester: First**
Course Title: Electrical Engineering		**Course Status: Theory**	

Rationale of the course:

To understand the behavior of linear electric circuits in their different regimes: direct current, alternating current and transient state

Course Content:

Circuit variables and elements: Voltage, current, power, energy, independent and dependent sources, and resistance. Basic laws: Ohm's law, Kirchoff's current and voltage laws. Simple resistive circuits: Series and parallel circuits, voltage and current division, wye-delta transformation. Techniques of circuit analysis: Nodal and mesh analysis including supernode and supermesh. Network theorems: Source transformation, Thevenin's, Norton's and superposition theorems with applications in circuits having independent and dependent sources, maximum power transfer condition and reciprocity theorem. Energy storage elements: Inductors and capacitors, series parallel combination of inductors and capacitors. Responses of RL and RC circuits: Natural and step responses.

Magnetic quantities and variables: Flux, permeability and reluctance, magnetic field strength, magnetic potential, flux density, magnetization curve. Laws in magnetic circuits: Ohm's law and Ampere's circuital law. Magnetic circuits: series, parallel and series-parallel circuits.

average power, phasors and complex quantities, impedance, real and reactive power, power factor. Analysis of single phase ac circuits: Series and parallel RL, RC and RLC circuits, nodal and mesh analysis, application of network theorems in ac circuit, circuits simultaneously excited by sinusoidal sources of several frequencies, transient response of RL and RC circuits with sinusoidal excitation. Resonance in ac circuits: Series and parallel resonance. Magnetically coupled circuits. Analysis of three phase circuits: Three phase supply, balanced and unbalanced circuits, and power calculation.

Text Books:

1. Alexander and Sadiku: Fundamentals of Electric Circuits
2. R L Boylestad: Introductory Circuit Analysis

Reference Books:

1. R C. DORF: Introduction to Electric Circuits
2. NILSSON: Introduction to Electrical Circuits
3. R M Kerchner and G F Corcoran: Alternating Current Circuits

Course Learning Outcomes (CLOs):

After completing the course, the students will be able to:

CLO 1: Understand fundamental concepts of DC and AC circuit behavior.

CLO 2: Develop and **solve** mathematical representations for simple RLC circuits.

CLO 3: Able to employ circuit analysis theorems and methods effectively.

Mapping of Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs):

Program Learning Outcomes (PLOs)	**CLO 1**	**CLO 2**	**CLO 3**
PLO 1: Engineering Knowledge	√		√
PLO 2: Problem Analysis		√	
PLO 3: Design/Development of Solutions		√	√
PLO 4: Investigation	√		
PLO 5: Modern Tool Usage			
PLO 6: The Engineer and Society			
PLO 7: Environment and Sustainability			
PLO 8: Ethics			
PLO 9: Communication			
PLO 10: Individual and Team Work			
PLO 11: Project Management and Finance			
PLO 12: Life-Long Learning	√		√

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	**Teaching-Learning Strategy**	**Assessment Strategy**
CLO1	Lecture	Assignment / Midterm Exam
CLO2	Lecture	Assignment / Class Test /Midterm Exam

CLO3	Lecture / Group Discussion	Presentation/ Class Test / Final Exam

Course Code: CSE 121	**Credits: 3.0**	**Year: First**	**Semester: Second**
Course Title: Structured Programming Language		**Course Status: Theory**	

Rationale of the course:
To give students the knowledge of basic programming skills of writing programs that will make him/her fit for applying structured language programming to develop programs.

Course Content:
Programming Concepts: Programming language, Language processors, Problem solving with computer problem definitions, analysis, algorithms, flowcharts, pseudo code, coding, running the programs, debugging, testing, documentations.
C Programming Language: Overview of C; C fundamentals; Operators and expressions; Data input and output; Control statements; program structure storage classes, automatic variables, external (global) variables, static variables, multifile programs; Character strings; Arrays; Functions; Structures and unions; Pointers; Data files.
Additional features of C: Enumerations, commend line parameters, macros, C preprocessor.

Text Books:
1. C The Complete Reference: Herbert Schildt
2. Schuam's Outlines Series: C Programming

Reference Book:
1. Programming in ANSI C: Balaguruswamy

Course Learning Outcomes (CLOs):
Upon completion of the course, the students will be able to:
CLO 1: Introduce students to high-level, general-purpose, structured programming languages and applications.
CLO 2: Understand and **explain** the basic principles of algorithm design.
CLO 3: Understand, **distinguish**, and **classify** the basic principles of computer programming.
CLO 4: Know and **use** the C programming language to write source code to solve problems.
CLO 5: Familiarize with and use C software development tools, analyze complex problems, solution (algorithm) construction and debug errors.

Mapping of Program Learning Outcomes (PLO) and Course Learning Outcomes (CLO):

Program Learning Outcomes (PLOs)	CLO 1	CLO 2	CLO 3	CLO 4	CLO 5
PLO 1: Engineering Knowledge	√	√	√	√	
PLO 2: Problem Analysis		√		√	
PLO 3: Design/Development of Solutions		√		√	

PLO 4: Investigation					√
PLO 5: Modern Tool Usage					
PLO 6: The Engineer and Society					
PLO 7: Environment and Sustainability					
PLO 8: Ethics					
PLO 9: Communication					
PLO 10: Individual and Team Work					
PLO 11: Project Management and Finance					
PLO 12: Life-Long Learning					

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lecture	Assignment / Midterm Exam
CLO2	Lecture / Case Study	Assignment / Class Test /Midterm Exam
CLO3	Lecture / Case Study	Assignment / Class Test /Midterm Exam
CLO4	Lecture / Group Discussion	Class Test / Midterm Exam/ Final Exam
CLO5	Lecture / Group Discussion	Presentation/ Class Test / Final Exam

Course Code: CSE 122	**Credits: 1.0**	**Year: First**	**Semester: Second**
Course Title: Structured Programming Language Lab		**Course Status: Lab**	

Rationale of the course:

To introduce students with introduction and development of basic principles and mechanisms of programming technology and basic programming skills for program design and development.

Course Content:

Operators and expressions; Data input and output; Control statements; program structure storage classes, automatic variables, external (global) variables, static variables, multifile programs; Character strings; Arrays; Functions; Structures and unions; Pointers; Data files, Enumerations, commend line parameters, macros, C preprocessor.

Course Learning Outcomes (CLOs):

Upon completion of the course, the students will be able to:

CLO 1: Use an integrated development environment (IDE) to write, compile, and execute C programs involving a small number of source files.

CLO 2: Utilize fundamental programming elements including: variable declaration, data types and simple data structures (arrays, strings, and structures), decision structures, loop structures, functions/methods, input and output for console and text files.
CLO 3: Adopt debugging and testing procedures to find and resolve mistakes and to decide the effectiveness of a program.
CLO 4: Write effective reports and documentation for the solution of the problems.

Mapping of Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs):

Program Learning Outcomes (PLOs)	CLO 1	CLO 2	CLO 3	CLO 4
PLO 1: Engineering Knowledge				
PLO 2: Problem Analysis		√		
PLO 3: Design/Development of Solutions	√	√		
PLO 4: Investigation				
PLO 5: Modern Tool Usage	√			
PLO 6: The Engineer and Society				
PLO 7: Environment and Sustainability				
PLO 8: Ethics				
PLO 9: Communication				
PLO 10: Individual and Team Work	√	√	√	√
PLO 11: Project Management and Finance				
PLO 12: Life-Long Learning	√	√		√

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lab demonstration/ Lecture	Assignment
CLO2	Lab demonstration/ Lecture	Assignment
CLO3	Lab demonstration/ Lecture	Presentation/Viva
CLO3	Lab demonstration/ Lecture	Lab Report /Final Exam

Course Code: EEE 127	Credits: 3.0	Year: First	Semester: Second
Course Title: Electronic Devices and Circuits		**Course Status: Theory**	

Rationale of the course:
This subject is part of the Applied Technology group and is designed to teach students about basic electronic circuit concepts, principles, and operation. Its goal is to provide a fundamental foundation for technology domains such as electronics devices, communication systems, industrial electronics, instrumentation, control systems, and diverse electronic circuit design.

Course Content:

Semiconductors, Junction diode characteristics, Bipolar Transistor: Characteristics, small signal low frequency h-parameter model, hybrid-pi model, Amplifiers: voltage and current amplifiers, Oscillators. Differential amplifiers, Operational Amplifiers, linear applications of op-amps, gain, input and output independence, off-set null adjustments, frequency response and noise. Introduction to JFET, MOSFET, PMOS, NMOS and CMOS: biasing and applications in switching circuits. SCR, TRIAC, DIAC, UJT: characteristic and applications. Introduction to rectifiers, active filters, regulated power supply.

Text Books:

1. Electronic Devices and Circuit Theory -Robert L. Boylestad and Louis Nashelsky
2. Electronic Principles – Albert P. Malvino.

Reference Books:

1. Micro Electronics Circuits-Adel S. Sedra & Keneth C. Smith
2. Operation Amplifiers and Linear Integrated Circuits-Robert F. Coughlin

Course Learning Outcomes (CLOs):

Upon completion of the course, the students will be able to:

CLO 1: Identify common electronic devices including diodes, BJTs, MOSFETs, and FETs, as well as more advanced electronic devices like operational amplifiers.

CLO 2: Analyze the characteristics and solving methods of the electronic devices

CLO 3: Demonstrate how feedback circuits work, how to model them, and how to use them.

Mapping of Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs):

Program Learning Outcomes (PLOs)	CLO 1	CLO 2	CLO 3
PLO 1: Engineering Knowledge			√
PLO 2: Problem Analysis		√	
PLO 3: Design/Development of Solutions	√		√
PLO 4: Investigation	√	√	
PLO 5: Modern Tool Usage			
PLO 6: The Engineer and Society			
PLO 7: Environment and Sustainability			
PLO 8: Ethics			
PLO 9: Communication			
PLO 10: Individual and Team Work			
PLO 11: Project Management and Finance			
PLO 12: Life-Long Learning			

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lecture	Assignment / Midterm Exam
CLO2	Lecture	Assignment / Class Test /Midterm Exam
CLO3	Lecture / Group Discussion	Presentation/ Class Test / Final Exam

Course Code: EEE 128	**Credits: 1.0**	**Year: First**	**Semester: Second**
Course Title: Electronic Devices and Circuits Lab		**Course Status: Lab**	

Rationale of the course:
Learn and become familiar with the fundamentals of electronic circuits, as well as how to use electronic devices in practical situations.

Course Content:
Kirchhoff's Voltage & Current law, Maxwell Loop Current Technique using 2-3 Meshes and 1-2 D.C. voltage sources, Verification of Supper Position Theorem using 1-2 D.C. sources, Thevenin's Theorem, Norton's Theorem, Calculation of Current, Voltage and Power using Multi-meter, AC Transient Analysis, R, R-L, R-C & R-L-C, Verification of Diode characteristics using Forward and Reversed biased, Applications of diodes in the Half Wave and Full wave circuits, Verification of Common Based Transistor Characteristics for PNP and NPN Transistor, Load-line and Output Characteristics using Common Emitter Connection, Design of Integrator and Differential Operational amplifiers using different gains, First order active filters circuit.

Text Books:

1. Electronic Devices and Circuit Theory -Robert L. Boylestad and Louis Nashelsky
2. Electronic Principles – Albert P. Malvino.

Reference Books:

1. Micro Electronics Circuits-Adel S. Sedra & Keneth C. Smith
2. Operation Amplifiers and Linear Integrated Circuits-Robert F. Coughlin

Course Learning Outcomes (CLOs):
Upon completion of the course, the students will be able to:
CLO 1: Gain knowledge about electronic circuits.
CLO 2: Acknowledge and **use** the electronic circuits and devices for theoretical and practical purposes.
CLO 3: Learn about operational amplifier circuits.
CLO 4: Overcome difficult design issues regarding electronics based on realistic aspects.

Mapping of Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs):

Program Learning Outcomes (PLOs)	CLO 1	CLO 2	CLO 3	CLO 4
PLO 1: Engineering Knowledge	√			
PLO 2: Problem Analysis				√
PLO 3: Design/Development of Solutions				
PLO 4: Investigation				
PLO 5: Modern Tool Usage			√	
PLO 6: The Engineer and Society				
PLO 7: Environment and Sustainability				
PLO 8: Ethics				
PLO 9: Communication				
PLO 10: Individual and Team Work		√		
PLO 11: Project Management and Finance				
PLO 12: Life-Long Learning				√

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lab demonstration/ Lecture	Assignment
CLO2	Lab demonstration/ Lecture	Assignment
CLO3	Lab demonstration/ Lecture	Presentation/Viva
CLO4	Lab demonstration/ Lecture	Lab Report /Final Exam

Course Code: CSE 213	Credits: 3.0	Year: Second	Semester: Third
Course Title: Digital Electronics		**Course Status: Theory**	

Rationale of the course:

To give a grasp of how to analyze and create, and digital circuits, as well as the underlying ideas about their physical functioning, analyze, build and digital circuit troubleshooting.

Course Content:

Diode logic gates, transistor switches, transistor gates, MOS gates; Logic Families: TTL, ECL, IIL and CMOS logic with operation details; Propagation delay, product and noise immunity; Open collector and high impedance gates; Electronic circuits for flip flops, counters and register, memory systems, PLA's; A/D and D/A converters with applications; S/H circuits, LED, LCD and optically coupled oscillators; Non-linear applications of OP AMPs; Analog switches. Linear

wave shaping: diode wave shaping techniques, clipping and clamping circuits, comparator circuits, switching circuits; Pulse transformers, pulse transmission, pulse generation; monostable, bistable and astable multivibrators, Schmitt trigger, blocking oscillators and time-base circuit; Timing circuits; Simple voltage sweeps, linear current sweeps.

Text Books:

1. Anil K. Maini: Digital Electronics
2. M. Moris Mano: Digital Logic and Computer Design

Reference Books:

1. Subir Kumar Sarkar, Asish Kumar De and Souvik Sarkar: Foundation of Digital Electronics and Logic Design

Course Learning Outcomes (CLO):

Upon completion of the course, the students will be able to:

CLO 1: Gather Knowledge of the operation and the structure of switching circuits.
CLO 2: Compare and **contrast** the employment of diodes and transistors in switching circuits.
CLO 3: Design and **build** the logic families, like TTL, ECL, MOSFET and CMOS.
CLO 4: Explain the usage of multi-vibrators circuits and memory elements.

Mapping of Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs):

Program Learning Outcomes (PLOs)	CLO 1	CLO 2	CLO 3	CLO 4
PLO 1: Engineering Knowledge	√	√	√	√
PLO 2: Problem Analysis				
PLO 3: Design/Development of Solutions			√	√
PLO 4: Investigation				
PLO 5: Modern Tool Usage				
PLO 6: The Engineer and Society				
PLO 7: Environment and Sustainability				
PLO 8: Ethics				
PLO 9: Communication				
PLO 10: Individual and Team Work				
PLO 11: Project Management and Finance				
PLO 12: Life-Long Learning				

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lecture	Assignment / Midterm Exam
CLO2	Lecture	Assignment / Class Test /Midterm Exam

CLO3	Lecture / Group Discussion	Presentation/ Final Exam
CLO4	Lecture / Group Discussion	Presentation/ Class Test / Final Exam

Course Code: CSE 214	**Credits: 1.0**	**Year: Second**	**Semester: Third**
Course Title: Digital Electronics Lab		**Course Status: Lab**	

Rationale of the course:
To be acquainted with integrated circuits, to design and implement digital circuits and to be able to troubleshoot implemented circuits.

Course Content:
Diode Logic (DL) Circuits, Transistor Gates and CMOS Logic Gates and Circuits, Electronic Circuits of Filp Flops, Registers and Counters, Clipping and Clamping circuits, Comparator and Multiplexer Circuits, Schmitt Trigger etc.

Course Learning Outcomes (CLOs):
Upon completion of the course, the students will be able to:
CLO 1: Use offline simulator such as DSCH and online simulator such as CircuitVerse and create a gate-level implementation of sequential and combinational logic functions.
CLO 2: Design and **implement** electronic circuits such as Flip-Flop, Registers, Counters, Comparator and Multiplexer.
CLO 3: Compare and **analyze** use of diodes and transistors as switching circuits.
CLO 4: Learn how to construct and troubleshoot simple electronic digital circuits.

Mapping of Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs):

Program Learning Outcomes (PLOs)	**CLO 1**	**CLO 2**	**CLO 3**	**CLO 4**
PLO 1: Engineering Knowledge				
PLO 2: Problem Analysis			√	
PLO 3: Design/Development of Solutions		√		
PLO 4: Investigation				√
PLO 5: Modern Tool Usage	√	√		√
PLO 6: The Engineer and Society				
PLO 7: Environment and Sustainability				
PLO 8: Ethics				
PLO 9: Communication				
PLO 10: Individual and Team Work	√	√	√	
PLO 11: Project Management and Finance				

PLO 12: Life-Long Learning	√	√		√

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lab demonstration/ Lecture	Assignment
CLO2	Lab demonstration/ Lecture	Assignment
CLO3	Lab demonstration/ Lecture	Presentation/Viva
CLO4	Lab demonstration/ Lecture	Lab Report /Final Exam

Course Code: CSE 215	**Credits: 3.0**	**Year: Second**	**Semester: Third**
Course Title: Data Structures		**Course Status: Theory**	

Rationale of the course:

Data structure is a logical & mathematical model of storing & organizing data in a particular way in a computer. In system programming application programming the methods & techniques of data structures are widely used. The study of data structure helps the students in developing logic & structured programs.

Course Content:

Data Structure : Concept of data types**,** data structure to suit software requirements; collection of similar pieces of data (sets, lists, tables of two or more dimensions, elements of lists and table, ordered lists); collections of similar pieces of data (records and fields); collections of pieces of data with particular forms of organizations (character strings, stacks, queues, binary trees, hash tables); Stack, Queues and Recursion: Fundamentals, different types of stack, queues, Recursion, Direct and indirect recursion, simulation of recursion, removal of recursion. Different types of linked lists and their operations.

Representation of data structures : Use of arrays of one or more dimensions and counters or markers; use of pointers in link list and other structures; use and management of free space.

Standard Algorithms : Methods of describing algorithms; expression of algorithms, making use of modules, selection, repetition, iteration and recursion, algorithm for accessing and updating files and for the management of indexes; algorithms for accessing and updating data structure; algorithms for merging two or more sequential files; algorithms for sorting; algorithms for searching ordered arrays (sequential, binary); hash table searching; algorithms for common tasks such as check digit calculation.

Text Books:

1. Tenenbaum, Aaron M.: Data Structure
2. Schuam Outlines Series: Data Structures With C

Referenced Books:

1. Reign Gold: Data Structure
2. Adam Drozdek : Data Structures and Algorithms in C++

Course Learning Outcomes (CLOs):

Upon completion of the course, the students will be able to:

CLO 1: Describe all types of data structures and know how to apply them effectively in program solving.

CLO 2: Summarize searching and sorting techniques.

CLO 3: Demonstrate skills in designing recursive algorithms and recursive methods.

CLO 4: Analyze and **select** the most suitable and effective algorithm for solving certain problem and case study.

CLO5: Determine and **demonstrate** bugs in program, recognize needed basic operations with algorithms and data structures.

Mapping of Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs):

Program Learning Outcomes (PLOs)	CLO 1	CLO 2	CLO 3	CLO 4	CLO 5
PLO 1: Engineering Knowledge					
PLO 2: Problem Analysis	√	√		√	
PLO 3: Design/Development of Solutions			√		√
PLO 4: Investigation				√	√
PLO 5: Modern Tool Usage					
PLO 6: The Engineer and Society					
PLO 7: Environment and Sustainability					
PLO 8: Ethics					
PLO 9: Communication					
PLO 10: Individual and Team Work					
PLO 11: Project Management and Finance					
PLO 12: Life-Long Learning					

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lecture	Assignment / Midterm Exam
CLO2	Lecture / Case Study	Assignment / Class Test /Midterm Exam
CLO3	Lecture / Case Study	Assignment / Class Test /Midterm Exam
CLO4	Lecture / Group Discussion	Class Test / Midterm Exam/ Final Exam
CLO5	Lecture / Group Discussion	Presentation/ Class Test / Final Exam

Course Code: CSE 216	Credits: 1.0	Year: Second	Semester: Third
Course Title: Data Structures Lab		Course Status: Lab	

Rationale of the course:

It strengthens the ability of the students to identify and apply the suitable data structure for the given real world problem. It enables them to gain knowledge in practical applications of data structures.

Course Content:

Array, Stack Queues and Recursion: Fundamentals, different types of stack, queues, Recursion, Direct and indirect recursion, simulation of recursion, removal of recursion. Different types of linked lists and their operations, algorithms for accessing and updating data structure; algorithms for merging two or more sequential files; algorithms for sorting; algorithms for searching ordered arrays (sequential, binary); hash table searching; algorithms for common tasks such as check digit calculation.

Course Learning Outcomes (CLOs):

Upon completion of the course, the students will be able to:

CLO 1: Write programs to solve problems.

CLO 2: Apply practical knowledge on the applications of data structures.

CLO 3: Design and **analyze** the time and space efficiency of the data structure.

Mapping of Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs):

Program Learning Outcomes (PLOs)	CLO 1	CLO 2	CLO 3
PLO 1: Engineering Knowledge			
PLO 2: Problem Analysis			√
PLO 3: Design/Development of Solutions			
PLO 4: Investigation			√
PLO 5: Modern Tool Usage			√
PLO 6: The Engineer and Society			
PLO 7: Environment and Sustainability			
PLO 8: Ethics			
PLO 9: Communication			
PLO 10: Individual and Team Work			√
PLO 11: Project Management and Finance			
PLO 12: Life-Long Learning		√	

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy

CLO1	Lab demonstration/ Lecture	Assignment
CLO2	Lab demonstration/ Lecture	Presentation/Viva
CLO3	Lab demonstration/ Lecture	Lab Report /Final Exam

Course Code: CSE 221	**Credits: 3.0**	**Year: Second**	**Semester: Fourth**
Course Title: Computer Algorithms		**Course Status: Theory**	

Rationale of the course:
Computer algorithms deals with efficient ways to solve different mathematical and real-life problems. The emphasis is on choosing appropriate data structures and designing correct and efficient algorithms to operate on these data structures. In addition, the analysis of the algorithm will be used to show the efficiency of the algorithm over the naive techniques.

Course Content:
Review of Standard Algorithms: Methods; expression, making use of modules, selection, repetition, iteration and recursion, algorithm for accessing and updating files and for the management of indexes; algorithms for accessing and updating data structure; algorithms for merging two or more sequential files; algorithms for sorting; algorithms for searching ordered arrays, hash table searching. **Structured Algorithm Design:** Modularity; top-down design, development of algorithms; appropriate use of algorithmic constricts of selection; repetitions, interaction, and recursion; algorithms for parallel tasks; methods of describing and documenting algorithms; testing, evaluation and analysis of algorithms. Lecture contains back tracking, branch and bound, Basic search and traversal techniques, graph algorithms, algebraic simplification and transformations, lower bound theory, NP-hard and NP-complete problems.

Text Books:
1. Sahni, Horowitz : Fundamentals of Computer Algorithms
2. Thomas H. Corman, Charle E.L & Ronald : Algorithms

Reference Books:
1. Jon Kleinberg, Eva Tardos: Algorithm Design

Course Learning Outcomes (CLOs):
Upon completion of the course, the students will be able to:
CLO 1: Analyze and **calculate** time complexity and space complexity of various algorithms or any written code using mathematical formula and comparison of algorithms.
CLO 2: Generate and **interpret** the output of iterative and recursive codes with the analysis of the problem definition.
CLO 3: Identify the characteristics of problems for which no computational solution exists.
CLO 4: Develop approximate solutions to NP and NP-complete problems.

CLO 5: Design and **apply** appropriate algorithms to solve real-life problems.

Mapping of Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs):

Program Learning Outcomes (PLOs)	CLO 1	CLO 2	CLO 3	CLO 4	CLO 5
PLO 1: Engineering Knowledge					
PLO 2: Problem Analysis	√		√		
PLO 3: Design/Development of Solutions		√		√	√
PLO 4: Investigation	√		√		
PLO 5: Modern Tool Usage					
PLO 6: The Engineer and Society					
PLO 7: Environment and Sustainability					
PLO 8: Ethics					
PLO 9: Communication					
PLO 10: Individual and Team Work					
PLO 11: Project Management and Finance					
PLO 12: Life-Long Learning					

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lecture	Assignment / Midterm Exam
CLO2	Lecture / Case Study	Assignment / Class Test /Midterm Exam
CLO3	Lecture / Case Study	Assignment / Class Test /Midterm Exam
CLO4	Lecture / Group Discussion	Class Test / Midterm Exam/ Final Exam
CLO5	Lecture / Group Discussion	Presentation/ Class Test / Final Exam

Course Code: CSE 222	**Credits: 1.0**	**Year: Second**	**Semester: Fourth**
Course Title: Computer Algorithms Lab		**Course Status: Lab**	

Rationale of the course:

It covers the common algorithms, algorithmic paradigms, and data structures used to solve computational problems. This course emphasizes the relationship between algorithms and programming and explores algorithms from the programmer's perspective for solving problems efficiently using various programming languages.

Course Content:

Searching and Sorting Algorithm Based on Brute Force Approach, Searching and Sorting Algorithm Based on Divide and Conquer Approach, Problem solving using Greedy Approach,

Problem solving using Dynamic Programming, Problem solving related to Graph and Graph Traversal, Problem solving using Backtracking, Problem solving using Branch and Bound technique.

Course Learning Outcomes (CLOs):

Upon completion of the course, the students will be able to:

CLO 1: Identify the problem given and design the algorithm using various algorithm design techniques.

CLO 2: Implement various algorithms in a high-level language.

CLO 3: Analyze the performance of various algorithms.

CLO 4: Compare the performance of different algorithms for same problem.

Mapping of Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs):

Program Learning Outcomes (PLOs)	CLO 1	CLO 2	CLO 3	CLO 4
PLO 1: Engineering Knowledge				
PLO 2: Problem Analysis	√		√	
PLO 3: Design/Development of Solutions		√		
PLO 4: Investigation				
PLO 5: Modern Tool Usage				√
PLO 6: The Engineer and Society				
PLO 7: Environment and Sustainability				
PLO 8: Ethics				
PLO 9: Communication				
PLO 10: Individual and Team Work		√		
PLO 11: Project Management and Finance				
PLO 12: Life-Long Learning				√

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lab demonstration/ Lecture	Assignment
CLO2	Lab demonstration/ Lecture	Assignment
CLO3	Lab demonstration/ Lecture	Presentation/Viva
CLO4	Lab demonstration/ Lecture	Lab Report /Final Exam

Course Code: CSE 223	Credits: 3.0	Year: Second	Semester: Fourth
Course Title: Object Oriented Programming		**Course Status: Theory**	

Rationale of the course:
To give students a thorough understanding about OOP principles, natural and pragmatic approach to effective problem solving.

Course Content:
Introduction: Object Orientation Development Themes.
Modeling Concepts: Modeling as a design Technique; Object modeling; Advanced Object modeling; Dynamic Modeling; Functional Modeling; Design Methodology - Analysis, System design; Object design; Comparison of Methodologies; Implementation.
Object Oriented languages: Non-object-oriented Languages, Relational Databases; Applications-object diagram computer; computer Animation
Beginning with Java: Object Oriented Systems Development.
Java: Java fundamentals, Java application, Java applets, Methods, Arrays, String & characters, Graphics & java 2D, Basic graphical user interface components, Multithreading, Multimedia, Files & streams, JDBC, Servlets, RMI, Networking, Java beans.

Text Book:

1. Mary Campione and Kathy Walrath: Java Tutorial

Reference Books:

1. Herbert Schildt: Complete Reference Java 2
2. Deitel and Deitel: Java How to Program

Course Learning Outcomes (CLOs):
Upon completion of the course, the students will be able to:
CLO 1: Explain essential concept of Object-orientation Technique and **carry** Object Oriented approach for programming.
CLO 2: Design an object-oriented program using object orientation model Technique.
CLO 3: Gather knowledge on the basics of Java programming like Packages and Interfaces.
CLO 4: Familiarize with Graphical User Interface and Event Handing using AWT and Applet in an Object Oriented Design.
CLO 5: Demonstrate Stream I/O and exception handling techniques to write programs.

Mapping of Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs):

Program Learning Outcomes (PLOs)	CLO 1	CLO 2	CLO 3	CLO 4	CLO 5
PLO 1: Engineering Knowledge	√		√	√	
PLO 2: Problem Analysis		√			√
PLO 3: Design/Development of Solutions		√			√
PLO 4: Investigation					
PLO 5: Modern Tool Usage					
PLO 6: The Engineer and Society					

PLO 7: Environment and Sustainability					
PLO 8: Ethics					
PLO 9: Communication					
PLO 10: Individual and Team Work					
PLO 11: Project Management and Finance					
PLO 12: Life-Long Learning					

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lecture	Assignment / Midterm Exam
CLO2	Lecture / Case Study	Assignment / Class Test /Midterm Exam
CLO3	Lecture / Case Study	Assignment / Class Test /Midterm Exam
CLO4	Lecture / Group Discussion	Class Test / Midterm Exam/ Final Exam
CLO5	Lecture / Group Discussion	Presentation/ Class Test / Final Exam

Course Code: CSE 224	**Credits: 1.0**	**Year: Second**	**Semester: Fourth**
Course Title: Object Oriented Programming Lab		**Course Status: Lab**	

Rationale of the course:

To introduce students with introduction and development of basic principles and mechanisms of programming technology and basic programming skills for program design and development.

Course Content:

Object Oriented Systems Development. Java fundamentals, Java application, Java applets, Methods, Arrays, String & characters, Graphics & java 2D, Basic graphical user interface components, Multithreading, Multimedia, Files & streams, JDBC, Servlets, RMI, Networking, Java beans.

Course Learning Outcomes (CLOs):

Upon completion of the course, the students will be able to:

CLO 1: Familiarize with and **use** Java software development tools, analyze complex problems, solution (algorithm) construction and coding in Java.

CLO 2: Understand the principles of object-oriented programming in Java, including how to define classes, invoke methods, and use class libraries, among other things.

CLO 3: Implement the concepts of data abstraction, data hiding, polymorphism and inheritance to build programs and applications in object oriented programming and **illustrate** the applications of Multithreading, Multimedia, Files & streams.

CLO 4: Develop Graphical User Interface utilizing AWT and applet.
CLO 5: Debug and **test** programs to solve problems and manipulate according to needs and write reports and design documentation.

Mapping of Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs):

Program Learning Outcomes (PLOs)	CLO 1	CLO 2	CLO 3	CLO 4	CLO 5
PLO 1: Engineering Knowledge					
PLO 2: Problem Analysis		√		√	
PLO 3: Design/Development of Solutions	√		√	√	
PLO 4: Investigation					
PLO 5: Modern Tool Usage	√				
PLO 6: The Engineer and Society					
PLO 7: Environment and Sustainability					
PLO 8: Ethics					
PLO 9: Communication					
PLO 10: Individual and Team Work					
PLO 11: Project Management and Finance					
PLO 12: Life-Long Learning	√	√			

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lab demonstration/ Lecture	Assignment
CLO2	Lab demonstration/ Lecture	Assignment
CLO3	Lab demonstration/ Lecture	Assignment
CLO4	Lab demonstration/ Lecture	Presentation/Viva
CLO5	Lab demonstration/ Lecture	Lab Report /Final Exam

Course Code: CSE 225	**Credits: 3.0**	**Year: Second**	**Semester: Fourth**
Course Title: Digital Logic Design		**Course Status: Theory**	

Rationale of the course:
To construct circuits and logic gates, students need to learn about different types of computer processors and how to use numbers to represent signals and sequences in a digital circuit.

Course Content:
Memory Devices: Memory Basics; RAM characteristics; Bipolar RAM; MOS static & MOS Dynamic RAM; ROM; EPROM; EEPROM; Flash memory.

Processor Logic Design: Processor organization; Arithmetic Logic Unit; Design o Arithmetic Circuit; Design of Logic circuit; Design of ALU; Design of Shifter & Accumulator.
Control Logic Design: Control organization; Hard-wired control; Micro program Control; Control of processor unit; PLA control; Micro program Sequencer.
Computer Design: System of configuration; Computer instructions; timing and control; Execution of Instruction; Design of computer Registers; Design of control; Computer console.
Microcomputer system Design: Microcomputer & microprocessor organization; Instruction & addressing mode; Stack, subroutines & interrupts; memory organization; direct memory address; Microprocessor based designs.

Text Books:

1. M.M. Mano : Digital Logic and Computer Design.
2. Ronald J. Tocci : Digital Systems principles and Application.

Reference Books:

1. L. Johnson, Mohammad A Karim : Digital Design.

Course Learning Outcomes (CLOs):
Upon completion of the course, the students will be able to:
CLO 1: Design with combinational and sequential digital logic and state machines.
CLO 2: Solve different switching algebra problems of logic functions.
CLO 3: Identify, **describe**, and **diagnose** the multi-variable Karnaugh map, as well as **execute** an algorithmic reduction of logic functions.
CLO 4: Identify, **recognize** and **diagnose** the combinational circuits: adder, subtractor, encoders/decoders, multiplexers, de-multiplexers, arithmetic-logic units.
CLO 5: Design digital circuits and systems.

Mapping of Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs):

Program Learning Outcomes (PLOs)	CLO 1	CLO 2	CLO 3	CLO 4	CLO 5
PLO 1: Engineering Knowledge					
PLO 2: Problem Analysis			√		
PLO 3: Design/Development of Solutions	√				
PLO 4: Investigation					
PLO 5: Modern Tool Usage					
PLO 6: The Engineer and Society					
PLO 7: Environment and Sustainability					
PLO 8: Ethics					
PLO 9: Communication					
PLO 10: Individual and Team Work					
PLO 11: Project Management and Finance					
PLO 12: Life-Long Learning					

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lecture	Assignment / Midterm Exam
CLO2	Lecture / Case Study	Assignment / Class Test /Midterm Exam
CLO3	Lecture / Case Study	Assignment / Class Test /Midterm Exam
CLO4	Lecture / Group Discussion	Class Test / Midterm Exam/ Final Exam
CLO5	Lecture / Group Discussion	Presentation/ Class Test / Final Exam

Course Code: CSE 226	**Credits: 1.0**	**Year: Second**	**Semester: Fourth**
Course Title: Digital Logic Design Lab		**Course Status: Lab**	

Rationale of the course:

This course will help students to solve problems using digital logic circuits and systems. The fundamental building blocks of combinational and sequential circuits are introduced in order for students to generate circuit solutions to issues and to comprehend the design and operation of digital hardware models.

Course Content:

Boolean algebra, De-Morgan`s theorems, logic gates and their truth tables, minimization techniques canonical forms, combinational logic circuits,; decoders and encoders, multiplexers and de-multiplexers; Flip-flops; Counters; Registers and basic memory unit; Design of sequential circuit: State diagram; State minimizations and assignments; PLA design using MSI and LSI components.

Course Learning Outcomes (CLOs):

Upon completion of the course, the students will be able to:

CLO 1: Demonstrate the fundamental techniques for creating state machines and combinational and sequential digital logic

CLO 2: Be prepared for computer engineering; apply practical knowledge to state simple digital circuits

CLO 3: Describe the various switching algebra theorems and how they can be used for logic functions

CLO 4: Correlate an issue in practice, and design with various ICs and equipment

CLO 5: Practically **predict** or synthesize simple projects and design those projects with different equipment

Mapping of Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs):

Program Learning Outcomes (PLOs)	CLO 1	CLO 2	CLO 3	CLO 4	CLO 5
PLO 1: Engineering Knowledge					
PLO 2: Problem Analysis					
PLO 3: Design/Development of Solutions		√			√
PLO 4: Investigation					
PLO 5: Modern Tool Usage				√	√
PLO 6: The Engineer and Society					
PLO 7: Environment and Sustainability					
PLO 8: Ethics					
PLO 9: Communication					
PLO 10: Individual and Team Work					√
PLO 11: Project Management and Finance					
PLO 12: Life-Long Learning		√			√

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lab demonstration/ Lecture	Assignment
CLO2	Lab demonstration/ Lecture	Assignment
CLO3	Lab demonstration/ Lecture	Assignment
CLO4	Lab demonstration/ Lecture	Presentation/Viva
CLO5	Lab demonstration/ Lecture	Lab Report /Final Exam

Course Code: CSE 227	**Credits: 3.0**	**Year: Second**	**Semester: Fourth**
Course Title: Data Communication		**Course Status: Theory**	

Rationale of the course:

The main goal of the course is to gain a working understanding of data transmission ideas, line control, and line sharing, as well as how to use compression optimizing data transfer techniques.

Course Content:

Introduction: Communication model, data communication task, data communication network, standard and organization, OSI and TCP/IP models. **Data Communications:** Analog and digital data, Spectrum and bandwidth, Transmission impairments, Data rate and channel capacity. Different techniques of modulation. Asynchronous and synchronous communications, Hardware interfaces, multiplexers, concentrators and buffers.

Transmission media: Characteristics and applications of twisted pair, coaxial and fiber optic cables, Satellite, microwave, radio wave, VSAT; Data Encoding, data and signal, NRZI, Manchester and differential Manchester encoding, ASK, FSK, PSK, QPSK, QAM encoding,

modems, Error detection techniques, Parity check and CRC, Error Correction and Hamming code, Interfacing and ELA 232D, null modem; Flow control and error control techniques, Sliding Window, stop and wait ARQ, Selective-Reject ARQ. **Frequency division :** multiplexing, international FDR carrier standards, synchronous TDM and international TDR carrier standards, statistical TDR and wavelength division multiplexing; Circuit switching, single node networks, digital private branch exchange, Packet switching, Diagram and virtual circuits.

Text Book:

1. Data Communications and Networking – Behrouz A. Forouzan

Reference Book:

1. Data Communications and Computer Networks –William Stallings Andrew S. Tanenbaum: Computer Networks

Course Learning Outcomes (CLOs):

Upon completion of the course, the students will be able to:

CLO 1: Describe the components of a data communications system and **identify** key considerations in selecting various transmission media in networks.

CLO 2: Explain the role of line codes and digital communications devices in a data communications network.

CLO 3: Describe the various types of signals and their features.

CLO 4: Identify and define roles and features of various data transmission protocols.

CLO 5: Describe the features and functions of multiplexing and modulation and various error detection and correction schemes.

Mapping of Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs):

Program Learning Outcomes (PLO)	CLO 1	CLO 2	CLO 3	CLO 4	CLO 5
PLO 1: Engineering Knowledge					
PLO 2: Problem Analysis	√				
PLO 3: Design/Development of Solutions					
PLO 4: Investigation				√	
PLO 5: Modern Tool Usage					
PLO 6: The Engineer and Society					
PLO 7: Environment and Sustainability					
PLO 8: Ethics					
PLO 9: Communication			√		
PLO 10: Individual and Team Work					
PLO 11: Project Management and Finance					
PLO 12: Life-Long Learning	√	√		√	

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lecture	Assignment / Midterm Exam
CLO2	Lecture / Case Study	Assignment / Class Test /Midterm Exam
CLO3	Lecture / Case Study	Assignment / Class Test /Midterm Exam
CLO4	Lecture / Group Discussion	Class Test / Midterm Exam/ Final Exam
CLO5	Lecture / Group Discussion	Presentation/ Class Test / Final Exam

Course Code: CSE 229	**Credits: 3.0**	**Year: Second**	**Semester: Fourth**
Course Title: Computer Architecture		**Course Status: Theory**	

Rationale of the course:

To study the basic organization and architecture of digital computers (CPU, memory, I/O, software). Such knowledge leads to better understanding and utilization of digital computers, and can be used in the design and application of computer systems or as foundation for more advanced computer-related studies.

Course Content:

Introduction: Generation and types of computer, languages, virtual machines, multilevel machines, hardware and software, milestones in computer architecture.

Computer organization : Processors architecture, functions, operations, instruction formats, addressing, instruction types, control flow, comparison between Intel and Motorola; memory - functions, classification of memory; Input/Output - keyboard, terminals, disk-drive, modem, mouse, printers; computer buses: control bus, data bus, address bus, synchronous bus, asynchronous bus, bus arbitration, interrupt handling, system bus - IBM PC XT/AT bus, VME bus, VESA bus, PCI bus.

Microprocessor : Types, overview of Intel and Motorola processor, instruction and addressing mode of Intel 8086 microprocessor, assembly language programming (any 8086 processor).

Microcomputer/Interfacing support chips: Interfacing technique, different support chips, bus controller, interrupt controller, DMA controller, key-board controller.

Micro Programmed Control: Basic concepts. Micro-instruction sequencing and execution; Grouping of signals; Bit-slice architecture. RISE & CISC machines, Parallel processing, multiprocessing, Vector computation. Fault tolerant systems: Instruction, execution Characteristics; Use of large register file; RISC architecture, RISC pipe lining, machine.

Text Book:

1. William Stalling: Computer Organizaiton and Architectrue.

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Reference Book:

1. Jim Ledin: Modern Computer Architecture and Organization.

Course Learning Outcomes (CLOs):

Upon completion of the course, the students will be able to:

CLO 1: Use various metrics to calculate the performance of a computer system.

CLO 2: Identify the addressing mode of instructions.

CLO 3: Determine which hardware blocks and control lines are used for specific instructions.

CLO 4: Demonstrate how to different arithmetic operations done by a digital computer

CLO 5: Analyze clock periods, performance, and instruction throughput of single-cycle, multi-cycle, and pipelined implementations.

Mapping of Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs):

Program Learning Outcomes (PLOs)	CLO 1	CLO 2	CLO 3	CLO 4	CLO 5
PLO 1: Engineering Knowledge	√				
PLO 2: Problem Analysis			√		√
PLO 3: Design/Development of Solutions					
PLO 4: Investigation		√			√
PLO 5: Modern Tool Usage					
PLO 6: The Engineer and Society					
PLO 7: Environment and Sustainability					
PLO 8: Ethics					
PLO 9: Communication					
PLO 10: Individual and Team Work					
PLO 11: Project Management and Finance					
PLO 12: Life-Long Learning					

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lecture	Assignment / Midterm Exam
CLO2	Lecture / Case Study	Assignment / Class Test /Midterm Exam
CLO3	Lecture / Case Study	Assignment / Class Test /Midterm Exam
CLO4	Lecture / Group Discussion	Class Test / Midterm Exam/ Final Exam
CLO5	Lecture / Group Discussion	Presentation/ Class Test / Final Exam

Course Code: CSE 311	**Credits: 3.0**	**Year: Third**	**Semester: Fifth**
Course Title: Theory of Computing		**Course Status: Theory**	

Rationale of the course:

Learn how to solve issues efficiently on a computer model utilizing algorithms and the fundamentals of how computers work.

Course Content:

Finite automata: deterministic finite automata, nondeterministic finite automata, equivalence and conversion of deterministic and nondeterministic finite automata, pushdown automata; Context free languages; Context free grammars, Chomsky normal form, Greibach Normal Form; Pushdown automata; Regular languages: regular expressions, nonregular languages, the pumping lemma; Turing Machines: basic machines, configuration, computing with Turing machines, combining Turing machines; Undecidability.

Text Books:

1. Jeffrey D. Ullman : Introduction to Automata Theory, Languages and Computation

Reference Books:

1. Adamek, Klywer: Automata and Algebra
2. Zvi Kohavi : Switching and finite automata theory

Course Learning Outcomes (CLOs):

Upon completion of the course, the students will be able to:

CLO 1: Recognize the mathematical fundamentals of computation, such as automata theory.

CLO 2: Be capable of defining the foundations of formal language theory, grammars, algorithmic ideas, decidability, complexity, and computability.

CLO 3: Create finite automata, pushdown automata, Turing machines, formal languages, and grammars by connecting and designing them.

CLO 4: Improve one's ability to describe and perform mathematical proofs for computation and algorithms

Mapping of Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs):

Program Learning Outcomes (PLOs)	CLO 1	CLO 2	CLO 3	CLO 4
PLO 1: Engineering Knowledge	√	√		
PLO 2: Problem Analysis			√	√
PLO 3: Design/Development of Solutions			√	
PLO 4: Investigation	√			
PLO 5: Modern Tool Usage				
PLO 6: The Engineer and Society				
PLO 7: Environment and Sustainability				
PLO 8: Ethics				
PLO 9: Communication				
PLO 10: Individual and Team Work				
PLO 11: Project Management and Finance				
PLO 12: Life-Long Learning			√	

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lecture	Assignment / Midterm Exam
CLO2	Lecture	Assignment / Class Test /Midterm Exam
CLO3	Lecture / Group Discussion	Presentation/ Final Exam
CLO4	Lecture / Group Discussion	Presentation/ Class Test / Final Exam

Course Code: CSE 313	**Credits: 3.0**	**Year: Third**	**Semester: Fifth**
Course Title: Database Management System		**Course Status: Theory**	

Rationale of the course:

This course is intended to introduce basic database ideas and study the foundations of database systems, with a focus on fundamentals such as relational algebra and data modeling, schema normalization, query optimization, and transactions.

Course Content:

Overview: Definition & types of DBMS, Data Model; ER & rational model, FD constraints, assertions, friggers. Database design; logical database design, physical database design, data storage device and data formats. **Basic Database Concepts :** Conventional file processing; three level architecture, Communication with database systems, database management systems. Distributed database, database machines, database administration; The hierarchical, network and relational database models and their languages, Data description and data manipulation languages. Structured query language (SQL), Storage representations and access methods. **Database Design:** Functional dependencies decomposition to normal forms, the entity relationship model. **Database Components:** Security, Data dictionary, transactions and recovery, concurrent processing. **Database Implementation:** Physical representation, query, optimization, projects using on RDMS; Logic for relational database; First order logic, relational calculus and algebra.

Intelligent Database : Deductive database with natural language interface, Specific database systems: FOXPRO / ORACL.

Text Book:

2. Abraham Silberschatz : Database System Concepts

Reference Books:

1. Raghu Ramakrishnan : Database Management System
2. Carlo Zaniolo : Advanced Database Systems

Course Learning Outcomes (CLOs):

Upon completion of the course, the students will be able to:

CLO 1: Explain the fundamental concepts of database systems and their applications.

CLO 2: Demonstrate the fundamentals of SQL and how to write SQL queries.

CLO 3: Write SQL using a commercial relational database system (Oracle) to become familiar with it.

CLO 4: Understand relational database theory and be able to write queries using relational algebra expressions.

Mapping of Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs):

Program Learning Outcomes (PLOs)	CLO 1	CLO 2	CLO 3	CLO 4
PLO 1: Engineering Knowledge	√		√	√
PLO 2: Problem Analysis		√		
PLO 3: Design/Development of Solutions				
PLO 4: Investigation				
PLO 5: Modern Tool Usage				
PLO 6: The Engineer and Society				
PLO 7: Environment and Sustainability				
PLO 8: Ethics				
PLO 9: Communication				
PLO 10: Individual and Team Work				
PLO 11: Project Management and Finance				
PLO 12: Life-Long Learning				

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lecture	Assignment / Midterm Exam
CLO2	Lecture	Assignment / Class Test /Midterm Exam
CLO3	Lecture / Group Discussion	Presentation/ Final Exam
CLO4	Lecture / Group Discussion	Presentation/ Class Test / Final Exam

Course Code: CSE 314	Credits: 1.0	Year: Third	Semester: Fifth
Course Title: Database Management System Lab		**Course Status: Lab**	

Rationale of the course:

This course is aimed to explain basic database principles, teach you how to build databases, and give you hands-on experience by having you develop a real-world e-commerce database application as part of a term project. Also, to gain an understanding of database architecture, from conceptual design to implementation of database schemas and user interfaces.

Course Content:

Introduction, Oracle Installation, Authentication, Security, Table Creation, SQL, Simple Query, Data Expressions, Join, Constraints, Advanced Query (GROUP Function etc.), Sub-queries, Single-row function, Numeric function, Manipulation function, Conversion function, Nesting of function, Abstract data type, Database Trigger/ Procedure, PL/SQL Packages, Indexing, View, Introduction to PL/SQL

Course Learning Outcomes (CLOs):

Upon completion of the course, the students will be able to:

CLO 1: Demonstrate the understanding in a team-based project using a commercial relational database system (Oracle).

CLO 2: Make use of SQL and PL SQL, as well as database architecture principles.

CLO 3: Demonstrate relational database theory and the ability to write queries using relational algebra expressions.

Mapping of Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs):

Program Learning Outcomes (PLOs)	CLO 1	CLO 2	CLO 3
PLO 1: Engineering Knowledge			
PLO 2: Problem Analysis			
PLO 3: Design/Development of Solutions	√		√
PLO 4: Investigation		√	
PLO 5: Modern Tool Usage	√	√	
PLO 6: The Engineer and Society			
PLO 7: Environment and Sustainability			
PLO 8: Ethics			
PLO 9: Communication			
PLO 10: Individual and Team Work			
PLO 11: Project Management and Finance			
PLO 12: Life-Long Learning	√		√

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lab demonstration/ Lecture	Assignment
CLO2	Lab demonstration/ Lecture	Presentation/Viva
CLO3	Lab demonstration/ Lecture	Lab Report /Final Exam

Course Code: CSE 315	Credits: 3.0	Year: Third	Semester: Fifth
Course Title: Microprocessors & Embedded System		Course Status: Theory	

Rationale of the course:
This course covers the fundamentals of assembly language, microprocessor architecture, and various interfaces as well as the creation of microprocessor and microcontroller-based systems.

Course Content:
Microprocessor: Types, overview of Intel and Motorola processor, architecture, interfacing, I/O operation, interrupt structure, DMA, instructions and addressing mode of Intel 8086 microprocessor. **Microcomputer/Interfacing support chips:** Interfacing technique, different support chips, bus controller, interrupt controller, DMA controller, key-board controller.
Micro Programmed Control: Basic concepts. Micro-instruction sequencing and execution; Grouping of signals; Bit-slice architecture. RISE & CISC machines, Parallel processing, multiprocessing, Vector computation. Fault tolerant systems: Instruction, execution Characteristics; Use of large register file; RISC architecture, RISC pipe lining, machine.
Assembly Language: Machine and Assembly instruction types and their formats, Character representation instructions, instruction execution, Machine language programming, instruction sets and their implementations, The Assembly process, Addressing methods, Subroutines, Macros and files, I/O programming, interrupts and concurrent processes.

Text Book:
1. Douglas V. Hall: Microprocessors and Interfacing

Reference Book:
1. Ytha Yu, C. Marut : Assembly Language Programming and Organization of IBM PC

Course Learning Outcomes (CLOs):
After completing the course, the students will be able to:
CLO 1: Learn how high-level languages are translated into low-level languages and how a processor runs a program line by line.
CLO 2: Use assembly language to work with simple microprocessors.
CLO 3: Understand the fundamentals of how a microprocessor and its associated components communicate and perform tasks.
CLO 4: Build different systems using 8086 microprocessor.

Mapping of Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs):

Program Learning Outcomes (PLOs)	CLO 1	CLO 2	CLO 3	CLO 4
PLO 1: Engineering Knowledge			√	
PLO 2: Problem Analysis				
PLO 3: Design/Development of Solutions		√		√
PLO 4: Investigation				
PLO 5: Modern Tool Usage				

PLO 6: The Engineer and Society				
PLO 7: Environment and Sustainability				
PLO 8: Ethics				
PLO 9: Communication				
PLO 10: Individual and Team Work				
PLO 11: Project Management and Finance				
PLO 12: Life-Long Learning				

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lecture	Assignment / Midterm Exam
CLO2	Lecture	Assignment / Class Test /Midterm Exam
CLO3	Lecture / Group Discussion	Presentation/ Final Exam
CLO4	Lecture / Group Discussion	Presentation/ Class Test / Final Exam

Course Code: CSE 316	**Credits: 1.0**	**Year: Third**	**Semester: Fifth**
Course Title: Microprocessors & Embedded System Lab		**Course Status: Lab**	

Rationale of the course:

This course covers the fundamentals of assembly language, microprocessor architecture, and various interfaces as well as the creation of microprocessor and microcontroller-based systems.

Course Content:

Machine and Assembly instruction types and their formats, Character representation instructions, instruction execution, Machine language programming, instruction sets and their implementations, The Assembly process, Addressing methods, Subroutines, Macros and files, I/O programming, interrupts and concurrent processes.

Course Learning Outcomes (CLOs):

After completing the course, the students will be able to:

CLO 1: Learn assembly language for basic microprocessors.

CLO 2: Understand the fundamentals of how a microcomputer and its associated components operate.

CLO 3: Understand the various microprocessor and microcontroller families and generations.

Mapping of Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs):

Program Learning Outcomes (PLOs)	CLO 1	CLO 2	CLO 3

PLO 1: Engineering Knowledge			
PLO 2: Problem Analysis	√		
PLO 3: Design/Development of Solutions	√	√	
PLO 4: Investigation			
PLO 5: Modern Tool Usage	√	√	
PLO 6: The Engineer and Society			
PLO 7: Environment and Sustainability			
PLO 8: Ethics			
PLO 9: Communication			
PLO 10: Individual and Team Work			
PLO 11: Project Management and Finance			
PLO 12: Life-Long Learning	√		√

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lab demonstration/ Lecture	Assignment
CLO2	Lab demonstration/ Lecture	Presentation/Viva
CLO3	Lab demonstration/ Lecture	Lab Report /Final Exam

Course Code: CSE 317	**Credits: 3.0**	**Year: Third**	**Semester: Fifth**
Course Title: Computer Networks		**Course Status: Theory**	

Rationale of the course:

This course helps students to excel in Designing, engineering, and installation of networks to connect digital computers. The key goals are resource sharing, high reliability, increased system performance, and security.

Course Content:

Introduction to Computer Networks: Basic concept of Computer networks and distributed systems; Goals of networking; General approaches of communication within a network. Network classification;. Uses & Network Software's. Network Topology : Bus, tree, ring, and star topology, transmission. Network Protocol : OSI model, TCP/IP protocol and the Internet, LAN and MAN-topologies, optical fiber bus, MAC protocols, LAN/MAN standards, FDDI and Fast Ethernet, LAN and bridges; Geosynchronous and 1ow orbit satellites, VSATS, protocols for multiple access unlink channel, digital cellular radio, GSM, personal communication services. Network Interconnections : bridges and gateway, connectionless and connection oriented internetworking, routing and fragmentation, firalls, IP protocol and IP addresses, internet Control

Protocols - ICMP, ARP, RARP, Domain name and Name servers; Overview of internet applications; Electronic Mail Systems and protocol SMIP and MIME, world wide web, URLS; ISDN user interface and channels, Broadband ISDN, Frame Relay services and protocols, ATM virtual Channels and paths, ATM cells, header error control, ATM switches. Network Security : Introduction to network security authentication protocols and digital signatures, simple network, management protocol.

Network Applications: Client/server programming, RPC techniques; applications- file server, print server, e-mail, directory services, remote login, HTTP, HTML, World Wide Web, web browser, applications server. **Network Management :** Internet management - management information base(MIB), simple network management protocol (SNMP). **The Network Layer:** The Physical Layer - The Data Link Layer - The Medium Access Control Sub-layer. Transport layer and the application of layers.

Test Books:

1. Computer Networks: Andrew S. Tanenbaum, David J. Wetherall
2. Internetworking with TCP/IP: Douglas E. Comer

Reference Book:

1. Computer Networking (A Top-Down Approach): James F. Kurose, Keith W. Ross

Course Learning Outcomes (CLOs):

Upon completion of the course, the students will be able to:

CLO 1: **Understand** and explain the layered protocol model

CLO 2: **Describe** what classful and classless addressing schemes are

CLO 3: Program network communication services for client/server and other application layouts

CLO 4: Use networking tools to observe and analyze behaviors of networking protocols

CLO 5: Describe, **analyze** and **evaluate** a number of datalink, network, and transport layer protocols

Mapping of Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs):

Program Learning Outcomes (PLOs)	CLO 1	CLO 2	CLO 3	CLO 4	CLO 5
PLO 1: Engineering Knowledge	√				√
PLO 2: Problem Analysis					√
PLO 3: Design/Development of Solutions					
PLO 4: Investigation					
PLO 5: Modern Tool Usage					
PLO 6: The Engineer and Society					
PLO 7: Environment and Sustainability					
PLO 8: Ethics					

PLO 9: Communication			√		
PLO 10: Individual and Team Work				√	
PLO 11: Project Management and Finance					
PLO 12: Life-Long Learning					

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lecture	Assignment / Midterm Exam
CLO2	Lecture	Assignment / Class Test /Midterm Exam
CLO3	Lecture / Group Discussion	Presentation/ Final Exam
CLO4	Lecture / Group Discussion	Presentation/ Class Test / Final Exam

Course Code: CSE 318	**Credits: 1.0**	**Year: Third**	**Semester: Fifth**
Course Title: Computer Networks Lab		**Course Status: Lab**	

Rationale of the course:

This course studies the standard models for the layered approach to communication between autonomous machines in a network, and the main characteristics of data transmission across various physical link types. It considers how to design networks and protocols for diverse situations, analyses several application and support protocols from a distributed systems viewpoint, and identifies significant problem areas in networked communications.

Course Content:

TCP/IP protocol and the Internet, LAN Configuration, bridges and gateway, connectionless and connection oriented internetworking, routing and fragmentation, firalls, IP protocol and IP addresses, internet Control Protocols - ICMP, ARP, RARP, Domain name and Name servers; Client/server programming, RPC techniques; applications- file server, print server, e-mail, directory services, remote login, HTTP, HTML, World Wide Web, web browser, applications server, DHCP, RIP, VLAN, VLSM, FLSM, NAT.

Course Learning Outcomes (CLOs):

Upon completion of the course, the students will be able to:

CLO 1: Apply VLSM using CISCO packet tracer.

CLO 2: Analyze application layer protocols.

CLO 3: Design and **implement** network protocols.

Mapping of Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs):

Program Learning Outcomes (PLOs)	CLO 1	CLO 2	CLO 3

PLO 1: Engineering Knowledge			
PLO 2: Problem Analysis		√	
PLO 3: Design/Development of Solutions			
PLO 4: Investigation			√
PLO 5: Modern Tool Usage			√
PLO 6: The Engineer and Society			
PLO 7: Environment and Sustainability			
PLO 8: Ethics			
PLO 9: Communication			
PLO 10: Individual and Team Work	√		√
PLO 11: Project Management and Finance			
PLO 12: Life-Long Learning			√

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lecture	Assignment / Midterm Exam
CLO2	Lecture	Assignment / Class Test /Midterm Exam
CLO3	Lecture / Group Discussion	Presentation/ Class Test / Final Exam

Course Code: CSE 321	**Credits: 3.0**	**Year: Third**	**Semester: Sixth**
Course Title: Operating System		**Course Status: Theory**	

Rationale of the course:

To have a clear idea about how operating systems manage process, hardware, file and memory, system configuration/optimization, and utilities.

Course Content:

Overview: Introduction, evaluation goals and components of OS, types of OS, Basic concept Batch processing, Multiprogramming, Time sharing, Distributed system, Real time systems, R/O structure, Dual-mode operation, OS services, system calls, system program, system design and implementation; Process control blocks, job and process scheduling, scheduling levels, objective and criteria.

CPU Scheduling: Scheduling concept, CPU scheduling, scheduling algorithm, multiprocessor scheduling, real time scheduling; process co-ordinations critical section problems semaphores, language constructs classical problems of process co-ordinations; under prices communication, message and mailbox.

I/O Systems: I/O devices, organization of the I/O function, operating system design issues, I/O buffering, disk I/O, I/O programming; device management techniques, Interrupt processing, Parallel processing, Deadlock, prevention, avoidance, detection and recovery.
Memory Management: Memory allocation schemes, paging and segmentation, virtual memory, page replacement strategies, working sets, demand paging; Secondary storage management, disk scheduling algorithms.
Distributed Process Management: Process migration, distributed global states, distributed process management- mutual exclusive, distributed deadlock.
File Management: File organization and access, file directories, file sharing, record blocking, secondary storage management, File system functions, file organization, logical and physical file maps, tree structured file systems, space allocation, file catalogs, file access control mechanisms, operating system security; Time sharing and virtual machines; Main features of MS-DOS, Unit and VSE. Loader and linker, Windows 95 and its function, Functions of Unix and Novell.
Protection and Security: Security, threats, protection, intruders, viruses and real threats, trusted systems, network security. **Process Description and Control**: Process concept, process states, process description, process control, processes and threads, concurrent process.

Text Book:

1. Greg Gagne, Abraham Silberschatz, Peter Baer Galvin: Operating System Concepts.

Reference Book:

1. Modern Operating Systems (4th ed) - Andrew S. Tanenbaum; Prentice Hall

Course Learning Outcomes (CLOs):
After completing the course, the students will be able to:
CLO 1: Demonstrate understanding of the concepts, structure and design of Operating Systems.
CLO 2: Demonstrate understanding of Operating System design and its impact on application system design and performance.
CLO 3: Demonstrate competence in recognizing and using Operating System features.
CLO 4: Understand and **analyze** process, threads, memory and file management system.
CLO 5: Able to implement algorithms for memory management and deadlock.

Mapping of Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs):

Program Learning Outcomes (PLOs)	CLO 1	CLO 2	CLO 3	CLO 4	CLO 5
PLO 1: Engineering Knowledge				√	
PLO 2: Problem Analysis			√		
PLO 3: Design/Development of Solutions					√
PLO 4: Investigation					
PLO 5: Modern Tool Usage					

PLO 6: The Engineer and Society					
PLO 7: Environment and Sustainability					
PLO 8: Ethics					
PLO 9: Communication					
PLO 10: Individual and Team Work					
PLO 11: Project Management and Finance					
PLO 12: Life-Long Learning				√	√

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lecture	Assignment / Midterm Exam
CLO2	Lecture / Case Study	Assignment / Class Test /Midterm Exam
CLO3	Lecture / Case Study	Assignment / Class Test /Midterm Exam
CLO4	Lecture / Group Discussion	Class Test / Midterm Exam/ Final Exam
CLO5	Lecture / Group Discussion	Presentation/ Class Test / Final Exam

Course Code: CSE 322	**Credits: 1.0**	**Year: Third**	**Semester: Sixth**
Course Title: Operating System Lab		**Course Status: Lab**	

Rationale of the course:

This course is designed to help students in learning operating system features. Students will be able to command the kernel using shell, monitor the activity of running processes. They will be trained to create multiple child processes from a single process, divide them in multiple threads to maintain concurrent operations to reduce execution time. They will also learn how to maintain synchronization among processes, CPU scheduling algorithms and reducing the access time of secondary storage devices.

Course Content:

Implementation of System Calls, I/O System Calls, Simulation of UNIX Commands, FCFS Scheduling SJF Scheduling, Priority Scheduling, Round Robin Scheduling, Inter Process Communication Using Shared Memory, Inter Process Communication Using Message Queues, Producer – Consumer Problem Using Semaphores, Memory Management Schemes.

Course Learning Outcomes (CLOs):

After completing the course, the students will be able to:

CLO 1: Demonstrate understanding about windows DOS, process activities from linux shell, linux shell commands.

CLO 2: Implement creating a child process from the parent process and dividing a process into multiple threads.
CLO 3: Understand and **analyze** processes, higher level synchronization among processes while communicating with each other.
CLO 4: Able to understand how processes are being selected from the ready queue as well as the seeking time of data from secondary storage devices.

Mapping of Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs):

Program Learning Outcomes (PLOs)	CLO 1	CLO 2	CLO 3	CLO 4
PLO 1: Engineering Knowledge				
PLO 2: Problem Analysis			√	
PLO 3: Design/Development of Solutions		√		
PLO 4: Investigation			√	√
PLO 5: Modern Tool Usage	√			
PLO 6: The Engineer and Society				
PLO 7: Environment and Sustainability				
PLO 8: Ethics				
PLO 9: Communication				
PLO 10: Individual and Team Work				
PLO 11: Project Management and Finance				
PLO 12: Life-Long Learning			√	

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lab demonstration/ Lecture	Assignment
CLO2	Lab demonstration/ Lecture	Assignment
CLO3	Lab demonstration/ Lecture	Presentation/Viva
CLO4	Lab demonstration/ Lecture	Lab Report /Final Exam

Course Code: CSE 323	Credits: 3.0	Year: Third	Semester: Sixth
Course Title: System Analysis & Design		**Course Status: Theory**	

Rationale of the course:
This course encourages students to think about information systems planning, analysis, design, and implementation; graphical methods for representing information structure, practical design methodologies, database design, and prototyping; communication skills, and project management in order to solve a variety of real-world problems.

Course Content:

The Context of Systems Development Project*:* the context of systems analysis and design methods, information system building blocks, information systems development, project management. **System Analysis Methods:** Systems analysis, fact-finding techniques for requirements discovery, modeling system requirements with use cases, data modeling and analysis, process modeling, feasibility analysis and the system proposal, object-oriented analysis and modeling using the UML. **System Design Methods:** Systems design, application architecture and modeling. Database design, output design and prototyping, input design and prototyping, user interface design, object-oriented design and modeling using the UML. **Beyond System Analysis and Design Methods:** Systems constructions and implementation, system operations and support.

Text Book:

1. Gerald M. Weinberg : Rethinking System Analysis & Design

Reference Books:

b. Gerald M. Weinberg, Daniela Weinberg : General Principles of Systems Design
c. Kenneth E. Kendall, Julie E. Kendall: System Analysis and Design (6th Edition)

Course Learning Outcomes (CLOs):

Upon completion of the course, the students will be able to:

CLO 1: Discuss realistic techniques to establishing and developing Information Systems for businesses and industries.

CLO 2: Learn about information technological ecosystems and how to become more aware of them.

CLO 3: Solve real-life challenges that are difficult to solve.

Mapping of Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs):

Program Learning Outcomes (PLOs)	CLO 1	CLO 2	CLO 3
PLO 1: Engineering Knowledge	√		
PLO 2: Problem Analysis			√
PLO 3: Design/Development of Solutions			√
PLO 4: Investigation		√	
PLO 5: Modern Tool Usage			
PLO 6: The Engineer and Society			
PLO 7: Environment and Sustainability			
PLO 8: Ethics			
PLO 9: Communication			
PLO 10: Individual and Team Work			
PLO 11: Project Management and Finance			
PLO 12: Life-Long Learning	√		√

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lecture	Assignment / Midterm Exam
CLO2	Lecture	Assignment / Class Test /Midterm Exam
CLO3	Lecture / Group Discussion	Presentation/ Class Test / Final Exam

Course Code: CSE 324	**Credits: 1.0**	**Year: Third**	**Semester: Sixth**
Course Title: System Analysis & Design Lab		**Course Status: Lab**	

Rationale of the course:

This course encourages students to learn about the architecture, different compression and coding standards for multimedia documents, database, network and operating system issues, traffic and service issues, and security issues, and then apply what they've learned to create various multimedia applications.

Course Content:

Introduction to System and Approaches to System development, Types of Information System Approaches to System development: Software Development Life cycle, Software Development Models: Waterfall model, Iterative Model, RAD model, Incremental model, Spiral model. Project management and Planning Project management Concepts: The management Spectrum: People, The Problem, The Process Software Project Planning: Project planning objectives, Software Scope, Resources, Software Project estimation, The Make-Buy decision, software risks. Analysis Concepts, Principles and Modeling Analysis Concepts and Principles: Requirement Analysis, Communication techniques: Initiating the Process, Facilitated Application Specification techniques, Quality Function development.

Analysis Principles: The Information Domain, Modeling, Partitioning, Software Requirement Specification. Analysis Modeling: Data Modeling: Data objects, Attributes and relationships, cardinality and modality. Data flow diagrams, Entity-Relationship Diagrams, The Data Dictionary. Design Concepts & Principles and Effective Modular design.

Design Concepts & Principles: Software Design and software Engineering, the design process, the design principles, Design Concepts: Abstraction, Refinement, Modularity, Software architecture, Control hierarchy, Structural Partitioning, Data Structure, Software procedure, Information Hiding. Effective Modular design: Functional independence, Cohesion, Coupling. User Interface Design: the Golden rules, User Interface and Design Process, Interface analysis.

Recommended Books:

1. Essential of Systems Analysis and Design (6th edition) by Joseph Valacich
2. Systems analysis and Design, Kendall and Kendall

3. Management Information Systems: Managing the Digital Firm (12th Edition), Pearson/Prentice Hall
4. Systems Analysis and Design Methods, Jeffrey Whitten, Lonnie Bentley

Course Learning Outcomes (CLOs):

Upon completion of the course, the students will be able to:

CLO 1: Identify and **articulate** system concepts, as well as walk through the creation and implementation of an information system at various stages of the system development life cycle.

CLO 2: Use flow diagrams, be able to visualize and model the design phase.

CLO 3: Become Capable of writing clear, succinct system requirements and translating them into technical specifications.

CLO 4: Design a user interface for input and output specifications, different types of data representation, and group distribution management with a focus on user interfaces

Mapping of Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs):

Program Learning Outcomes (PLOs)	CLO 1	CLO 2	CLO 3	CLO 4
PLO 1: Engineering Knowledge				
PLO 2: Problem Analysis				
PLO 3: Design/Development of Solutions		√		√
PLO 4: Investigation	√			
PLO 5: Modern Tool Usage				
PLO 6: The Engineer and Society				
PLO 7: Environment and Sustainability				
PLO 8: Ethics				
PLO 9: Communication				
PLO 10: Individual and Team Work				
PLO 11: Project Management and Finance				
PLO 12: Life-Long Learning	√		√	

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lab demonstration/ Lecture	Assignment
CLO2	Lab demonstration/ Lecture	Assignment
CLO3	Lab demonstration/ Lecture	Presentation/Viva
CLO4	Lab demonstration/ Lecture	Lab Report /Final Exam

Course Code: CSE 325	Credits: 3.0	Year: Third	Semester: Sixth
Course Title: Computer Peripherals & Interfacing		Course Status: Theory	

Rationale of the course:
This course is designed to help students gather a sufficient knowledge of computers. Hardware (PC, peripherals and networks), Microprocessor peripheral chips and Interfacing of the peripherals

Course Content:
Character Peripherals: Keyboards, printers (dot-matrix, laser, link-jet) VDUS, computer graphics hardware, plotters, disc-drivers, CD - ROM. Optical displays and sensors, High power interface devices, transducers, stepper motors and peripheral devices,
Microprocessor peripheral chips: application to peripheral sub-systems PPI, DMAC, PCI. Interfacing data converters, general purpose programmable peripheral devices, serial I/O and data communication.
Interfacing: Design and operation of interface between computer and the outside world, sensors, transducers and signal conditioning circuits, interfacing memory, and I/O devices such as monitors, printers, disk drivers, optical displays, some special interface cards, stepper motors and other peripheral devices. IEEE488, RS-232 and other buses; Study and applications of peripheral chips including 8212, 8155, 8255, 8251.

Text Book:
1. Barry Wilkinson and David Horrocks: Computer Peripherals

Reference Books:
1. Ramesh S. Gaonkar: Microprocessor Architecture, Programming and Applications.
2. Barry M Cook and Neil H White: Computer Peripherals.

Course Learning Outcomes (CLOs):
Upon completion of the course, the students will be able to:
CLO 1: Gather Knowledge about the input devices, output devices, display devices, hard copy devices, memory devices, multimedia devices.
CLO 2: Exemplify the use complex computer systems such as high-performance peripheral interfaces, significant signal processing, and graphics applications.
CLO 3: Understand and **explain** the basic principles of instructional set design.
CLO 4: Familiarize yourself with all types of semiconductor storage devices and Memory interface requirements.
CLO 5: Gather ideas on various Microcontroller's variety of devices and the Interfacing of these devices

Mapping of Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs):

Program Learning Outcomes (PLOs)	CLO 1	CLO 2	CLO 3	CLO 4	CLO 5
PLO 1: Engineering Knowledge	√				√
PLO 2: Problem Analysis					
PLO 3: Design/Development of Solutions					

PLO 4: Investigation					√
PLO 5: Modern Tool Usage					
PLO 6: The Engineer and Society					
PLO 7: Environment and Sustainability					
PLO 8: Ethics					
PLO 9: Communication					
PLO 10: Individual and Team Work					
PLO 11: Project Management and Finance					
PLO 12: Life-Long Learning					

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lecture	Assignment / Midterm Exam
CLO2	Lecture / Case Study	Assignment / Class Test /Midterm Exam
CLO3	Lecture / Case Study	Assignment / Class Test /Midterm Exam
CLO4	Lecture / Group Discussion	Class Test / Midterm Exam/ Final Exam
CLO5	Lecture / Group Discussion	Presentation/ Class Test / Final Exam

Course Code: CSE 326	**Credits: 1.0**	**Year: Third**	**Semester: Sixth**
Course Title: Computer Peripherals & Interfacing Lab		**Course Status: Lab**	

Rationale of the course:

This course introduces students with the necessary knowledge and skills of the working principles and interfacing aspects of peripherals.

Course Content:

Introduction to EDSIM51 Simulator, Proteus Simulator and Arduino IDE, Basic Operation on EDSIM51, Interfacing LED and Up Down Counter, Creating LED Counter and Performing Sequence Generation, Interfacing and Controlling ADC, DAC, Switch and Keypad, Scrolling Text on LCD, Interfacing and Multiplexing 7 segment displays, Creating and Moving Stepper Motor in Different Directions, Build a Capacitive Touch Sensor, Counter from 0 to 99, Build Digital Thermometer (LM35), DHT11 Humidity and Temperature Sensor, Capacitance Measuring, Serial Data transmission to Arduino.

Course Learning Outcomes (CLOs):

Upon completion of the course, the students will be able to:

CLO 1: Use automation tool like Proteus to draw circuits and an integrated development environment (IDE) to write, compile, and execute programs involving a small number of source files.
CLO 2: Use EDSIM 51 to scan a keypad, multiplex 7-segment displays, controls a motor and counts its revolutions, etc.
CLO 3: Interface with the peripheral I/O chips like Timer, PPI, Interrupt controller, etc.
CLO 4: Design and **construct** data logging, simple control systems to take input from and produce output to peripheral devices.
CLO 5: Learn interfacing with USART, PPI, PIC, and DMA Controller.

Mapping of Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs):

Program Learning Outcomes (PLOs)	CLO 1	CLO 2	CLO 3	CLO 4	CLO 5
PLO 1: Engineering Knowledge					
PLO 2: Problem Analysis	√	√			
PLO 3: Design/Development of Solutions					
PLO 4: Investigation					
PLO 5: Modern Tool Usage	√	√			
PLO 6: The Engineer and Society					
PLO 7: Environment and Sustainability					
PLO 8: Ethics					
PLO 9: Communication					
PLO 10: Individual and Team Work			√	√	√
PLO 11: Project Management and Finance					
PLO 12: Life-Long Learning		√	√		

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lab demonstration/ Lecture	Assignment
CLO2	Lab demonstration/ Lecture	Assignment
CLO3	Lab demonstration/ Lecture	Assignment
CLO4	Lab demonstration/ Lecture	Presentation/Viva
CLO5	Lab demonstration/ Lecture	Lab Report /Final Exam

Course Code: CSE 411	Credits: 3.0	Year: Fourth	Semester: Seventh
Course Title: Artificial Intelligence		**Course Status: Theory**	

Rationale of the course:

The objective of the course is to present an overview of artificial intelligence (AI) principles and approaches. Develop a basic understanding of the building blocks of AI as presented in terms of intelligent agents: Search, Knowledge representation, inference, logic, and learning.

Course content:

Introduction : Introduction and concepts of artificial intelligence, symbolic computing using functional and logic programming language; Concept of declarative programming, Classification and evaluation of declarative language; Knowledge organization and manipulation; search strategies and game planning; Matching techniques, Knowledge organization and management.

Functional Programming: Mathematical background. Data flow analysis; Programming in LIPS; Program design and Styles, flow of control; Axioms and list structures and processing properties, cell notations and pointers.

Logic Programming : Mathematical background, Representation and reasoning Logic programs and programming styles.

Programming in PROLOG : Prolong environment, List processing, arithmetic, I/O and memory operations and databases in PROLOG.

Object Oriented Programming: Introduction to object-oriented programming style. Encapsulation, inheritance polymorphism and overloading; Object-orientation. Classes, Declaring object, Member functions, Constructor and destructor; Extending classes, Multiple inheritance, Virtual functions, Dynamics objects, case tools.

Knowledge Engineering : Representing knowledge; Logic, procedural representation, semantic networks, productions systems, semantic primitives, frames and scripts.

Search: Goal, data driven, top-down / bottom-up control, automatic backtracking, Knowledge-based system design. Expert systems, Basic concepts of computer vision, robotics, voice processing.

Functional concepts : derived from the biological neuron; Characteristics of a single neuron, the perception and the construction of neuron network. Implementation of neural networks by hardware and software, Properties of neural networks; Learning in neural networks, Applications of neural networks.

Text Books:

1. Stuart J. Russel and Peter Norving: Artificial Intelligence: A Modern Approach

Reference Books:

1. Nills J. Nilson: Principles of Artificial Intelligence
2. J. M. Zurada: Introduction to Artificial Neural Systems

Course Learning Outcomes (CLOs):

Upon completion of the course, the students will be able to:

CLO 1: Acknowledge what constitutes "Artificial" Intelligence and how to identify systems with Artificial Intelligence.

CLO 2: Explain how Artificial Intelligence enables capabilities that are beyond conventional technology, for example, chess-playing computers, self-driving cars, robotic vacuum cleaners.
CLO 3: Implement common Artificial Intelligence techniques, such as search algorithms, minimax algorithm, and neural networks.
CLO 4: Ability to apply Artificial Intelligence techniques for problem solving.
CLO 5: Explain the limitations of current Artificial Intelligence techniques.

Mapping of Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs):

Program Learning Outcomes (PLOs)	CLO 1	CLO 2	CLO 3	CLO 4	CLO 5
PLO 1: Engineering Knowledge	√				
PLO 2: Problem Analysis				√	
PLO 3: Design/Development of Solutions					
PLO 4: Investigation					
PLO 5: Modern Tool Usage					
PLO 6: The Engineer and Society			√	√	
PLO 7: Environment and Sustainability					
PLO 8: Ethics					
PLO 9: Communication					
PLO 10: Individual and Team Work					
PLO 11: Project Management and Finance					
PLO 12: Life-Long Learning					

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lecture	Assignment / Midterm Exam
CLO2	Lecture / Case Study	Assignment / Class Test /Midterm Exam
CLO3	Lecture / Case Study	Assignment / Class Test /Midterm Exam
CLO4	Lecture / Group Discussion	Class Test / Midterm Exam/ Final Exam
CLO5	Lecture / Group Discussion	Presentation/ Class Test / Final Exam

Course Code: CSE 412	**Credits: 1.0**	**Year: Fourth**	**Semester: Seventh**
Course Title: Artificial Intelligence Lab		**Course Status: Lab**	

Rationale of the course:
To study first-order predicate logic and logical reasoning using Prolog language and implementing various techniques and algorithms of AI used in general problem solving, optimization problems, and also to familiarize students with various scopes of AI, such as neural network and machine learning.

Course Content

Programming in LIPS; Program design and Styles, flow of control; Axioms and list structures and processing properties, cell notations and pointers, Prolong environment, List processing, arithmetic, I/O and memory operations and databases in PROLOG, Logic, procedural representation, semantic networks, productions systems, semantic primitives, frames and scripts. automatic backtracking, Knowledge-based system design. Expert systems, biological neuron; Implementation of neural networks by hardware and software, Properties of neural networks; Learning in neural networks, Applications of neural networks.

Course Learning Outcomes (CLOs):

Upon completion of the course, the students will be able to:

CLO 1: Explain artificial intelligence, its features, and its potential applications.

CLO 2: Express real-world issues can be express as state space problems, optimization problems, or constraint satisfaction problems.

CLO 3: Solve complex issues, choose and use relevant algorithms and AI techniques.

CLO 4: Use relevant tools and approaches design and **create** an expert system.

CLO 5: Analyze the performance of different optimizer algorithms.

Mapping of Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs):

Program Learning Outcomes (PLOs)	CLO 1	CLO 2	CLO 3	CLO 4	CLO 5
PLO 1: Engineering Knowledge					
PLO 2: Problem Analysis					√
PLO 3: Design/Development of Solutions			√		
PLO 4: Investigation					
PLO 5: Modern Tool Usage				√	
PLO 6: The Engineer and Society					
PLO 7: Environment and Sustainability					
PLO 8: Ethics					
PLO 9: Communication					
PLO 10: Individual and Team Work					
PLO 11: Project Management and Finance					
PLO 12: Life-Long Learning		√			√

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lab demonstration/ Lecture	Assignment
CLO2	Lab demonstration/ Lecture	Assignment
CLO3	Lab demonstration/ Lecture	Assignment
CLO4	Lab demonstration/ Lecture	Presentation/Viva

CLO5	Lab demonstration/ Lecture	Lab Report /Final Exam

Course Code: CSE 413	**Credits: 3.0**	**Year: Fourth**	**Semester: Seventh**
Course Title: Computer Graphics		**Course Status: Theory**	

Rationale of the course:
This course encourages students to improve their capacity to conceptualize, develop, and adapt various shapes, structures, and images in real time, which is critical in the field of engineering and imaging technology.

Course Content:
Introduction: Basic concept of Graphical data processing, fundamentals of standard graphics primitive, graphical user interface, graphics hardware, display devices, raster refresh graphics display, use of frame buffer and look-up table, co-ordinate convention, device co-ordinate; Raster scan graphics, line and circle generation, algorithms character generation different types of polygon, point location, polygon filling, triangulation, window and clipping, viewpoint, zooming, planning, line, text and polygon clipping. **Language for Graphics:** Display file compilers, system software for interaction co-ordinate vector and matrix, representation of 3-D objects transformation; Architecture of display device and connective to a computer, implementation of graphics contents of 2-D and 2-D viewing; Windowing, clipping, and transformation; Hidden surfaces and lines algorithms; Light, color and shading, Curves and fractals; Adding faces for visual realism. **Raster graphics**: concept; architecture, algorithms, and other images synthesis methods; Design of interactive graphic conversion; Future trends in computer graphics. **Video :** Digital video fundamentals, configuring the video drivers, Video Capture techniques. Video editing techniques; Video editing software.
Text: Digital text, Fonts management, Types of Font, 2 & 3 Dimensional text, Text animation, Hypermedia, Hypertext, Special effect to text. **Animation :** Concept of animation, Classification, 2 & 3 Dimensional animation, Key framing & animating, Rendering, Optimizing Animation software tools; Cartoon & Fun Animation.

Text Books:
1. Roy A Plastock, Zhigang Xiang : Computer Graphics
2. Foley, Van Dam, Feiner, Hughes: Computer Graphics Principles and Practice

Reference Book:
1. Stuart Mealing : The art and science of computer animation .

Course Learning Outcomes (CLOs):
Upon completion of the course, the students will be able to:

CLO 1: Provide a thorough introduction to computer graphics, allowing students to comprehend current terminology, progress, difficulties, and trends.
CLO 2: Discuss the computational and mathematical tools that are utilized to create various digital images and effects.
CLO 3: Demonstrate three major computer graphics concepts: modeling, rendering, and animation.
CLO 4: Choose and **analyze** fundamentals such as digital image representation, color perception, picture generation, and image processing.

Mapping of Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs):

Program Learning Outcomes (PLOs)	CLO 1	CLO 2	CLO 3	CLO 4
PLO 1: Engineering Knowledge				
PLO 2: Problem Analysis				√
PLO 3: Design/Development of Solutions				
PLO 4: Investigation			√	
PLO 5: Modern Tool Usage				
PLO 6: The Engineer and Society				
PLO 7: Environment and Sustainability				
PLO 8: Ethics				
PLO 9: Communication				
PLO 10: Individual and Team Work				
PLO 11: Project Management and Finance				
PLO 12: Life-Long Learning				

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lecture	Assignment / Midterm Exam
CLO2	Lecture	Assignment / Class Test /Midterm Exam
CLO3	Lecture / Group Discussion	Presentation/ Final Exam
CLO4	Lecture / Group Discussion	Presentation/ Class Test / Final Exam

Course Code: CSE 414	Credits: 1.0	Year: Fourth	Semester: Seventh
Course Title: Computer Graphics Lab		**Course Status: Lab**	

Rationale of the course:
This course encourages students to use a graphics library to create and change 2D and 3D visualizations and transformations of any geometric object, as well as work with texturing,

lighting, and coloring of such objects to create many sorts of digital images with diverse effects. This course will help students to solve problems using digital logic circuits and systems. The fundamental building blocks of combinational and sequential circuits are introduced in order for students to generate circuit solutions to issues and to comprehend the design and operation of digital hardware models.

Course Content:

Introduction to OpenGL, Drawing 2D geometric objects, Simple 2D animation and modeling transformation, Drawing 3D geometric objects, Viewing transformation and Texturing and lighting.

Course Learning Outcomes (CLOs):

Upon completion of the course, the students will be able to:

CLO 1: Create design drawings that display your knowledge of computer graphics and design.
CLO 2: Gain a fundamental understanding of the OpenGL graphics library.
CLO 3: Design and **construct** 2D and 3D graphical geometric objects.
CLO 4: Learn simple animation, coloring, texturing and lighting,.

Mapping of Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs):

Program Learning Outcomes (PLOs)	CLO 1	CLO 2	CLO 3	CLO 4
PLO 1: Engineering Knowledge				
PLO 2: Problem Analysis		√		
PLO 3: Design/Development of Solutions	√		√	
PLO 4: Investigation		√		
PLO 5: Modern Tool Usage			√	√
PLO 6: The Engineer and Society				
PLO 7: Environment and Sustainability				
PLO 8: Ethics				
PLO 9: Communication				
PLO 10: Individual and Team Work				
PLO 11: Project Management and Finance				
PLO 12: Life-Long Learning				√

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lab demonstration/ Lecture	Assignment
CLO2	Lab demonstration/ Lecture	Assignment
CLO3	Lab demonstration/ Lecture	Presentation/Viva
CLO4	Lab demonstration/ Lecture	Lab Report /Final Exam

Course Code: CSE 415	Credits: 3.0	Year: Fourth	Semester: Seventh
Course Title: Digital Signal Processing		Course Status: Theory	

Rationale of the Course:

Digital Signal Processors (DSP) take real-world signals like voice, audio, video, temperature, pressure, or position that have been digitized and then mathematically manipulate them. Signal processing is integral in searching for life beyond Earth. An important aspect to effective communications across satellite, video, radio and wireless systems, signal processing makes the processing and transmission of data more efficient.

Digital signal processing (DSP) is the use of digital processing, such as by computers or more specialized digital signal processors, to perform a wide variety of signal processing operations. The digital signals processed in this manner are a sequence of numbers that represent samples of a continuous variable in a domain such as time, space, or frequency.

Course Content:

Introduction to Digital Signal Processing: Signal, System, Signal processing, Basic Elements of DSP, Advantages, Drawbacks and Application areas of Digital Signal Processing, Classification of Signals with Explanation & Example, Concept of frequency in continuous time sinusoidal signals and Characteristics of Analog Sinusoidal signal, Concept of frequency in Discrete time sinusoidal signals and Properties of Discrete time Sinusoids, ADC, Sampling, Aliasing effect, Anti-aliasing filtering, related mathematical problems.

Discrete-Time Signals and Systems: Representation of discrete-time signal, Basic Elementary Signals, Classification of Discrete-Time Signals, Signal Operations, Block Diagram, Representation and Classification of Discrete-Time Systems, Analysis of Discrete-Time Linear Time-Invariant Systems, Response of LTI Systems to Arbitrary Inputs: The Convolution Sum, Mathematical and graphical representation of linear convolution, Circular convolution using matrix method, Correlation, Cross correlation coefficient, Auto correlation.

The z-Transform and Its Application to the Analysis of LTI Systems: z-Transform, Properties of z-Transform, Relation between Laplace & z-transform, One-sided z-Transform and Related Mathematical Problems, Rational z-Transform, Pole-Zero Locations & Plot and Related Mathematical Problems.

Frequency Analysis of Signals: The Fourier Series for Continuous-Time Periodic Signals.

The Discrete Fourier Transform: Its Properties and Applications: IDFT, Properties of DFT, DFT as Linear Transformation, Fast Fourier Transform (FFT).

Frequency-Domain Analysis of LTI Systems: LTI Systems as Frequency Selective Filters, Digital Resonators, Notch Filters Design Digital Resonators.

Design of Digital Filters: Introduction to Digital filter, Structure of realization of LTI system, Filter design steps, FIR filter calculation: Window method, Optimal method, Frequency sampling method, Basic features of IIR filter.

Text Books:

1. Proakis and Manolakis: Digital Signal Processing - Principles, Algorithms and Applications
2. A.V.Oppenheim & R.W.Schafer : Digital Signal Processing

Reference Books:

1. E Efeachor: Digital Signal Processing – A Practical Approach
2. A.V.Oppenheim & R.W.Schafer: Discrete Time Signal Processing J.R.Johnson : Introduction to Digital Signal Processing

Course Learning Outcomes (CLOs):

After completing the course, the students will be able to:

CLO 1: Remember the definitions of signal, systems and their discrete-time and continuous time representation using the concepts of trigonometry, complex algebra and acquire knowledge about Systems.

CLO 2: Apply Fourier transform and z-transform in solving real life problems.

CLO 3: Apply different properties of the transform equations to design, implement, analyze and compare the digital filters for processing of discrete-time signals.

CLO 4: Analyze various transfer domain representation in designing filters and integrate computer-based tools for engineering applications and employing signal processing strategies.

CLO 5: Develop creative and innovative designs that achieve desired performance criteria within specified objectives and constraints.

Mapping of Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs):

CLOs	Program Learning Outcomes (PLOs)											
	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√											
CLO2		√										
CLO3			√									
CLO4			√	√								
CLO5			√						√	√		

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lecture	Class Test/ Assignment / Midterm Exam
CLO2	Lecture	Assignment / Class Test /Midterm Exam
CLO3	Lecture / Group Discussion	Presentation/ Midterm Exam/ Final Exam
CLO4	Lecture / Group Discussion	Presentation/ Class Test / Final Exam
CLO5	Lecture/ Group Discussion	Assignment/ Mini Project/ Presentation

Course Code: CSE 416	Credits: 1.0	Year: Fourth	Semester: Seventh
Course Title: Digital Signal Processing Lab		Course Status: Lab	

Rationale of the course:

The area of digital signal processing has advanced rapidly over the last four decades. This advancement is attributed to the progress in digital computer technology and integrated circuit fabrications. This lab introduces students to DSP design and analysis techniques that are core knowledge for DSP engineers, and which serve as solid grounding for advanced level work in DSP. This lab aims at supporting the teaching and research activities in the area of DSP, which are either currently being.

Course Content:

Introduction to MATLAB and Generation of Continuous time signal and Discrete time signal, Generation of different types of basic signals (Unit step signal, Impulse signal, Ramp signal, Exponential signal), Verification of Sampling Theorem, Aliasing and Spectrum analysis of sampled signals, Different types of signal operation of any sinusoidal signal, DTFT of finite-length signals in MATLAB and their spectra, Impulse response of a given system, Linear Convolution of two given systems, Auto correlation of a given sequence and verification its properties, z-Transformation and LTI system analysis in MATLAB, Write the MATLAB code for Fast Fourier Transform of any sinusoidal signal, FIR/IIR Filters using pole-zero placement, IIR filter design using impulse invariance, Perform Fast Fourier Transform of any sinusoidal signal using FPGA board.

Course Learning Outcomes (CLOs):

After completing the course, the students will be able to:

CLO 1: Write comprehensive reports on the experiments done in laboratory or in a group project, and demonstrate the ability to work as an individual and within a team.

CLO 2: Apply the theoretical knowledge for signal generation, processing and filter design using MATLAB and also by hardware implementation.

Mapping of Course Learning Outcomes (CLOs) to Program Learning Outcomes PLOs:

CLOs	Program Learning Outcomes (PLOs)											
	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1									√	√		
CLO2			√		√							

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lab demonstration/ Lecture	Lab report/Final exam
CLO2	Lab demonstration/ Lecture	Lab report/Final exam

Course Code: CSE 417	Credits: 3.0	Year: Fourth	Semester: Seventh
Course Title: Compiler Design		Course Status: Theory	

Rationale of the course:

The purpose of this course is to educate students the fundamental techniques that underpin the practice of various phases of compiler construction

Course Content:

Introduction: Introduction to Compiler – Analysis-Synthesis model of Compilation – Phases of a Compiler. **Lexical Analysis:** Role of a lexical analyzer; Token, lexeme and patterns – Lexical errors – Input buffering – sentinels. Specification of Tokens; Regular Expressions – Recognition of Tokens – Transition diagrams. Finite Automata – NFA – DFA – Simulating a DFA/NFA – Conversion of an NFA into a DFA – NFA from Regular Expressions – Regular Expression to a DFA. **Syntax Analysis:** Role of the Parser – Context-Free Grammars – Parse Trees & Derivations – Ambiguity – Regular expressions vs. context-free grammars – Elimination of left recursion – Top down parsing – predictive parsing – bottom up parsing – LR parsers. **Syntax-Directed Translation:** Inherited and Synthesized attributes – Form of a Syntax-Directed Definition – Dependency Graph – Construction of syntax trees – Directed Acyclic Graph – Bottom up and Top down evaluation of attributes – L- and S-attributed definitions. **Type Checking:** Static checking – Type systems – Specification of a simple type checker – Type checking of Expressions – Type checking of Statements – Type checking of Functions. **Intermediate Code Generation:** Intermediate language – Types of Three-Address Statements – Implementations of Three-Address Statements – Assignment statements – Boolean expressions. **Code Generation:** Issues – Basic blocks and Flow graphs. **Code Optimization:** Principal sources of optimization – Optimization of basic blocks.

Text Books:

1. Jeffrey D.Ullman : Compilers: Principles, Techniques and Tools

Reference Books:

1. Holub : Compiler design
2. Trembly and Sorensen : Theory and practices of Compiler writing

Course Learning Outcomes (CLOs):

Upon completion of the course, the students will be able to:

CLO 1: Explain the role and purpose of compilers in programming languages.

CLO 2: Discuss the steps of the compilation procedure.

CLO 3: Separate the lexical, syntactic, and semantic analysis into meaningful phases so that a compiler can perform language translation and specify and **evaluate** advanced language characteristics' lexical, syntactic, and semantic structures.

CLO 4: Create scanners and parsers, as well as abstract syntax trees with the help of tools.

Mapping of Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs):

Program Learning Outcomes (PLOs)	CLO 1	CLO 2	CLO 3	CLO 4
PLO 1: Engineering Knowledge				
PLO 2: Problem Analysis	√	√	√	
PLO 3: Design/Development of Solutions				√
PLO 4: Investigation			√	
PLO 5: Modern Tool Usage				
PLO 6: The Engineer and Society				
PLO 7: Environment and Sustainability				
PLO 8: Ethics				
PLO 9: Communication				
PLO 10: Individual and Team Work				
PLO 11: Project Management and Finance				
PLO 12: Life-Long Learning				

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lecture	Assignment / Midterm Exam
CLO2	Lecture	Assignment / Class Test /Midterm Exam
CLO3	Lecture / Group Discussion	Presentation/ Final Exam
CLO4	Lecture / Group Discussion	Presentation/ Class Test / Final Exam

Course Code: CSE 418	Credits: 1.0	Year: Fourth	Semester: Seventh
Course Title: Compiler Design Lab		**Course Status: Lab**	

Rationale of the course:
To develop a tokenizer and an arithmetic calculator, as well as the ability to write code in Flex and Bison.

Course Content:
Symbol Table, Tokenizer, Tokenizer Using Flex, Arithmetic Calculator using Bison, Intermediate Code Generator (Flex + Bison).

Course Learning Outcomes (CLOs):
Upon completion of the course, the students will be able to:

CLO 1: Use basic compiler construction techniques and tools to convert a high-level programming language into executable code using syntax-directed translation.
CLO 2: Explain how the lex and yacc compilers work to debug programs.
CLO 3: Discuss the new tools and technology used in compiler development.

Mapping of Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs):

Program Learning Outcomes (PLOs)	CLO 1	CLO 2	CLO 3
PLO 1: Engineering Knowledge			
PLO 2: Problem Analysis	√	√	
PLO 3: Design/Development of Solutions			√
PLO 4: Investigation			
PLO 5: Modern Tool Usage		√	√
PLO 6: The Engineer and Society			
PLO 7: Environment and Sustainability			
PLO 8: Ethics			
PLO 9: Communication			
PLO 10: Individual and Team Work			
PLO 11: Project Management and Finance			
PLO 12: Life-Long Learning	√		√

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lab demonstration/ Lecture	Assignment
CLO2	Lab demonstration/ Lecture	Presentation/Viva
CLO3	Lab demonstration/ Lecture	Lab Report /Final Exam

Course Code: CSE 419	Credits: 3.0	Year: Fourth	Semester: Seventh
Course Title: Communication Engineering		Course Status: Theory	

Rationale of the course:
This course motivates students to design and install equipment for transmitting wired phone, cellular, cable, and broadband data, as well as working with copper or fiber optic cabling, complex networks, and switching systems, in order for businesses to effectively communicate with customers and provide high-quality service.

Course Content:

Synchronous and asynchronous communications; Hardware interfaces, multiplexers, concentrators and buffers; Communication mediums and their characteristics; Data communication services: SMDS and ATM; Error control codes: linear block codes, cyclic codes, MLDC codes, convolution codes, Trellis code modulation; Digital switching: space and time division switching; Radio system design; Fiber optics communication: transmitter, receivers, network components, WDM; Line coding, trunks, multiplexing, switching, ATM switches; Satellite communications: frequency bands and characteristics, types of satellites, multiple access techniques; Cellular communications: GSM, CPDP.

Text Books:

1. Data Communications and Computer Networks –William Stallings
2. Data Communications and Networking – Behrouz A. Forouzan
3. Computer Networking: A Top Down Approach Featuring the Internet, J. Kurose & Keith Ross, Addison Wesley

Reference Books:

1. Andrew S. Tanenbaum: Computer Networks
2. Douglas E. Comer : Internetworking with TCP/IP
3. Larry L. Peterson, Bruce S. Davie : Computer Networks

Course Learning Outcomes (CLOs):

Upon completion of the course, the students will be able to:

CLO 1: Analyze communication systems in both the time and frequency domains and describe the principles of amplitude modulated and angle modulated communication systems and **analyze** their performance in the presence of noise.

CLO 2: Understand source coding, information theory and Shannon's theorem.

CLO 3: Describe the principles of various digital modulation systems and their properties, including bandwidth, channel capacity, transmission over bandlimited channels, inter-symbol interference (ISI), demodulation methods, and error performance in the presence of noise.

CLO 4: Explain and **analyze** error correcting codes, including block codes.

CLO 5: Understand engineering fundamentals of photogeneration, photodetection, lightwave propagation, for optical communications and be able to implement noise and error analysis of an analogue or digital telecommunication system.

Mapping of Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs):

Program Learning Outcomes (PLOs)	**CLO 1**	**CLO 2**	**CLO 3**	**CLO 4**	**CLO5**
PLO 1: Engineering Knowledge					
PLO 2: Problem Analysis	√			√	
PLO 3: Design/Development of Solutions					
PLO 4: Investigation	√				
PLO 5: Modern Tool Usage					

PLO 6: The Engineer and Society					
PLO 7: Environment and Sustainability					
PLO 8: Ethics					
PLO 9: Communication					
PLO 10: Individual and Team Work					
PLO 11: Project Management and Finance					
PLO 12: Life-Long Learning					

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lecture	Assignment / Midterm Exam
CLO2	Lecture / Case Study	Assignment / Class Test /Midterm Exam
CLO3	Lecture / Case Study	Assignment / Class Test /Midterm Exam
CLO4	Lecture / Group Discussion	Class Test / Midterm Exam/ Final Exam
CLO5	Lecture / Group Discussion	Presentation/ Class Test / Final Exam

Course Code: CSE 421	**Credits: 3.0**	**Year: Fourth**	**Semester: Eighth**
Course Title: Software Engineering		**Course Status: Theory**	

Rationale of the course:

This course is intended to give students a broad overview of software engineering and design. It will cover key topics such as software processes and agile approaches, as well as critical software development tasks ranging from basic software specification to system evolution.

Course Content:

The Software Process: Generic view of process, process models, an agile view of process. Software Engineering Practice: Software engineering, system engineering, requirement engineering, building the analysis model, design engineering, creating an architectural design, modeling component –level design, performing user interface design, testing strategies, testing tactics, product metrics. Applying Web Engineering: web engineering, initiating a Web App project, Analysis for Web App project, Design for Web App project, Testing Web App project. Managing Software Projects: Project management, metrics for process and project, estimation, project scheduling, risk management, quality management, change management. Advanced Topics in Software Engineering: formal methods, clean room software engineering, component-based development, reengineering, and the root ahead.

Text Books:

1. Ian Sommerville : Software Engineering

Reference Books:

1. Roger S Pressman : Software Engineering: A Practitioner's Approach
2. Richard D. Stutzke : Estimating Software Intensive Systems: Projects, Products and Process

Course Learning Outcomes (CLOs):

Upon completion of the course, the students will be able to:

CLO 1: create a model correctly of the structure and behavior of a software system.

CLO 2: Design and **build** an executable solution to a given problem using common software concepts and best practices in a programming language.

CLO 3: Use proper software testing methodologies to assess a software product's quality at the module, integration, and system levels and

CLO 4: Choose and **adapt** appropriate aspects from existing and growing software development life-cycle procedures, then apply the new process to a software project.

CLO5: Collaborate in groups to create a large-scale software system from conception to completion and **communicate** effectively design and development decisions through written and graphical demonstration.

Mapping of Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs):

Program Learning Outcomes (PLOs)	CLO 1	CLO 2	CLO 3	CLO 4	CLO5
PLO 1: Engineering Knowledge					
PLO 2: Problem Analysis		√			
PLO 3: Design/Development of Solutions	√	√	√		√
PLO 4: Investigation			√	√	
PLO 5: Modern Tool Usage					
PLO 6: The Engineer and Society					
PLO 7: Environment and Sustainability					
PLO 8: Ethics					
PLO 9: Communication					√
PLO 10: Individual and Team Work					√
PLO 11: Project Management and Finance					
PLO 12: Life-Long Learning					√

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lecture	Assignment / Midterm Exam
CLO2	Lecture / Case Study	Assignment / Class Test /Midterm Exam
CLO3	Lecture / Case Study	Assignment / Class Test /Midterm Exam
CLO4	Lecture / Group Discussion	Class Test / Midterm Exam/ Final Exam
CLO5	Lecture / Group Discussion	Presentation/ Class Test / Final Exam

Course Code: CSE 423	Credits: 3.0	Year: Fourth	Semester: Eighth
Course Title: Digital Image Processing		Course Status: Theory	

Rationale of the course:
Introduce the principles of image processing and manipulation, medical imaging modalities like X-ray or ultrasound, photography, security, astronomy, and remote sensing.

Course Content:
Digital Image fundamentals, Image transforms Different image enhancement techniques, histogram equalization and modification for effective visualization spatial filtering, median filtering, edge enhancement and detection.
Image restoratives, Image compression, Image segmentation, Representation and description, Recognition and interpretation. Mathematical Morphology: Erosion, Dilation, opening closing; Application of mathematical morphology in digital image processing.

Text Book:
1. Rafael C. Gonzalez, Richard E. Woods; Digital Image Processing

Reference Book:
1. Phillips; Image Processing in C

Course Learning Outcomes (CLOs):
Upon completion of the course, the students will be able to:
CLO 1: Explain image formation and the role human visual system plays in perception of gray and color image data.
CLO 2: Describe the basic elements and applications of image processing.
CLO 3: Choose and **examine** the requirements for picture sampling and quantization, as well as their implications.
CLO 4: Evaluate Gray level transformations for Image enhancement.

Mapping of Program Learning Outcomes (PLO) and Course Learning Outcomes (CLO):

Program Learning Outcomes (PLOs)	CLO 1	CLO 2	CLO 3	CLO 4
PLO 1: Engineering Knowledge				
PLO 2: Problem Analysis			√	
PLO 3: Design/Development of Solutions				
PLO 4: Investigation			√	
PLO 5: Modern Tool Usage				
PLO 6: The Engineer and Society				
PLO 7: Environment and Sustainability				

PLO 8: Ethics				
PLO 9: Communication				
PLO 10: Individual and Team Work				
PLO 11: Project Management and Finance				
PLO 12: Life-Long Learning		√		

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lecture	Assignment / Midterm Exam
CLO2	Lecture	Assignment / Class Test /Midterm Exam
CLO3	Lecture / Group Discussion	Presentation/ Final Exam
CLO4	Lecture / Group Discussion	Presentation/ Class Test / Final Exam

Course Code: CSE 424	**Credits: 1.0**	**Year: Fourth**	**Semester: Eighth**
Course Title: Digital Image Processing Lab		**Course Status: Lab**	

Rationale of the course:

Students are introduced to contemporary analytical tools and methodologies in digital image processing as they apply to picture data for human viewing. They will be able to use these tools for picture restoration, enhancement, and compression.

Course content:

Sampling and Quantization, Imaging Geometry, Image Acquisition Systems, Medial Axis Transform, Component Labeling, Thinning, Morphological Processing, Extension to Grey Scale Morphology, Optimal Thresholding using Bayesian Classification, Multilevel Thresholding, first Order and Second Order Edge Operators, Multi-Scale Edge Detection, Canny's Edge Detection Algorithm, Hough Transform for Detecting Lines and Curves, Edge Linking, Point Processing, Spatial Filtering, Frequency Domain Filtering, Multi-Spectral Image Enhancement, Image Restoration, Segmentation of Grey Level Images, Water Shade Algorithm for Segmenting Grey Level Image. Image Representation and Description, Recognition and Interpretation, Lossy and Lossless Compression Schemes, Prediction based Compression Schemes, Vector Quantization, Sub-Band Encoding Schemes, JPEG Compression Standard, Fractal Compression Scheme, Wavelet Compression Scheme.

Text Book:

1. Rafael C. Gonzalez, Richard E. Woods; Digital Image Processing

Reference Book:

1. Phillips; Image Processing in C

Course Learning Outcomes (CLOs):

Upon completion of the course, the students will be able to:

CLO 1: Describe the basic elements and applications of image processing, different techniques employed for the enhancement of images.

CLO 2: Understand the need for image compression and to learn the spatial and frequency domain techniques of image compression.

CLO 3: Learn different causes for image degradation and overview of image restoration techniques.

CLO 4: Implement different feature extraction techniques for image analysis and recognition.

Mapping of Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs):

Program Learning Outcomes (PLOs)	**CLO 1**	**CLO 2**	**CLO 3**	**CLO 4**
PLO 1: Engineering Knowledge				
PLO 2: Problem Analysis			√	
PLO 3: Design/Development of Solutions				√
PLO 4: Investigation			√	
PLO 5: Modern Tool Usage				√
PLO 6: The Engineer and Society				
PLO 7: Environment and Sustainability				
PLO 8: Ethics				
PLO 9: Communication				
PLO 10: Individual and Team Work				
PLO 11: Project Management and Finance				
PLO 12: Life-Long Learning	√			

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	**Teaching-Learning Strategy**	**Assessment Strategy**
CLO1	Lab demonstration/ Lecture	Assignment
CLO2	Lab demonstration/ Lecture	Assignment
CLO3	Lab demonstration/ Lecture	Presentation/Viva
CLO4	Lab demonstration/ Lecture	Lab Report /Final Exam

Course Code: CSE 429	Credits: 1.0	Year: Fourth	Semester: Eighth
Course Title: Project/Thesis/Internship			

Rationale of the course:

This course will help Students building high-quality software in a variety of languages, either in groups or independently. Students will write well-structured programs that are well-documented.

Course Content:

Under the guidance of a faculty member, students will be allocated a thesis or project topic from their major group. Students will be placed in an organization that does technical work in the field of their major group for a one-semester internship. A faculty supervisor will be appointed to the student to oversee the internship work.

d) Optional Courses:

Course Code: CSE 431	Credits: 3.0	Year: Fourth	Semester: Eighth
Course Title: Multimedia Systems Design		Course Status: Theory	

Rationale of the course:
This course encourages students to learn about the architecture, different compression and coding standards for multimedia documents, database, network and operating system issues, traffic and service issues, and security issues, and then apply what they've learned to create various multimedia applications.

Course Content:
Organization and structure of modern multimedia systems; text, audio and video encoding; Data compression: lossless and lossy techniques; Multimedia networking: Quality of Service management and multimedia protocols; Streaming multimedia: peer-to-peer, video-on-demand, live streaming; Multimedia storage: data placement and scheduling, caching, and data retrieval; Scheduling algorithms for multimedia within OS; Synchronization schemes: in-band and out-band, synchronization skews and specification; Design of real-world multimedia solution.

Recommended Books:
1. Multimedia: Computing, Communications & Applications (US Edition) - Ralf Steinmetz, Klara Nahrstedt; Prentice Hall (1995).
2. Fundamentals of Multimedia byZ. M. Li andM. S. Drew, Prentice Hall (2004), ISBN: 0-13-127256-X
3. Digital Multimedia by N. Chapman and J. Chapman. 2ndEdition, Wiley 2004, ISBN: 0-470-85890-7.
4. The Technology of Video and Audio Streaming by David Austerberry, Focal Press; 2nd Edition (2004). ISBN-10: 0240805801
5. Multimedia Security: Watermarking, Steganography, and Forensics by Frank Y. Shih, CRC Press; 1st Edition (2012), ISBN-10: 1439873313.
6. Multimedia Computing by Daniel Cunliffe and Geoff Elliott, Lexden Publishing Ltd (2005). ISBN-10: 1904995055.
7. Multimedia Foundations: Core Concepts for Digital Design by Vic Costello, Ed Youngblood and Susan Youngblood, Focal Press; 1st Edition (2012). ISBN-10: 0240813944.

Course Learning Outcomes (CLOs):
Upon completion of the course, the students will be able to:
CLO 1: Discuss indexing and storing multimedia data for multimedia document.
CLO 2: Explain disk placement, disk scheduling, searching techniques for multimedia document.

CLO 3: Discover and apply the knowledge acquired in developing multimedia applications – audio and video conferencing, video on demand, and voice over IP.

Mapping of Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs):

Program Learning Outcomes (PLOs)	CLO 1	CLO 2	CLO 3
PLO 1: Engineering Knowledge	√	√	
PLO 2: Problem Analysis			
PLO 3: Design/Development of Solutions			
PLO 4: Investigation			
PLO 5: Modern Tool Usage			√
PLO 6: The Engineer and Society			
PLO 7: Environment and Sustainability			
PLO 8: Ethics			
PLO 9: Communication			
PLO 10: Individual and Team Work			
PLO 11: Project Management and Finance			
PLO 12: Life-Long Learning			

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lecture	Assignment / Midterm Exam
CLO2	Lecture	Assignment / Class Test /Midterm Exam
CLO3	Lecture / Group Discussion	Presentation/ Class Test / Final Exam

Course Code: CSE 433	Credits: 3.0	Year: Fourth	Semester: Eighth
Course Title: Telecommunication Engineering		**Course Status: Theory**	

Rationale of the course:

The goal of this course is to teach CSE students about the fundamentals of telecommunication engineering. This course covers analog modulation methods, the performance of various modulation schemes in the presence of noise, and conversion from analog to digital communication systems. In addition, this course covers tele traffic systems and digital modulation algorithms.

Course content:

Introduction: Elements of a communication systems; Rotary dialing telephone, pulse and mid-frequency dialing, Communication model, data communication tasks, data communication

network standards and organization; Design parameters of a switching system; cross bar switching, stored program control, enhanced services, two-stage and three stage networks, n-stage networks, time division switching, introduction to OSI and TCP/IP models, exchange and international regulator bodies.

Data Transmission Basics: Analog and digital data spectrum and bandwidth, transmission impairments, data rate and channel capacity.

Transmission Media: Characteristics and applications of twisted pairs, coaxial cables and optical fibers, terrestrial and satellite microwave, radio waves, VSAT; Data encoding and signals.

Telephone Apparatus: Microphone, speakers, ringer, pulse and tone dialing mechanism, side-tone mechanism, local and central batteries and advanced features.

Switching System: Introduction to analog system, digital switching systems – space division switching, blocking probability and multistage switching, time division switching and two dimensional switching.

Traffic Analysis: Traffic characterization, grades of service, network blocking probabilities, delay system and queuing.

Modern Telephone Services and Network: Internet telephony, facsimile, integrated services digital network, asynchronous transfer mode and intelligent networks. Introduction to cellular telephony.

Text Books:

1. Alberto Leon-Garcia, Indra Widjaja: Communication Networks-Fundamental Concepts and Key Architectures
2. Annabel Z. Dodd: Essential Guide to Telecommunications
3. Lillian Goleniewski: Telecommunications Essentials

Reference Books:

1. John Bellamy: Digital Telephony
2. James Martin: Telecommunications and the Computer
3. M T Hills: Telecommunication Switching Principle

Course Learning Outcomes (CLOs):

Upon completion of the course, the students will be able to:

CLO 1: Demonstrate technical and theoretical knowledge of telecommunications systems associated with LANs, MANs, and WANs.

CLO 2: Learn to design, implement, and manage telecommunications systems using voice and data.

CLO 3: Model and **simulate** telecommunications systems and networks in order to identify and solve these problems.

CLO 4: Acquire the knowledge and expertise in the field of telecommunication hardware.

Mapping of Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs):

Program Learning Outcomes (PLOs)	CLO 1	CLO 2	CLO 3	CLO 4
PLO 1: Engineering Knowledge	√			√
PLO 2: Problem Analysis				
PLO 3: Design/Development of Solutions		√		
PLO 4: Investigation				
PLO 5: Modern Tool Usage				
PLO 6: The Engineer and Society				
PLO 7: Environment and Sustainability				
PLO 8: Ethics				
PLO 9: Communication				
PLO 10: Individual and Team Work				
PLO 11: Project Management and Finance				
PLO 12: Life-Long Learning		√		√

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lecture	Assignment / Midterm Exam
CLO2	Lecture	Assignment / Class Test /Midterm Exam
CLO3	Lecture / Group Discussion	Presentation/ Final Exam
CLO4	Lecture / Group Discussion	Presentation/ Class Test / Final Exam

Course Code: CSE 435	Credits: 3.0	Year: Fourth	Semester: Eighth
Course Title: Web Programming		**Course Status: Theory**	

Rationale of the course:

Web Engineering introduces a structured methodology utilized in software engineering to Web development projects. The course addresses the concepts, methods, technologies, and techniques of developing Web sites that collect, organize and expose information resources. Topics covered include requirements engineering for Web applications, design methods and technologies, interface design, usability of web applications, accessibility, testing, metrics, operation and maintenance of Web applications, security, and project management. Specific technologies covered in this course include client-side (HTML, JavaScript, and CSS) and server-side (ASP.NET/PHP).

Course Content:

Review: Short lecture on how a Web server processes Web pages and Active Server Pages including programming scripts, Tutorial on additional HTML coding including links, images, lists, and tables.

Visual Basic Programming: Short lecture on programming concepts: variables and memory, replacement statement, expressions, conditional processing, looping, Web pages as output.

Interacting with Databases in Visual Basic Programs: Short lecture on application programming, the ASP object model, and overview of middleware for connecting to databases

Web Programming: Introduction to the World Wide Web, HTTP, Markup with HTML5, and Revision Control with Git, Design and Layout with Cascading Style Sheets (CSS), Client-Side Programming with JavaScript and the Document Object Model (DOM), More JavaScript, More HTML5, APIs, Server-Side Programming with PHP, Server-Side Programming with PHP (continued); Pattern Matching with Regular Expressions, Ajax; Cookies, Sessions, and Local Storage, Databases with MySQL, Web Security, JavaScript Libraries

Text Books:

1. Steflik and Shridharen; *Advanced Java Networking*; Prentice-Hall
2. Essential Javascript; ; Prentice-Hall
3. Goodman - JavaScript Bible; Goodman, IDG
4. Cosentino - essential PHP, Prentice-Hall

Reference Books:

1. Atkinson - Core PHP
2. Weissinger - ASP in a Nutshel, O'Reilly
3. Garfinkle and Spafford- Web Security and Commerc, O'Reilly
4. Menasce'- Web Performance, Prentice-Hall
5. Philip and Alex's Guide to Web Publishing, Greenspun; Morgen Kaufmann

Course Learning Outcomes (CLOs):

Upon successful completion of this course, the students able to:

CLO 1: **Analyze** and **design** comprehensive systems for the creation, dissemination, storage, retrieval, and use of electronic records and documents

CLO 2: **Learn** and **use** some of the client-side and server-side languages used to manipulate information on the World Wide Web – i.e. PHP, and Javascript.

CLO 3: **Learn** techniques and evaluation metrics for ensuring the proper operability, maintenance and security of a web application.

CLO 4: **Apply** the web engineering methodologies for Web application development

CLO 5: Identify and **discuss** the security risk of a Web application.

Mapping of Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs):

Program Learning Outcomes (PLO)	**CLO 1**	**CLO 2**	**CLO 3**	**CLO 4**	**CLO 5**
PLO 1: Engineering Knowledge	√			√	
PLO 2: Problem Analysis	√				

PLO 3: Design/Development of Solutions	√	√			
PLO 4: Investigation					√
PLO 5: Modern Tool Usage		√			
PLO 6: The Engineer and Society					
PLO 7: Environment and Sustainability					
PLO 8: Ethics					
PLO 9: Communication					
PLO 10: Individual and Team Work					
PLO 11: Project Management and Finance					
PLO 12: Life-Long Learning					

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lecture	Assignment / Midterm Exam
CLO2	Lecture / Case Study	Assignment / Class Test /Midterm Exam
CLO3	Lecture / Case Study	Assignment / Class Test /Midterm Exam
CLO4	Lecture / Group Discussion	Class Test / Midterm Exam/ Final Exam
CLO5	Lecture / Group Discussion	Presentation/ Class Test / Final Exam

Course Code: CSE 436	**Credits: 1.0**	**Year: Fourth**	**Semester: Eighth**
Course Title: Web Programming Lab		**Course Status: Lab**	

Rationale of the course:

Web Engineering introduces a structured methodology utilized in software engineering to Web development projects. The course addresses the concepts, methods, technologies, and techniques of developing Web sites that collect, organize and expose information resources. Topics covered include requirements engineering for Web applications, design methods and technologies, interface design, usability of web applications, accessibility, testing, metrics, operation and maintenance of Web applications, security, and project management. Specific technologies covered in this course include client-side (HTML, JavaScript, and CSS) and server-side (ASP.NET/PHP).

Course Content:

HTML Basic ,HTML Elements, HTML Attributes, HTML Headings, HTML Paragraphs, HTML Styles, HTML Formatting, HTML Quotations, HTML Comments, HTML Colors, HTML CSS, HTML Links, HTML Images, HTML Favicon, HTML Tables, HTML Lists, HTML Block & Inline HTML Classes, HTML Layout; CSS Tutorial, CSS HOME, CSS Introduction, CSS Syntax, CSS Selectors, CSS Comments, CSS Colors, CSS Backgrounds, CSS Borders, CSS Margins CSS Padding, CSS Height/Width, CSS Box Model, CSS Outline, CSS Text, CSS Fonts,

CSS Icons, CSS Links, CSS Lists, CSS Tables, CSS Display, CSS Navigation Bar, CSS Forms, PHP Basics, Form validation, Regular expression, Database connection with PHP server, Object oriented php, Bootstrap functions, JavaScript, JQuery, Web Based Application Development.

Course Learning Outcomes (CLOs):

Upon successful completion of this course, the students able to:

CLO 1: **Analyze** and **design** comprehensive systems for the creation, dissemination, storage, retrieval, and use of electronic records and documents

CLO 2: **Learn** and **use** some of the client-side and server-side languages used to manipulate information on the World Wide Web – i.e. PHP, and Javascript.

CLO 3: **Learn** techniques and evaluation metrics for ensuring the proper operability, maintenance and security of a web application.

CLO 4: **Apply** the web engineering methodologies for Web application development

CLO 5: Identify and **discuss** the security risk of a Web application.

Mapping of Program Learning Outcomes (PLO) and Course Learning Outcomes (CLO):

Program Learning Outcomes (PLO)	**CLO 1**	**CLO 2**	**CLO 3**	**CLO 4**	**CLO 5**
PLO 1: Engineering Knowledge					
PLO 2: Problem Analysis					
PLO 3: Design/Development of Solutions	√			√	
PLO 4: Investigation					√
PLO 5: Modern Tool Usage		√		√	
PLO 6: The Engineer and Society			√		
PLO 7: Environment and Sustainability					
PLO 8: Ethics					
PLO 9: Communication					
PLO 10: Individual and Team Work					
PLO 11: Project Management and Finance					
PLO 12: Life-Long Learning		√			√

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	**Teaching-Learning Strategy**	**Assessment Strategy**
CLO1	Lab demonstration/ Lecture	Assignment
CLO2	Lab demonstration/ Lecture	Assignment
CLO3	Lab demonstration/ Lecture	Assignment
CLO4	Lab demonstration/ Lecture	Presentation/Viva
CLO5	Lab demonstration/ Lecture	Lab Report /Final Exam

Course Code: CSE 437	Credits: 3.0	Year: Fourth	Semester: Eighth
Course Title: Software Project Management		Course Status: Theory	

Rationale of the course:

Studying Software Project Management can lead to a rewarding profession or simply help you improve your organizational abilities. Knowing how to manage a project from beginning to end might mean the differences between success and failure. It is increasingly in-demand ability, and learning it can help you advance your career in any field.

Course Content:

The project as a system, management control, requirement specification, information and control in the organization, project selection, project scope and objectives, project infrastructure, and project characteristics analysis Cost-benefit analysis, selection of a suitable project report, and process model selection Activity throat, network planning model, identifying key path, and activity sequencing and scheduling Risk Management, Resource Allocation, and Control Monitoring ISO 9126: Software Quality Management.

Text Books:

1. Bob Hughes and Mike Cotterell, "Software Project Management ", Tata McGraw Hill, 4th edition, 2006

Reference Books:

1. Royce, "Software Project Management", Pearson Education, 2005
2. Kieron Conway, "Software Project Management", Dreamtech Press, 2001
3. S. A. Kelkar, "Software Project Management", PHI Publication, 15th edition, 2013.
4. Roger S. Pressman , "Software Engineering – A Practitioner's approach", Tata McGraw Hill, 2009

Course Learning Outcomes (CLOs):

Upon completion of the course, the students will be able to:

CLO 1: Learn different software quality assurance and quality control activities and standards for software projects.

CLO 2: Understand review and inspection techniques and formulate appropriate test plan and test cases based on system specifications.

CLO 3: Analyze different testing techniques and apply and **identify** appropriate testing for real-life complex software projects.

CLO 4: Apply and **examine** automated testing tools; **demonstrate** and **adapt** those skills; **justify** and **compare** them for optimized quality control.

Mapping of Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs):

Program Learning Outcomes (PLOs)	CLO 1	CLO 2	CLO 3	CLO 4
PLO 1: Engineering Knowledge	√	√		

PLO 2: Problem Analysis			√	
PLO 3: Design/Development of Solutions				
PLO 4: Investigation				
PLO 5: Modern Tool Usage				√
PLO 6: The Engineer and Society				
PLO 7: Environment and Sustainability				
PLO 8: Ethics				
PLO 9: Communication				
PLO 10: Individual and Team Work				
PLO 11: Project Management and Finance				
PLO 12: Life-Long Learning			√	

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lecture	Assignment / Midterm Exam
CLO2	Lecture	Assignment / Class Test /Midterm Exam
CLO3	Lecture / Group Discussion	Presentation/ Final Exam
CLO4	Lecture / Group Discussion	Presentation/ Class Test / Final Exam

Course Code: CSE 439	**Credits: 3.0**	**Year: Fourth**	**Semester: Eighth**
Course Title: Distributed Systems		**Course Status: Theory**	

Rationale of the course:

Distributed systems enable scalability and increased performance in ways that monolithic systems cannot, and they can offer features that would be difficult or impossible to design on a single system because they can depend on the capabilities of other computing devices and processes.

Course Content:

Introduction, interconnections, distributed system taxonomy, service models client-server computing, network protocols, IP and ATM networking communication models naming and binding. Sockets programming: C/Unix, Java. remote procedure calls (RPC), object brokers distributed file system design distributed file system case studies: NFS, AFS, logical clocks, vector clocks synchronization group communication: message ordering and message delivery ,IP multicasting mutual exclusion, election algorithms, distributed shared memory and memory consistency models authentication: protocols, digital certificates, Kerberos, SSL smart cards, biometrics, authenticating humans. Steganography. Firewalls and Virtual Private Networks fault

tolerance clustering: scalable performance, load balancing, and reliability, storage virtualization processor virtualization

Text Book:

1. A S Tanenbaum and MV Steen : Distributed Systems-Principles and paradigms

Reference Books:

1. R Adler : Distributed Co-ordination Models for Client/Server Computing-IEEE Computer
2. G Blair and J B Stefani : Open Distributed Processing and Multimedia.
3. L Gilman and R Schreiber : Distributed Computing with IBM MQSeries.
4. Jie Wu*:* Distributed system

Course Learning Outcomes (CLOs):

Upon completion of the course, the students will be able to:

CLO 1: Apply knowledge of distributed systems techniques and methodologies.

CLO 2: Explain the design and development of distributed systems and distributed systems applications.

CLO 3: Use the application of fundamental Computer Science methods and algorithms in the development of distributed systems and distributed systems applications.

CLO 4: Discuss the design and testing of a large software system, and to be able to communicate that design to others.

Mapping of Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs):

Program Learning Outcomes (PLOs)	**CLO 1**	**CLO 2**	**CLO 3**	**CLO 4**
PLO 1: Engineering Knowledge	√			
PLO 2: Problem Analysis				
PLO 3: Design/Development of Solutions				
PLO 4: Investigation				
PLO 5: Modern Tool Usage				
PLO 6: The Engineer and Society				
PLO 7: Environment and Sustainability				
PLO 8: Ethics				
PLO 9: Communication				
PLO 10: Individual and Team Work				√
PLO 11: Project Management and Finance				
PLO 12: Life-Long Learning				

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	**Teaching-Learning Strategy**	**Assessment Strategy**
CLO1	Lecture	Assignment / Midterm Exam

CLO2	Lecture	Assignment / Class Test /Midterm Exam
CLO3	Lecture / Group Discussion	Presentation/ Final Exam
CLO4	Lecture / Group Discussion	Presentation/ Class Test / Final Exam

Course Code: CSE 441	**Credits: 3.0**	**Year: Fourth**	**Semester: Eighth**
Course Title: Network Programming		**Course Status: Theory**	

Rationale of the course:

This course aims to teach students the fundamentals of computer networks and Internet programming. The TCP/IP protocol stack and some of its key protocols will be introduced to students.

Course Content:

Introduction to networking and Internet protocols via programming and hands-on labs. TCP/IP protocol architecture; user datagram protocol (UDP); multicasting; transmission control protocol (TCP); standard Internet services, and protocol usage by common Internet applications. Sockets programming; client/server; peer-to-peer; Internet addressing; TCP sockets; UDP sockets; raw sockets. Multithreading and exception handling. Finger, DNS, HTTP, and ping clients and servers. Routers and architectures, routing protocols. Router and switch configurations, Internet operating systems. Internet work setup, network topology, wireless internetworking. Network protocol analyzers; traffic generation.

Recommended Books:

1. J. F. Kurose and K. W. Ross, *Computer Networking: A Top-Down Approach Featuring the Internet*, Addison-Wesley Publishing, 2nd edition, 2002
2. W. R. Stevens, *UNIX Network Programming*, Prentice Hall PTR, 2nd edition, January 1998

Course Learning Outcomes (CLOs):

Upon completion of the course, the students will be able to:

CLO 1: Understand the basics of computer networks and the OSI reference model.

CLO 2: Understand the operation of multi-access and point-to-point link layer technologies.

CLO 3: Acknowledge the basics of the TCP/IP protocol suite and some of its widely used protocols

CLO 4: Experiment with network routing (static and dynamic) and understand the process of implementing simple routed inter-networks

CLO 5: Learn the 3-tier software architecture (presentation/client tier, application tier, data tier)

Mapping of Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs):

Program Learning Outcomes (PLOs)	CLO 1	CLO 2	CLO 3	CLO 4	CLO 5
PLO 1: Engineering Knowledge	√	√			√
PLO 2: Problem Analysis					
PLO 3: Design/Development of Solutions					
PLO 4: Investigation					
PLO 5: Modern Tool Usage					
PLO 6: The Engineer and Society					
PLO 7: Environment and Sustainability					
PLO 8: Ethics					
PLO 9: Communication					
PLO 10: Individual and Team Work					
PLO 11: Project Management and Finance					
PLO 12: Life-Long Learning			√		√

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lecture	Assignment / Midterm Exam
CLO2	Lecture / Case Study	Assignment / Class Test /Midterm Exam
CLO3	Lecture / Case Study	Assignment / Class Test /Midterm Exam
CLO4	Lecture / Group Discussion	Class Test / Midterm Exam/ Final Exam
CLO5	Lecture / Group Discussion	Presentation/ Class Test / Final Exam

Course Code: CSE 443	**Credits: 3.0**	**Year: Fourth**	**Semester: Eighth**
Course Title: Pattern Recognition		**Course Status: Theory**	

Rationale of the course:

Pattern recognition techniques are used to design automated systems that improve their own performance through experience. It is used to give human recognition intelligence to machines which are required in image processing. This course covers the methodologies, technologies, and algorithms of statistical pattern recognition from a variety of perspectives.

Course Content:

Introduction: Basic concept to pattern recognition: features, classification, learning, statistical methods, structural methods and hybrid method. **Application of Pattern Recognition**: Different application of pattern recognition, remote sensing and biomedical area. **Algorithm**: Learning algorithm, Syntactic approach. **Pattern grammars and language:** Introduction to pattern grammars and language, parsing techniques, pattern recognition in computer aided design.

Text Books:

1. Julius T. Tou and Rafael C: Gonzalez, Pattern Recognition Principles.
2. S.Theodoridis and K.Koutroumbas, Pattern Recognition, 4th Ed., Academic Press, 2009

Reference Book:

1. Patrick, E: Fundamentals of Pattern Recognition.

Course Learning Outcomes (CLOs):

Upon completion of the course, the students will be able to:

CLO 1: Explain and **compare** a variety of pattern classification, structural pattern recognition, and pattern classifier combination techniques.

CLO 2: Apply performance evaluation methods for pattern recognition, and critique comparisons of techniques made in the research literature.

CLO 3: Apply pattern recognition techniques to real-world problems such as document analysis and recognition.

CLO 4: Implement simple pattern classifiers, classifier combinations, and structural pattern recognizers.

Mapping of Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs):

Program Learning Outcomes (PLOs)	CLO 1	CLO 2	CLO 3	CLO 4
PLO 1: Engineering Knowledge			√	
PLO 2: Problem Analysis				
PLO 3: Design/Development of Solutions		√	√	√
PLO 4: Investigation	√		√	
PLO 5: Modern Tool Usage		√	√	√
PLO 6: The Engineer and Society				
PLO 7: Environment and Sustainability				
PLO 8: Ethics				
PLO 9: Communication				
PLO 10: Individual and Team Work		√		√
PLO 11: Project Management and Finance				
PLO 12: Life-Long Learning			√	

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lecture	Assignment / Midterm Exam
CLO2	Lecture	Assignment / Class Test /Midterm Exam
CLO3	Lecture / Group Discussion	Presentation/ Final Exam
CLO4	Lecture / Group Discussion	Presentation/ Class Test / Final Exam

Course Code: CSE 445	Credits: 3.0	Year: Fourth	Semester: Eighth
Course Title: VLSI Design		Course Status: Theory	

Rationale of the course:
The study of the fundamental concepts and structures of designing digital VLSI systems include CMOS devices and circuits, standard CMOS fabrication processes, CMOS design rules, static and dynamic logic structures interconnect analysis, CMOS chip layout, simulation and testing, low power techniques, design tools and methodologies, VLSI architecture.

Course Content:
VLSI design methodology: top-down design approach, technology trends. NMOS, CMOS inverters. **Pass transistor and pass gates:** DC and transient characteristics. Brief overview of fabrication process: NMOS, CMOS, Bi-CMOS process. NMOS and CMOS layout, stick diagram and design rules. **CMOS circuit characteristics and performance estimation:** resistance and capacitance, rise and fall time, power estimation. Buffer circuit design. Introduction to Bi-CMOS circuits. Complex CMOS gates. **CMOS building block:** multiplexer, barrel shifter, adder, counter, multipliers. Data Path and memory structures. Design style: FPGA and PLDs. **Introduction to HDL**: basic digital design using VHDL.

Text Books:

1. David A. Hodges, Horace G. Jackson, and Resve A. Saleh: Analysis and Design of Digital Integrated Circuits, Third Edition McGraw-Hill, 2004.
2. David A. Johns and Ken Martin: Analog Integrated Circuit Design,Wiley, 1997.

Reference Books:

1. Jan M. Rabaey, Anantha P. Chandrakasan, and Borivoje Nikolic: Digital Integrated Circuits, Second Edition, , Prentice-Hall, 2002.
2. Roubik Gregorian: Introduction to CMOS Op-Amps and Comparators, Wiley, 1999.

Course Learning Outcomes (CLOs):
Upon completion of the course, the students will be able to:
CLO 1: Create models of moderately sized CMOS circuits that realize specified digital functions.
CLO 2: Have an understanding of the characteristics of CMOs circuit construction.
CLO 3: Able to complete a significant VLSI design project having a set of objective criteria and design constraints.
CLO 4: Estimate and **optimize** combinational circuit delay using RC delay models and logical effort.

Mapping of Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs):

Program Learning Outcomes (PLOs)	CLO 1	CLO 2	CLO 3	CLO 4
PLO 1: Engineering Knowledge		√	√	
PLO 2: Problem Analysis				
PLO 3: Design/Development of Solutions	√		√	
PLO 4: Investigation				
PLO 5: Modern Tool Usage	√		√	
PLO 6: The Engineer and Society				
PLO 7: Environment and Sustainability				
PLO 8: Ethics				
PLO 9: Communication				
PLO 10: Individual and Team Work				
PLO 11: Project Management and Finance				
PLO 12: Life-Long Learning				

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lecture	Assignment / Midterm Exam
CLO2	Lecture	Assignment / Class Test /Midterm Exam
CLO3	Lecture / Group Discussion	Presentation/ Final Exam
CLO4	Lecture / Group Discussion	Presentation/ Class Test / Final Exam

Course Code: CSE 447	Credits: 3.0	Year: Fourth	Semester: Eighth
Course Title: Digital Forensics and Investigation		**Course Status: Theory**	

Rationale of the course:

To introduce students with the recovery and investigation techniques of evidence during a criminal investigation involving either cyber or computer-related crimes to ensure the overall integrity and survivability of both the evidence and network infrastructures.

Course Content:

An overview, Forensics basic and criminalities, Forensic modeling and principles. Understanding Computer Investigations, Basics of Operating system: A review, Data Acquisition File carving and testing, Cyber forensics tools and testing, processing crime and incident scenes, Recovering Graphics Files, Computer Forensics Analysis and validation, Cell Phone and mobile device forensics, Virtual Machine, Network forensics and Live acquisition, network attach trace back and attribution, E-mail Investigation multicast finger printing, multimedia forensics, Intrusion and online frauds detection; Court testimony and report writing skills; Digital Evidence control

Text Books:

1. Bill Nelson, Amelia Phillips, Christopher Steuart: Guide to Digital Forensics and Investigation
2. Darren R. Hayes: Practical Guide to Digital Forensics Investigations

Reference Books:

1. Gerard Johansen: Digital Forensics and Incident Response
2. Greg Gogolin: Digital Forensics Explained

Course Learning Outcomes (CLOs):

Upon completion of the course, the students will be able to:

CLO 1: **Explain** and properly **document** the process of digital forensics analysis.

CLO 2: Learn about the tradeoffs and differences between various forensic tools.

CLO 3: Describe how data and metadata are represented and organized in modern computer systems.

CLO 4: Understand the inner workings of file systems.

CLO 5: Create disk images, **recover** deleted files and extract hidden information.

Mapping of Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs):

Program Learning Outcomes (PLOs)	CLO 1	CLO 2	CLO 3	CLO 4
PLO 1: Engineering Knowledge		√		
PLO 2: Problem Analysis				
PLO 3: Design/Development of Solutions	√			
PLO 4: Investigation			√	
PLO 5: Modern Tool Usage				√
PLO 6: The Engineer and Society				
PLO 7: Environment and Sustainability				
PLO 8: Ethics				
PLO 9: Communication				
PLO 10: Individual and Team Work				
PLO 11: Project Management and Finance				
PLO 12: Life-Long Learning	√		√	

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lecture	Assignment / Midterm Exam
CLO2	Lecture	Assignment / Class Test /Midterm Exam
CLO3	Lecture / Group Discussion	Presentation/ Final Exam

CLO4	Lecture / Group Discussion	Presentation/ Class Test / Final Exam

Course Code: CSE 449	**Credits: 3.0**	**Year: Fourth**	**Semester: Eighth**
Course Title: Digital Communication		**Course Status: Theory**	

Rationale of the course:

Digital communication involves the transmission of information in digital form from the source of the information to single or multiple destination(s). The block that connects the source of information and the destination is the channel. The physical communication channels have different characteristics that influence the performance of the system. The course gives a fundamental introduction to the principles and systems for digital transmission of information over channels with Gaussian noise, including basic transmission and reception strategies, as well as an understanding of expected performance and performance limits.

Course Content:

Introduction: Communication channels, mathematical model and characteristics. Probability and Stochastic processes. **Source coding:** Mathematical models of information, entropy, Huffman code and linear predictive coding. **Digital transmission system:** Base band digital transmission, inter-symbol interference, bandwidth, power efficiency, modulation and coding trade-off. **Receiver for AWGN channels**: Correlation demodulator, matched filter demodulator and maximum likelihood receiver. **Channel capacity and coding:** Channel models and capacities and random selection of codes. Block codes and conventional codes: Linear block codes, convolution codes and coded modulation. Spread spectrum signals and system.

Text Book:

1. S Haykin: Digital Communication

Reference Books:

1. K Feher: Wireless Digital Communication
2. Zieer & Peterson: Introduction to Digital Communications
3. E. A. Lee and D.G. Messerschmitt: Digital Communications

Course Learning Outcomes (CLOs):

Upon completion of the course, the students will be able to:

CLO 1: Explain the most important properties of channel codes, and be able to choose an appropriate code for a given application.

CLO 2: Analyze the error performance of digital modulation techniques and determine its power spectral density.

CLO 3: Design optimum receivers for digital modulation techniques.

CLO 4: Construct proposals for radio resource management in cellular systems under prescribed system requirements.
CLO 5: Design digital communication systems under given bandwidth, power, data rate, spectral, fidelity, complexity and error performance constrains.

Mapping of Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs):

Program Learning Outcomes (PLOs)	CLO 1	CLO 2	CLO 3	CLO 4	CLO 5
PLO 1: Engineering Knowledge	√				
PLO 2: Problem Analysis		√			
PLO 3: Design/Development of Solutions			√	√	√
PLO 4: Investigation					
PLO 5: Modern Tool Usage					
PLO 6: The Engineer and Society					
PLO 7: Environment and Sustainability					
PLO 8: Ethics				√	
PLO 9: Communication					
PLO 10: Individual and Team Work					
PLO 11: Project Management and Finance					
PLO 12: Life-Long Learning					

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lecture	Assignment / Midterm Exam
CLO2	Lecture / Case Study	Assignment / Class Test /Midterm Exam
CLO3	Lecture / Case Study	Assignment / Class Test /Midterm Exam
CLO4	Lecture / Group Discussion	Class Test / Midterm Exam/ Final Exam
CLO5	Lecture / Group Discussion	Presentation/ Class Test / Final Exam

Course Code: CSE 451	**Credits: 3.0**	**Year: Fourth**	**Semester: Eighth**
Course Title: Wireless & Mobile Communication		**Course Status: Theory**	

Rationale of the course:
Wireless communication networks offer transmission of signals, such as voice, data, and multimedia, without using wires, which is the crucial part of mobile communications. It begins with the basic cellular system modeling and then proceeds towards the characterization and modeling of radio fading channels, multiplexing techniques in wireless communications and major standards of mobile radio systems.

Course Content

Wireless: Introduction: Introduction to wireless communication systems: fixed wireless access, cellular, paging, turnkey mobile systems. **Basics of wireless access:** Overview of wireless access networks, base and subscriber stations, frequency planning, multiple access, Noise and interference in wireless communication systems.

Mobile: Accessing: Introduction, FDMA, TDMA, CDMA **Cordless Telephone:** CTO, CT1, CT2, DECT, Cellular mobile communication, Cellular network architecture, Radio network planning, C-network, D-network, GSM, Personal communication network (PCN), DCS L800, E1. **Paging System:** Beam communication (Tracking systems), Mobile data transmission, Modacom, IRIDIUM, NMARSAT, GPS, and EMC – F. **Cellular Systems:** Evolution of cellular systems, operation, Capacity considerations, Standards. **Propagation and System Planning:** Radio wave propagation in the mobile environment- fading, interference, Mobile radio link design. **Mobile Satellite Systems:** Introduction to mobile satellite system operation, Illustrative systems.

Satellite Communication: Evaluation and growth of Communication satellite, Kepler's Laws of motion, orbits, altitude control, Satellite launch vehicles, sub-systems of communication satellite, spectrum allocation and bandwidth, propagation characteristics, satellite transponders and earth station technology, link design, multi access techniques. **Digital Satellite Communication**: Satellite channel, frequency, switch, time slot, frame, scanning, use of orbit and spectrum, satellite switching, speech interpolation, echo and delay cancellation, tracking integrated satellite networks.

Text Books:

1. W. C. Y. Lee: Mobile Cellular Communications
2. Pratt & Bostian: Satellite Communication
3. Merri & Skolnik: Introduction to Radar System

Reference Books:

1. W. C. Y. Lee: Mobile Communication Design Fundamentals
2. Andreas S. Molisch: Wideband Wireless Digital Communication

Course Learning Outcomes (CLOs):

Upon completion of the course, the students will be able to:

CLO 1: Explain various wireless communication systems and recognize their multiple access technologies in accordance with their use in current and future wireless and cellular communications standards.

CLO 2: Explain cellular radio concepts such as frequency reuse, handover, capacity and traffic and relate interference between mobiles and/or base stations with the capacity of cellular systems (C2).

CLO 3: Apply the knowledge of multiple access techniques, cellular concept, truncking, system capacity and large- and small-scale fading to solve for various performance parameters.

CLO 4: Analyze propagation effects such as fading, time delay spread and Doppler spread, and analyze their impact on the instantaneous received signal strength in multipath channels.
CLO 5: Develop advanced knowledge and analyze techniques on how to monitor network threads/processes, as well as identify inter-operational characteristics in a wireless network system.

Mapping of Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs):

Program Learning Outcomes (PLOs)	CLO 1	CLO 2	CLO 3	CLO 4	CLO 5
PLO 1: Engineering Knowledge		√			
PLO 2: Problem Analysis					
PLO 3: Design/Development of Solutions					√
PLO 4: Investigation					
PLO 5: Modern Tool Usage					
PLO 6: The Engineer and Society					
PLO 7: Environment and Sustainability					
PLO 8: Ethics					
PLO 9: Communication					
PLO 10: Individual and Team Work					√
PLO 11: Project Management and Finance					
PLO 12: Life-Long Learning					

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lecture	Assignment / Midterm Exam
CLO2	Lecture / Case Study	Assignment / Class Test /Midterm Exam
CLO3	Lecture / Case Study	Assignment / Class Test /Midterm Exam
CLO4	Lecture / Group Discussion	Class Test / Midterm Exam/ Final Exam
CLO5	Lecture / Group Discussion	Presentation/ Class Test / Final Exam

Course Code: CSE 453	**Credits: 3.0**	**Year: Fourth**	**Semester: Eighth**
Course Title: Mobile Application Development		**Course Status: Theory**	

Rationale of the course:
This course introduces students to mobile application programming technology, design, and development. Students learn about current mobile platforms, mobile application development environments, mobile device input techniques, and how to design applications for various mobile

platforms. Throughout the course, students will create and construct a variety of apps to reinforce learning and acquire actual expertise.

Course Content:
Mobile Phones and Network Technologies, Android Programming, Android Application Frameworks, Building a Simple User Interface, Activities and Intents, Services, Broadcast Receivers, Data Persistence, Processes and Threads, Asynchronous Tasks, Internet Resources, Apps Publishing and Business Models, iOS platform, Objective-C, Application development in iOS.

Recommended Books:

1. Professional Mobile Application Development - Jeff McWherter, Scott Gowell
2. Mobile Applications Development: with Python in Kivy Framework - Kamal Kant Hiran, Ruchi Doshi

Course Learning Outcomes (CLOs):
Upon completion of the course, the students will be able to:
CLO 1: Design mobile software applications, **describe** and **apply** several mobile application models and patterns.
CLO 2: Describe, **implement**, and **evaluate** a mobile development framework for the development of a realistic mobile application.
CLO 3: Use, analyze, and **justify** advanced programming competency for developing a maintainable and efficient cloud based mobile application.
CLO 4: Choose software tools, **create** mobile programming software, **demonstrate** these skills, and write a report to design, build, and test mobile programming software.

Mapping of Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs):

Program Learning Outcomes (PLOs)	CLO 1	CLO 2	CLO 3	CLO 4
PLO 1: Engineering Knowledge				
PLO 2: Problem Analysis			√	
PLO 3: Design/Development of Solutions		√		
PLO 4: Investigation				
PLO 5: Modern Tool Usage	√	√		
PLO 6: The Engineer and Society				
PLO 7: Environment and Sustainability				
PLO 8: Ethics				
PLO 9: Communication				
PLO 10: Individual and Team Work				√
PLO 11: Project Management and Finance				
PLO 12: Life-Long Learning				

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lecture	Assignment / Midterm Exam
CLO2	Lecture	Assignment / Class Test /Midterm Exam
CLO3	Lecture / Group Discussion	Presentation/ Final Exam
CLO4	Lecture / Group Discussion	Presentation/ Class Test / Final Exam

Course Code: CSE 455	**Credits: 3.0**	**Year: Fourth**	**Semester: Eighth**
Course Title: Internet of Things		**Course Status: Theory**	

Rationale of the course:
This course will go over the components and characteristics of the Internet of Things (IoTs). The development of domain-specific IoTs and their real-world applications will take place here. This course will also cover the programming components of IoTs, with the goal of allowing students to quickly prototype sophisticated IoT applications.

Course Content:
Introduction to IoT, Basics of Networking, Communication Protocols, Sensor Networks: Machine-to-Machine Communications, Interoperability in IoT: Integration of Sensors and Actuators, Implementation of IoTwith programmable devices, Software Defined Networking *(*SDN) for IoT: Data Handling and Analytics, Cloud Computing, Fog Computing: Smart Cities and Smart Homes, Connected Vehicles: Smart Grid, Industrial IoT: Agriculture, Healthcare, Activity Monitoring and related case studies,

Recommended Books:

1. The Internet of Things (The MIT Press Essential Knowledge series) – Samuel Greengard
2. The Internet of Things – Michael R. Miller
3. Analytics for the internet of things (IoT) – Andrew Minteer

Course Learning Outcomes (CLOs):
Upon completion of the course, the students will be able to:
CLO 1: Understand the building blocks of IoTs and their characteristics with real-world applications.
CLO 2: Understand, **apply**, and **examine** different designs for different levels of complicated IoT applications
CLO 3: Examine and **justify** IoT data analytics and various tools for IoT,
perform and **demonstrate** skills, and **write** report on realistic data analytics.

CLO 4: Select and **justify** software and hardware tools, **generate** source code for various IoT domains, demonstrate and perform skills, and produce a report on actual IoT development.

Mapping of Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs):

Program Learning Outcomes (PLOs)	CLO 1	CLO 2	CLO 3	CLO 4
PLO 1: Engineering Knowledge	√			
PLO 2: Problem Analysis				
PLO 3: Design/Development of Solutions			√	
PLO 4: Investigation				
PLO 5: Modern Tool Usage				√
PLO 6: The Engineer and Society				
PLO 7: Environment and Sustainability				
PLO 8: Ethics				
PLO 9: Communication				
PLO 10: Individual and Team Work				
PLO 11: Project Management and Finance				
PLO 12: Life-Long Learning	√		√	

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lecture	Assignment / Midterm Exam
CLO2	Lecture	Assignment / Class Test /Midterm Exam
CLO3	Lecture / Group Discussion	Presentation/ Final Exam
CLO4	Lecture / Group Discussion	Presentation/ Class Test / Final Exam

Course Code: CSE 457	**Credits: 3.0**	**Year: Fourth**	**Semester: Eighth**
Course Title: Data Mining		**Course Status: Theory**	

Rationale of the course:

The course helps to educate students the fundamental concepts of data mining techniques. The course examines the types of data that will be mined and how to apply preprocessing procedures to raw data. The focus of the course is on finding intriguing patterns, assessing supervised and unsupervised models, and estimating algorithm accuracy. Various Data Mining tools will also be introduced to the students.

Course Content:

Introduction to Data Mining, Data Mining Goals, Stages of the Data Mining Process, Data Mining Techniques, Knowledge Representation Methods, Data preprocessing: Data cleaning,

Data transformation, Data reduction, Data mining knowledge representation, Representing input data and output knowledge, Association rules, Correlation analysis, Classification, Basic learning/mining tasks, Decision trees, Covering rules, Prediction, The prediction task, Statistical classification, Instance-based methods, Linear models, Clustering, Basic issues in clustering, First conceptual clustering system, Hierarchical methods, Conceptual clustering., Advanced techniques, Data Mining software and applications Text mining, Web mining, Assignment and Mini Project with Report and Presentation.

Recommended Books:

1. Data Mining. Concepts and Techniques (The Morgan Kaufmann Series)- Jiawei Han, Micheline Kamber, Jian Pei
2. Data Mining: The Textbook - Charu C. Aggarwal

Course Learning Outcomes (CLOs):

Upon completion of the course, the students will be able to:

CLO 1: Offer adequate input for a variety of data mining methods, **process** raw input data.

CLO 2: Discover and **quantify** intriguing patterns in a variety of databases**.**

CLO 3: Evaluate and **choose** relevant data-mining algorithms**,** then **implement, interpret,** and **report** the results**.**

CLO 4: Develop and **execute** a data-mining application utilizing sample, actual data sets, **choose** and **justify** software tools, **perform** and **exhibit skills, choose** and **justify** software tools, **perform** and **demonstrate** skills, and **write** a report.

Mapping of Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs):

Program Learning Outcomes (PLOs)	CLO 1	CLO 2	CLO 3	CLO 4
PLO 1: Engineering Knowledge		√		
PLO 2: Problem Analysis			√	
PLO 3: Design/Development of Solutions				
PLO 4: Investigation				
PLO 5: Modern Tool Usage				
PLO 6: The Engineer and Society				
PLO 7: Environment and Sustainability				
PLO 8: Ethics				
PLO 9: Communication				
PLO 10: Individual and Team Work		√	√	
PLO 11: Project Management and Finance				
PLO 12: Life-Long Learning				

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy

CLO1	Lecture	Assignment / Midterm Exam
CLO2	Lecture	Assignment / Class Test /Midterm Exam
CLO3	Lecture / Group Discussion	Presentation/ Final Exam
CLO4	Lecture / Group Discussion	Presentation/ Class Test / Final Exam

Course Code: CSE 459	**Credits: 3.0**	**Year: Fourth**	**Semester: Eighth**
Course Title: Embedded Network Systems		**Course Status: Theory**	

Rationale of the course:
To introduce the fundamental hardware components related to network system and the software including routing protocols to establish a complete data transmission system.

Course Content:
Wireless Embedded Networks: Introduction, Applications. Embedded Processors. The Embedded Computing Platform IP- and Platform-Based SoC Designs. Component Technologies: Sensors, Physical Layer, Sensor Platforms, Link and Routing Protocols. System Infrastructure and Development: Clock Synchronization, Power Management, Security, Topology Control, Programming Abstractions, Storage. Scheduling Theory and Algorithms: Real-Time Operating Systems, Fixed Priority and Dynamic Scheduling, Aperiodic and Sporadic Task Scheduling, Priority Inversion and Task Synchronization Applications: Localization, Data Dissemination, Mobility, Distributed Control. Power issues in embedded systems.

Text Book:
1. Gregory J. Pottie , William J. Kaiser : Principles of Embedded Networked Systems Design

Reference Book:
1. Graham Wilson**:** Embedded Systems and Computer Architecture

Course Learning Outcomes (CLOs):
After completing the course, the students will be able to:
CLO 1: Learn architectures and protocols for embedded network system.
CLO 2: Understand the impact of protocol design on energy-consumption and energy-efficiency.
CLO 3: Discuss issues relating to embedded network security.
CLO 4: Explain the principles of layered networking models.

Mapping of Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs):

Program Learning Outcomes (PLOs)	**CLO 1**	**CLO 2**	**CLO 3**	**CLO 4**
PLO 1: Engineering Knowledge	√			

PLO 2: Problem Analysis			√	
PLO 3: Design/Development of Solutions				
PLO 4: Investigation				
PLO 5: Modern Tool Usage				
PLO 6: The Engineer and Society				
PLO 7: Environment and Sustainability				
PLO 8: Ethics				
PLO 9: Communication				
PLO 10: Individual and Team Work				
PLO 11: Project Management and Finance				
PLO 12: Life-Long Learning	√			

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lecture	Assignment / Midterm Exam
CLO2	Lecture	Assignment / Class Test /Midterm Exam
CLO3	Lecture / Group Discussion	Presentation/ Final Exam
CLO4	Lecture / Group Discussion	Presentation/ Class Test / Final Exam

Course Code: CSE 461	**Credits: 3.0**	**Year: Fourth**	**Semester: Eighth**
Course Title: Client-Server Computing		**Course Status: Theory**	

Rationale of the course:

To introduce the fundamental principles and the mechanisms responsible for the communication between client and the server database system.

Course Content:

Overview: Definition, history, myths, transition to client server computing, database architectures, advantages and disadvantages of client server architecture.

Components: Client, server, network, role and services of client-server, selection of operating system as client & server, types of client & servers, connectivity, messaging and peer-to-peer communication, role of OSI layer in client server, TCP/IP networks.

Middle-ware: Definition, role, 2 tier v/s 3 tiers, network file system, network operating system, API, RPC model & implementation, IPC.

Database Server: Memory and process architecture, SQL and PL SQL.

Client/Server processing: Transaction processing, remote processing, distributed processing. Developing and Managing C/S Applications: development tools, network management, backup and recovery, data integrity & security.

Distributed objects: CORBA, COM, DCOM, OLE, Java-RMI, etc., ODBC & JDBC connectivity. Data warehousing: operational data & analytical data, architecture.

Text Book:

1. Doug Lowe, David J. Helda: *Client*/*Server Computing* for Dummies

Reference Book:

1. Eric Johnson: The Complete Guide to Client/Server Computing

Course Learning Outcomes (CLOs):

After completing the course, the students will be able to:

CLO 1: Define the underlying concepts in client server development using common access databases.

CLO 2: Design and **Set up** a client /server environment using LAN and WAN Scenarios.

CLO 3: Describe the concept of middleware, and communication protocols.

CLO 4: Design and **build** client server applications with network programming exposure. Understand basic networking concepts using sockets.

Mapping of Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs):

Program Learning Outcomes (PLOs)	CLO 1	CLO 2	CLO 3	CLO 4
PLO 1: Engineering Knowledge			√	
PLO 2: Problem Analysis				
PLO 3: Design/Development of Solutions		√		√
PLO 4: Investigation			√	
PLO 5: Modern Tool Usage				
PLO 6: The Engineer and Society				
PLO 7: Environment and Sustainability				
PLO 8: Ethics				
PLO 9: Communication				
PLO 10: Individual and Team Work				
PLO 11: Project Management and Finance				
PLO 12: Life-Long Learning				

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lecture	Assignment / Midterm Exam
CLO2	Lecture	Assignment / Class Test /Midterm Exam

CLO3	Lecture / Group Discussion	Presentation/ Final Exam
CLO4	Lecture / Group Discussion	Presentation/ Class Test / Final Exam

Course Code: CSE 463	**Credits: 3.0**	**Year: Fourth**	**Semester: Eighth**
Course Title: Decision Support System		**Course Status: Theory**	

Rationale of the course:
This course is designed to give students a basic understanding of decision support systems for managers and IT developers. This course covers subjects in computer-based Decision Support Systems, with a practical emphasis on using information technology to solve management problems.

Course Content:
Introduction: Decision Support and Business Intelligence
Computerized Decision Support: Decision Making, Systems, Modeling and Support, Decision Support Systems Concepts, Methodologies, and Technologies: An Overview, Modeling and Analysis
The Essentials of Business Intelligence: Data Warehousing, Business Analytics and Data Visualization, Data, Text, and Web Mining, Neural Networks for Data Mining, Business Performance Management
Collaboration, Communication, Group Support Systems, and Knowledge Management: Collaborative Computing-Supported Technologies and Group Support Systems, Knowledge Management
Intelligent Systems: Artificial Intelligence and Expert Systems, Advanced Intelligent Systems, Intelligent Systems over the Internet
Implementing Decision Support Systems: Systems Development and Acquisition, Integration, Impacts, and the Future of Management Support Systems
Text Book:

1. Efraim Turban, E. Aronson, Ting-Peng Liang & Ramesh Sharda; Decision Support and Business Intelligence Systems, 8th Edition, Prentice Hall, 2007.

Reference Book:

1. Daellen Bach, H. G. -- Mcnickle, D. C.; Management science : decision making through systems thinking, Houndmills, Palgrave Macmillan, 2005

Course Learning Outcomes (CLOs):
After completing the course, the students will be able to:
CLO 1: Learn the fundamentals of decision support systems (DSS) and management sciences.
CLO 2: Examine the components of the DSS as well as the key individuals involved in the decision-making process.

CLO 3: Implement different types of modeling and analysis.
CLO 4: Explain how essential disciplines including information acquisition, expert systems, and knowledge base systems contribute to DSS.

Mapping of Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs):

Program Learning Outcomes (PLO)	CLO 1	CLO 2	CLO 3	CLO 4
PLO 1: Engineering Knowledge	√			
PLO 2: Problem Analysis				
PLO 3: Design/Development of Solutions			√	
PLO 4: Investigation				
PLO 5: Modern Tool Usage				
PLO 6: The Engineer and Society				
PLO 7: Environment and Sustainability				
PLO 8: Ethics				
PLO 9: Communication				
PLO 10: Individual and Team Work				
PLO 11: Project Management and Finance	√			
PLO 12: Life-Long Learning				

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lecture	Assignment / Midterm Exam
CLO2	Lecture	Assignment / Class Test /Midterm Exam
CLO3	Lecture / Group Discussion	Presentation/ Final Exam
CLO4	Lecture / Group Discussion	Presentation/ Class Test / Final Exam

Course Code: CSE 465	Credits: 3.0	Year: Fourth	Semester: Eighth
Course Title: Wireless Network		**Course Status: Theory**	

Rationale of the course:
This course covers wireless data communications, notably wireless Internet. The students learn about the current and future state-of-the-art in wireless communications technology.

Course Content:
Wired transmission techniques: design of wireless modems, power efficiency, out of band radiation, applied wireless transmission techniques, short distance base band transmission, VWB pulse transmission, broad Modems for higher speeds, diversity and smart receiving techniques, random access for data oriented networks, integration of voice and data traffic.

Wireless networks topologies, cellular topology, cell fundamentals signal to interference ratio calculation, capacity expansion techniques, cell splitting, use of directional antennas for cell sectoring, micro cell method, overload cells, channels allocation techniques and capacity expansion FCA, channel borrowing techniques, DCA, mobility management, radio resources and power management securities in wireless networks.
Mechanism to support a mobile environment, communication in the infrastructure, IS-95 CDMA forward channel, IS – 95 CDMA reverse channel, pallert and frame formats in IS – 95, IMT – 2000; forward channel in W-CDMA and CDMA 2000, reverse channels in W-CDMA and CDMA-2000, GPRS and higher data rates, short messaging service in GPRS mobile application protocols.
Historical overviews of the LAN industry, evolution of the WLAN industry, wireless home networking, IEEE 802.11. The PHY Layer, MAC Layer, wireless ATM, HYPER LAN, HYPER LAN – 2.
IEEE 802.15 WPAN, Home RF, Bluetooth, interface between Bluetooth and 802.11, wireless geolocation technologies for wireless geolocation, geolocation standards for E.911 service.

Text Book:

1. Kaveh Pahlavan, Prashant Krishnamoorthy, Principles of Wireless Networks, - A united approach - Pearson Education, 2002.

Reference Books:

1. Jochen Schiller, Mobile Communications, Person Education – 2003, 2nd Edn.
2. X.Wang and H.V.Poor, Wireless Communication Systems, Pearson education, 2004.
3. M.Mallick, Mobile and Wireless design essentials, Wiley Publishing Inc. 2003.
4. P.Nicopolitidis, M.S.Obaidat, G.I. papadimitria, A.S. Pomportsis, Wireless Networks, John Wiley & Sons, 2003.

Course Learning Outcomes (CLOs):
After completing the course, the students will be able to:
CLO 1: Explain how wireless systems should be designed to successfully carry various information.
CLO 2: Recognize wireless internetworking concepts, architecture and protocols.
CLO 3: Explain the fundamentals about wireless security
CLO 4: Describe the main technologies used in modern wireless network to provide QoS and fairness
CLO 5: Describe the most important standardization and spectrum regulation bodies, how they interact, and how they impact the design of technical systems

Mapping of Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs):

Program Learning Outcomes (PLOs)	CLO 1	CLO 2	CLO 3	CLO 4	CLO 5

PLO 1: Engineering Knowledge	√	√			
PLO 2: Problem Analysis					
PLO 3: Design/Development of Solutions					
PLO 4: Investigation					√
PLO 5: Modern Tool Usage					
PLO 6: The Engineer and Society					
PLO 7: Environment and Sustainability					
PLO 8: Ethics					
PLO 9: Communication					
PLO 10: Individual and Team Work					
PLO 11: Project Management and Finance					
PLO 12: Life-Long Learning			√	√	

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lecture	Assignment / Midterm Exam
CLO2	Lecture	Assignment / Class Test /Midterm Exam
CLO3	Lecture / Group Discussion	Presentation/ Final Exam
CLO4	Lecture / Group Discussion	Presentation/ Class Test / Final Exam

Course Code: CSE 467	**Credits: 3.0**	**Year: Fourth**	**Semester: Eighth**
Course Title: Simulation and Modeling		**Course Status: Theory**	

Rationale of the course:

This course aims to enable a substitute for physical experimentation, which is frequently used when conducting experiments on a real system is impossible or impractical, often due to cost or time constraints, and instead uses mathematical knowledge and computing power to solve real-world problems reasonably and quickly.

Course Content:

Simulation modeling basics: systems, Models and simulation, classification of simulation models, steps in simulation Study, concepts in discrete event simulation , event scheduling and process interaction approaches, time advance mechanism, organization of a discrete event Simulation model, continuous simulation models, combined discrete continuous Models, Monte-Carlo simulation, simulation of queuing system. Building credible and credible simulation models: validation principle principles and techniques, Statistical procedures for comparing real world observation and simulation outputs, input modeling, generating random

numbers and random vitiates, output analysis. Simulation languages; analysis and modeling of some practical systems.

Text Books:

1. Simulation Modeling and Analysis: Averill M. Law.
2. Discrete Event System Simulation: Banks, Carson, Nelson, Nicol, Prentice, Hall.
3. Discrete Event Systems: Modeling and Performance Analysis", Christos G. Cassandras, Aksen Associates.

Reference Book:

1. Parallel and Distributed Simulation Systems: Richard M. Fujimoto, Wiley-Interscience.

Course Learning Outcomes (CLOs):

Upon completion of the course, the students will be able to:

CLO 1: Understand the fundamentals of computer simulation and how it may be used to solve engineering problems.

CLO 2: Perceive the advantages of utilizing simulation and modeling to make engineering decisions.

CLO 3: Classify different simulation models and provide examples for each one.

CLO 4: Create a model for a set of data and justify its validity.

Mapping of Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs):

Program Learning Outcomes (PLO)	CLO 1	CLO 2	CLO 3	CLO 4
PLO 1: Engineering Knowledge	√		√	
PLO 2: Problem Analysis			√	
PLO 3: Design/Development of Solutions				√
PLO 4: Investigation		√		
PLO 5: Modern Tool Usage				
PLO 6: The Engineer and Society				
PLO 7: Environment and Sustainability				
PLO 8: Ethics				
PLO 9: Communication				
PLO 10: Individual and Team Work				
PLO 11: Project Management and Finance				
PLO 12: Life-Long Learning				

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lecture	Assignment / Midterm Exam

CLO2	Lecture	Assignment / Class Test /Midterm Exam
CLO3	Lecture / Group Discussion	Presentation/ Final Exam
CLO4	Lecture / Group Discussion	Presentation/ Class Test / Final Exam

Course Code: CSE 468	**Credits: 1.0**	**Year: Fourth**	**Semester: Eighth**
Course Title: Simulation and Modeling Lab		**Course Status: Lab**	

Rationale of the course:
This course encourages students to create various models to solve real-world problems using mathematics, computer programming languages, computation power, and other methods, as well as to analyze the behavior of a system for various types of datasets in order to make an informed decision about a system's performance in a cost- and time-effective manner.

Course Content:
Chi-square goodness-of-fit test, One-sample Kolmogorov-Smirnov test, One-sample Kolmogorov-Smirnov test, Test for Standard Normal Distribution, Testing Random Number Generators, Monte-Carlo Simulation, Simulation of Single Server Queuing System, Simulation of Two-Server Queuing System, **Simulate and control a conveyor belt system** and Two-sample Kolmogorov-Smirnov test.

Course Learning Outcomes (CLOs):
Upon completion of the course, the students will be able to:
CLO 1: Analyze the generation Various Signals and Sequences in MATLAB, including the operations on Signals and Sequences.
CLO 2: Verification of Sampling Theorem, Linearity and Time Invariance Properties of a given Signals/ Systems.
CLO 3: Analyze the Fourier Transform of a given signal and plotting its magnitude and phase spectrum and also plot Pole-Zero Maps in Z-Plane..
CLO 4: Analyze and **design** various combinational using VHDL like Adder, Subtractor, Multiplexers, etc.
CLO 5. Analyze and **design** various sequential circuits using VHDL like Flip flop, Counters, etc.

Mapping of Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs):

Program Learning Outcomes (PLO)	**CLO 1**	**CLO 2**	**CLO 3**	**CLO 4**	**CLO 5**
PLO 1: Engineering Knowledge					
PLO 2: Problem Analysis	√		√	√	√
PLO 3: Design/Development of Solutions		√			
PLO 4: Investigation		√			

PLO 5: Modern Tool Usage				√	√
PLO 6: The Engineer and Society					
PLO 7: Environment and Sustainability					
PLO 8: Ethics					
PLO 9: Communication					
PLO 10: Individual and Team Work					
PLO 11: Project Management and Finance					
PLO 12: Life-Long Learning			√		√

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lab demonstration/ Lecture	Assignment
CLO2	Lab demonstration/ Lecture	Assignment
CLO3	Lab demonstration/ Lecture	Assignment
CLO4	Lab demonstration/ Lecture	Presentation/Viva
CLO5	Lab demonstration/ Lecture	Lab Report /Final Exam

Course Code: CSE 469	**Credits: 3.0**	**Year: Fourth**	**Semester: Eighth**
Course Title: Computer and Network Security		**Course Status: Theory**	

Rationale of the course:

The purpose of this course is to educate students of the increasing demands for combining Computer Networks, Security, and Programming expertise raised by the growing adoption of the Next-Generation Network (NGN) services, e.g., VoIP, and Next Generation Network Model.

Course Content:

Cryptography and cryptography algorithms: Traditional cryptograph. Cryptanalysis, Private-key (Symmetric-key) and public-key (asymmetric-key) cryptographic algorithms. DES and block cipher modes, Advanced encryption standard. RSA and other public key cryptosystems, key management , Diffie-Hellman key exchange, Elliptic curve cryptography. Cryptographic hash functions, Secure Hash Algorithms. Message authentication codes, Digital signatures and digital signature standard.

Cryptography and Network Security: Data origin authentication and data integrity. Key distribution, management, Kerberos and X.509 authentication service, Certificate authority and public key infrastructure, e-mail security, PGP and S/MIME, IP security, Authentication header and ESP, Security associations, key management, Oakley key determination protocol and

ISAKMP. Web security considerations, secure socket layer and transport layer security. Secure electronic transactions (SET).

Cryptography for antivirus software and password management: Generic Decryption, Digital Immune System and UNIX password Scheme.

Text Book:

1. Cryptography & Network Security: Behrouz A Forouzan

Reference Books:

1. Cryptography & Network Security: Principles & Practice (5th Edition) by W. Stallings
2. Introduction to Modern Cryptography: Principles & Protocols by J. Katz, Y.Lindell

Course Learning Outcomes (CLOs):

Upon completion of the course, the students will be able to:

CLO1: Understand the evolution of security, classical encryption, security assaults, and basic security goals.

CLO2: Examine the challenges with a real-world security solution's design and implementation.

CLO3: Determine the security objectives, attacks, and models so that real-world security requirements may be recognized.

Mapping of Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs):

Program Learning Outcomes (PLOs)	CLO 1	CLO 2	CLO 3
PLO 1: Engineering Knowledge			
PLO 2: Problem Analysis		√	
PLO 3: Design/Development of Solutions			
PLO 4: Investigation			
PLO 5: Modern Tool Usage			
PLO 6: The Engineer and Society			
PLO 7: Environment and Sustainability			
PLO 8: Ethics	√		
PLO 9: Communication			
PLO 10: Individual and Team Work			√
PLO 11: Project Management and Finance			
PLO 12: Life-Long Learning			

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lecture	Assignment / Midterm Exam
CLO2	Lecture	Assignment / Class Test /Midterm Exam
CLO3	Lecture / Group Discussion	Presentation/ Class Test / Final Exam

Course Code: CSE 471	Credits: 3.0	Year: Fourth	Semester: Eighth
Course Title: Parallel Processing		Course Status: Theory	

Rationale of the course:

This course encourages students to familiarize with the fundamental concepts, techniques and tools of parallel computing. Participation in this course will enable you to better use parallel computing in your application area, and will prepare you to take advanced courses in more specific areas of parallel computing.

Course Content:

Parallel Computing

Introduction: Need for Parallel Computing, Scope of Parallel Computing, Issues in Parallel Computing

Parallel Algorithms: Models of Parallel Computing, Interconnection networks, Basic Computing and Communication Primitives, Performance Modeling, Dense Matrix Algorithms, Searching and Optimization: Dynamic Programming, Branch and Bound, Backtrack, Sorting, Graph Algorithms, Fast Fourier Transforms

Parallel Architecture

Introduction: Flynn's classification, SIMD and MIMD operations, Shared Memory vs. message passing multiprocessors, Distributed shared memory, Hybrid multiprocessors

Shared Memory Multiprocessors: SMP and CC-NUMA architectures, Cache coherence protocols, Consistency protocols, Data pre-fetching, CC-NUMA memory management, SGI 4700 multiprocessor, Network Processors

Interconnection Networks: Static and Dynamic networks, switching techniques, Routers, Internet techniques

Message Passing Architectures: Message passing paradigms, Grid architecture, Workstation clusters, User level software

Scheduling: Multiprocessor Programming Technique, Scheduling and mapping, Internet web servers, P2P, Content aware load balancing

Text Books

1. Ananth Grama, Anshul Gupta, George Karypis, Vipin Kumar; *Introduction to Parallel Computing*, 2nd Edition, The Addison Wesley Publishing Company
2. John Hennessy and David Patterson, *Computer Architecture: A Quantitative Approach*, Morgan Kauffman Publisher

Reference Books

1. T. H. Cormen, C. E. Leiserson and R. L. Rivest, Introduction to Algorithms, MIT Press, 1990.

Course Learning Outcomes (CLOs):

Upon completion of the course, the students will be able to:

CLO1: Understand the key parallel computational models.

CLO2: Implement key algorithms using fine grained computation models and High Level Design Tools.

CLO3: Write parallel programs using message passing and shared memory paradigms.

CLO4: Learn How to develop computer programs for different types of parallel computers.

Mapping of Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs):

Program Learning Outcomes (PLOs)	CLO 1	CLO 2	CLO 3	CLO 4
PLO 1: Engineering Knowledge	√			√
PLO 2: Problem Analysis				
PLO 3: Design/Development of Solutions		√	√	
PLO 4: Investigation				
PLO 5: Modern Tool Usage				
PLO 6: The Engineer and Society				
PLO 7: Environment and Sustainability				
PLO 8: Ethics				
PLO 9: Communication				
PLO 10: Individual and Team Work				
PLO 11: Project Management and Finance				
PLO 12: Life-Long Learning		√		√

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lecture	Assignment / Midterm Exam
CLO2	Lecture	Assignment / Class Test /Midterm Exam
CLO3	Lecture / Group Discussion	Presentation/ Final Exam
CLO4	Lecture / Group Discussion	Presentation/ Class Test / Final Exam

Course Code: CSE 473	Credits: 3.0	Year: Fourth	Semester: Eighth
Course Title: Robotics Technology		**Course Status: Theory**	

Rationale of the course:

This course is an overview of robotic and automated systems technology. The student will be introduced to basic manufacturing techniques, robot terminology and different types of automation. Safety, basic robotic programming, interfacing robotic communications, and robotic applications. Robot operations and programming fundamentals will be applied by the students.

Course Content:

Introduction to robotics: History of Robotics, Basics of Electronics, Resistance and its types, Capacitors and its types, Different mechanism of Robots

Type of Robots: Manual Robots, Robots for Wars, Pick and place type Robots, Robotic Arm, Wireless Controlled robots

Introduction to Autonomous Robotics: Basics of Microcontroller, ATmega 16, Programming of Microcontroller, Line follower and Grid solving concepts, ADC, Motor control DC, Steeper Motor & Interfacing.

Text Book:

1. Introduction to Robotics: Analysis, Systems, and Applications, Saeed B. Niku, 2001, Person Education, ISBN: 9780130613097

Reference Books:

1. Autonomous Mobile Robots, Roland Siegwart and Ihhah R. Nourbakhsh, 2004, MIT Press, ISBN: 9780262195027
2. Robot Modeling and Control, Mark W. Spong, Seth Hutchinson, and M. Vidyasagar, 2006, Wiley, ISBN: 9780471649908

Course Learning Outcomes (CLOs):

Upon completion of the course, the students will be able to:

CLO1: Describe current status of robotics technologies and new developments.

CLO2: Understand the fundamental concepts of mobile robots and robotic manipulators.

CLO3: Use sensors, actuators, microcontrollers to design, construct, program, and operate a robotic system.

Mapping of Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs):

Program Learning Outcomes (PLO)	CLO 1	CLO 2	CLO 3
PLO 1: Engineering Knowledge	√	√	
PLO 2: Problem Analysis			
PLO 3: Design/Development of Solutions		√	√
PLO 4: Investigation			
PLO 5: Modern Tool Usage			√
PLO 6: The Engineer and Society			
PLO 7: Environment and Sustainability			
PLO 8: Ethics			
PLO 9: Communication			
PLO 10: Individual and Team Work			
PLO 11: Project Management and Finance			
PLO 12: Life-Long Learning			

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lecture	Assignment / Midterm Exam
CLO2	Lecture	Assignment / Class Test /Midterm Exam
CLO3	Lecture / Group Discussion	Presentation/ Class Test / Final Exam

Course Code: CSE 475	**Credits: 3.0**	**Year: Fourth**	**Semester: Eighth**
Course Title: E-Commerce & E-Business		**Course Status: Theory**	

Rationale of the course:
Students will be able to develop the skills in the field of Electronic Commerce and to keep students at the forefront of the Global Digital Economy via facilitating innovation, creating values in the Electronic marketplace with the Cooperation of leading Edge organizations.

Course Content:
History, business models; Ecommerce channels: Portals, auctions, communities, marketplace; Managing the marketplace: Demographics and advertising; Customer relationship management, web services for B2B and B2C ecommerce, electronic payment systems; Network security, cryptography, digital certificates; Markup for ecommerce: ebXML, M-commerce, wireless and U-commerce, digital money and electronic banking; Ethical, legal, and regulatory environment: Intellectual property, copyright, trademark, patents.

Text Book:
1. The Complete E-Commerce Book, Second Edition: Design, Build & Maintain a Successful Web-based Business by **Janice Reynolds**

Reference Books:
1. E-commerce: Get It Right! By Ian Daniel
2. E-Commerce 2009: Business, Technology, and Society by Kenneth Laudon and Carol Guercio Traver

Course Learning Outcomes (CLOs):
Upon completion of the course, the students will be able to:
CLO 1: Become familiar with state of the art Electronic Model, Payment Mechanisms
CLO 2: Understand the basic principle of E-Government, Securities, Supply Chain Management
CLO 3: Evaluate and **observe** various online businesses

Mapping of Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs):

Program Learning Outcomes (PLOs)	CLO 1	CLO 2	CLO 3
PLO 1: Engineering Knowledge	√	√	
PLO 2: Problem Analysis			
PLO 3: Design/Development of Solutions			√
PLO 4: Investigation			
PLO 5: Modern Tool Usage			
PLO 6: The Engineer and Society			
PLO 7: Environment and Sustainability			
PLO 8: Ethics			
PLO 9: Communication			
PLO 10: Individual and Team Work			√
PLO 11: Project Management and Finance			
PLO 12: Life-Long Learning			

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lecture	Assignment / Midterm Exam
CLO2	Lecture	Assignment / Class Test /Midterm Exam
CLO3	Lecture / Group Discussion	Presentation/ Class Test / Final Exam

Course Code: CSE 477	**Credits: 3.0**	**Year: Fourth**	**Semester: Eighth**
Course Title: Cloud Computing		**Course Status: Theory**	

Rationale of the course:

To provide students with fundamental concepts of cloud computing as well as how to centralize the setup, implementation, maintenance, and management of integrated computing services for individual and corporate end users.

Course Content:

Grid computing: definition of grid, infrastructure of hardware and software, applications, Grid architecture: overview of resource managers, overview of grid systems, Grid application management: Grid Application Description Languages, application partitioning, Meta-scheduling, mapping, monitoring, Web services, grid portals; Distributed computing: trends of distributed computing, evolution of Cloud computing; Cloud computing basics: properties and characteristics, different service models, deployment models; Infrastructure as a Service (IaaS): IaaS basics, resource virtualization, server, storage, network; Platform as a Service (PaaS): PaaS basics, Cloud platform and management, computation, storage; Software as a Service (SaaS):

SaaS basics, Web services, Web 2.0, Web OS; Cloud issues and challenges: Cloud provider lock-in, security, trust model; Case studies: Google Cloud infrastructure, Amazon Web services, Elastic Cloud, Storage Services, Microsoft Cloud infrastructure.

Text Books:

1. Anthony T.Velte , Toby J. Velte Robert Elsenpeter : Cloud computing a practical approach
2. Michael Miller :Cloud Computing: Web-Based Applications That Change the Way You Work and Collaborate Online

Reference Books:

1. Judith Hurwitz , Robin Bloor , Marcia Kaufman ,Fern Halper: Cloud computing for dummies
2. Cloud Computing (Principles and Paradigms): RajkumarBuyya, James Broberg, Andrzej Goscinski

Course Learning Outcomes (CLOs):

Upon completion of the course, the students will be able to:

CLO 1: Articulate the fundamental concepts, essential technologies, strengths, and limitations of cloud computing, as well as the potential uses for cutting-edge cloud computing.

CLO 2: Identify cloud computing architecture and infrastructure, such as SaaS, PaaS, IaaS, public cloud, private cloud, hybrid cloud, and so on, and differentiate between them.

CLO 3: Explain emerging cloud trends such as hybrid multicloud, microservices, serverless, cloud native, devops, and application modernization.

CLO 4: Identify problems, and **explain**, **analyze**, and **evaluate** various cloud computing solutions.

CLO 5: Create a hypervisor to use various cloud computing services.

Mapping of Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs):

Program Learning Outcomes (PLOs)	**CLO 1**	**CLO 2**	**CLO 3**	**CLO 4**	**CLO 5**
PLO 1: Engineering Knowledge	√	√	√		
PLO 2: Problem Analysis					
PLO 3: Design/Development of Solutions		√		√	
PLO 4: Investigation					
PLO 5: Modern Tool Usage					√
PLO 6: The Engineer and Society					
PLO 7: Environment and Sustainability					
PLO 8: Ethics					
PLO 9: Communication					

PLO 10: Individual and Team Work					
PLO 11: Project Management and Finance					
PLO 12: Life-Long Learning	√	√	√	√	

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lecture	Assignment / Midterm Exam
CLO2	Lecture / Case Study	Assignment / Class Test /Midterm Exam
CLO3	Lecture / Case Study	Assignment / Class Test /Midterm Exam
CLO4	Lecture / Group Discussion	Class Test / Midterm Exam/ Final Exam
CLO5	Lecture / Group Discussion	Presentation/ Class Test / Final Exam

Course Code: CSE 479	**Credits: 3.0**	**Year: Fourth**	**Semester: Eighth**
Course Title: Advanced Database Management System		**Course Status: Theory**	

Rationale of the course:

To introduce students to current database system and data management techniques, methods, and results as well as design, development, and use of a relational database. Topics such as query planning and optimization, transaction processing and concurrency control, big data management, data warehousing and OLAP, and database theory are common.

Course Content:

Database system architecture; managing primary and secondary storage; query processing; metadata and catalog management; language processing; query optimization and plan generation; concurrency; failures and recovery; extensibility; client-server interactions. Object-oriented database systems, XML, database and the web, data management in distributed mobile computing environment, data broadcasting, text database, digital library design and implementation, multimedia database: Basic concept, design and optimization of access strategies; parallel database, spatial database, temporal database.

Text Books:

1. Munindar P. Singh and Michael N. Huhns: Transparencies for Advanced Database Management (Cooperative Information Systems).
2. Omran Bukhres and Ahmed K. Elmagarmid (eds.): Object-Oriented Multidatabase Systems: A Solution for Advanced Applications.

Reference Books:

1. Silberschatz, Korth, Sudarshan: Database System Concepts.

2. Won Kim (ed.): Modern Database Systems: The Object Model, Interoperability, and Beyond, Addison-Wesley, 1995. ISBN: 0-201-59098-0.

Course Learning Outcomes (CLOs):

Upon completion of the course, the students will be able to:

CLO 1: Identify and be able to employ advanced database approaches (e.g. concurrency control, buffer management and recovery).

CLO 2: Enable the Ability to use learned information to generate comprehensive solutions using database systems and procedures.

CLO 3: Interpret and **explain** the significance of emerging database standards.

CLO 4: Possess the ability to create new database methods based on knowledge of existing techniques.

Mapping of Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs):

Program Learning Outcomes (PLO)	CLO 1	CLO 2	CLO 3	CLO 4
PLO 1: Engineering Knowledge	√		√	
PLO 2: Problem Analysis				√
PLO 3: Design/Development of Solutions		√		√
PLO 4: Investigation				
PLO 5: Modern Tool Usage	√			√
PLO 6: The Engineer and Society				
PLO 7: Environment and Sustainability				
PLO 8: Ethics				
PLO 9: Communication				
PLO 10: Individual and Team Work				
PLO 11: Project Management and Finance				
PLO 12: Life-Long Learning				

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lecture	Assignment / Midterm Exam
CLO2	Lecture	Assignment / Class Test /Midterm Exam
CLO3	Lecture / Group Discussion	Presentation/ Final Exam
CLO4	Lecture / Group Discussion	Presentation/ Class Test / Final Exam

Course Code: CSE 481	Credits: 3.0	Year: Fourth	Semester: Eighth
Course Title: Machine Learning		Course Status: Theory	

Rationale of the course:
To provide students with an in-depth introduction to two main- areas of Machine Learning: supervised and unsupervised.

Course Content:
Introduction, Supervised and Unsupervised learning in propositional logic, Induction of decision trees, Noise and over-fitting issues, Minimum description length principle, Conceptual clustering, Version space, Nearest neighbor classifier, Genetic algorithm, Computational learning theory. Learning in first order logic, Top-down and Bottom-up approaches for inducing first order theory, Handling noise, First order theory revision, Predicate invention, Application of Inductive Logic Programming, Multiple predicate learning, Different types of language bias, PAC Learnability, knowledge discovery in database and data mining, Text and image retrieval.

Text Books:

3. Marco Gori: Machine Learning: A Constraint-Based Approach
4. Ethem Alpaydin: Machine Learning
5. Pattern Recognition and Machine Learning - Christopher M. Bishop; Springer

Reference Books:

1. Ryszard, S., Michalski, J. G.Carbonell and Tom M. Mitchell: Machine Learning: An Artificial Intelligence Approach
2. Stephen Marsland: Machine Learning: An Algorithmic Perspective

Course Learning Outcomes (CLOs):
Upon completion of the course, the students will be able to:
CLO 1: Understand the significance of visualization in data analytics.
CLO 2: Gather knowledge about a wide range of machine learning challenges and algorithms.
CLO 3: Apply the algorithms to a real-world situation, improve the learnt models, and report on the expected accuracy of the models.
CLO 4: Develop an understanding of what it takes to learn from data.

Mapping of Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs):

Program Learning Outcomes (PLOs)	CLO 1	CLO 2	CLO 3	CLO 4
PLO 1: Engineering Knowledge	√	√		√
PLO 2: Problem Analysis			√	
PLO 3: Design/Development of Solutions			√	
PLO 4: Investigation			√	
PLO 5: Modern Tool Usage				

PLO 6: The Engineer and Society				
PLO 7: Environment and Sustainability				
PLO 8: Ethics				
PLO 9: Communication				
PLO 10: Individual and Team Work				
PLO 11: Project Management and Finance				
PLO 12: Life-Long Learning				

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lecture	Assignment / Midterm Exam
CLO2	Lecture	Assignment / Class Test /Midterm Exam
CLO3	Lecture / Group Discussion	Presentation/ Final Exam
CLO4	Lecture / Group Discussion	Presentation/ Class Test / Final Exam

Course Code: CSE 482	**Credits: 1.0**	**Year: Fourth**	**Semester: Eighth**
Course Title: Machine Learning Lab		**Course Status: Lab**	

Rationale of the course:

To familiarize students with various datasets and handling methods for implementing machine learning algorithms and implementing machine learning concepts and algorithms in any suitable language.

Course Content:

Decision Trees, Linear regression, Logistic Regression, Support Vector Machines, feature selection methods, Clustering (k-means, soft k-means), Ensemble methods such as Random Forest and Ada Boost, Probabilistic methods (Bayesian view), model evaluation and model selection, Neural Networks, Convolutional Neural Networks, Autoencoders.

Course Learning Outcomes (CLOs):

Upon completion of the course, the students will be able to:

CLO 1: Familiarize with different tools to be applied in machine learning and apply appropriate datasets and employ different data handling techniques.

CLO 2: Understand the mathematical and statistical aspects of machine learning algorithms using Python programming.

CLO 3: Design various supervised and unsupervised algorithms in machine learning.

CLO 4: Evaluate the performance of different machine learning algorithms.

CLO 5: Apply algorithms to a specific problem, optimize learned patterns, and report the expected accuracy that can be achieved using models.

Mapping of Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs):

Program Learning Outcomes (PLOs)	CLO 1	CLO 2	CLO 3	CLO 4	CLO 5
PLO 1: Engineering Knowledge					
PLO 2: Problem Analysis	√				√
PLO 3: Design/Development of Solutions	√		√		√
PLO 4: Investigation				√	√
PLO 5: Modern Tool Usage	√				
PLO 6: The Engineer and Society					
PLO 7: Environment and Sustainability					
PLO 8: Ethics					
PLO 9: Communication					
PLO 10: Individual and Team Work					√
PLO 11: Project Management and Finance					
PLO 12: Life-Long Learning	√	√	√		√

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lab demonstration/ Lecture	Assignment
CLO2	Lab demonstration/ Lecture	Assignment
CLO3	Lab demonstration/ Lecture	Assignment
CLO4	Lab demonstration/ Lecture	Presentation/Viva
CLO5	Lab demonstration/ Lecture	Lab Report /Final Exam

Course Code: CSE 483	**Credits: 3.0**	**Year: Fourth**	**Semester: Eighth**
Course Title: Big Data Analytics		**Course Status: Theory**	

Rationale of the course:

This course helps students learn the fundamentals of big data platforms, problems, and applications. The focus of the course will be on identifying and analyzing large-scale machine learning approaches, as well as current tools like Hadoop, in the context of big data research.

Course Content:

Introduction to Big Data: characteristics of Big Data and dimensions of scalability; Data Science: getting value out of Big Data, foundations for Big Data systems and programming; Big Data

Modeling and Management Systems: Big Data modeling, Big Data management, designing a Big Data management system; Big Data Integration and Processing: retrieving Big Data, Big Data integration, processing Big Data, Big Data analytics using Spark; Machine Learning with Big Data: introduction to machine learning with Big Data, data exploration, classification, evaluation of machine learning models, regression, cluster analysis, and association analysis; Graph Analytics for Big Data: introduction to graphs, graph Analytics, graph analytics techniques, computing platforms for graph analytics.

Recommended Books:

1. Big Data Analytics- Venkat Ankam
2. High-Performance Big Data Computing (Scientific and Engineering Computation) - Dhabaleswar K. Panda
3. Big Data: Principles and best practices of scalable realtime data systems - Nathan Marz and James Warren

Course Learning Outcomes (CLOs):

Upon completion of the course, the students will be able to:

CLO 1: Describe the landscape and the V's of Big Data for real world Big Data problems.

CLO 2: Apply and investigate various tools and strategies for storing, managing, and analyzing big data; adapt skills and create a report to use the essential tools for a variety of big data analytics.

CLO 3: Apply and **study** the key ideas of machine learning techniques in the context of big data for real-world problem solving; **adapt** the skills and **create** a report to apply the essential tools for a variety of big data analytics.

CLO 4: Understand, **apply**, **investigate**, and **justify** the relationship between big data components; adapt skills and create a report to use the essential tools for a variety of big data analytics.

Mapping of Program Learning Outcomes (PLOs) and Course Learning Outcomes (CLOs):

Program Learning Outcomes (PLOs)	**CLO 1**	**CLO 2**	**CLO 3**	**CLO 4**
PLO 1: Engineering Knowledge	√			
PLO 2: Problem Analysis				
PLO 3: Design/Development of Solutions		√	√	√
PLO 4: Investigation		√		√
PLO 5: Modern Tool Usage				
PLO 6: The Engineer and Society				
PLO 7: Environment and Sustainability				
PLO 8: Ethics				
PLO 9: Communication				
PLO 10: Individual and Team Work				
PLO 11: Project Management and Finance				

PLO 12: Life-Long Learning	√		√	

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning & Assessment Strategy:

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Lecture	Assignment / Midterm Exam
CLO2	Lecture	Assignment / Class Test /Midterm Exam
CLO3	Lecture / Group Discussion	Presentation/ Final Exam
CLO4	Lecture / Group Discussion	Presentation/ Class Test / Final Exam

20. Grading/Evaluation

20. a) Grading Scale/Point System

Numerical Grades	Letter Grades		Grade Point
80%-100%	A+	(A Plus)	4.0
75%-79.99%	A	(A Regular)	3.75
70%-74.99%	A-	(A Minus)	3.50
65%-69.99%	B+	(B Plus)	3.25
60%-64.99%	B	(B Regular)	3.0
55%-59.99%	B-	(B Minus)	2.75
50%-54.99%	C+	(C Plus)	2.50
45%-49.99%	C	(C Regular)	2.25
40%-44.99%	D	-	2.0
Less than 40%	F	Fail	0.0
	W	Withdrawn	
	I	Incomplete	

Note: Grades without numerical value

I: Incomplete

W: Withdrawal

20. b) Grades

Hamdard University Bangladesh (HUB) will follow the Uniform Grading System (As directed by University Grants Commission of Bangladesh).

20. c) Grade Point Average (GPA) and Cumulative Grade Point Average (CGPA)

Two kinds of grade point average will be use at Hamdard University Bangladesh (UB). The CGPA is calculated by dividing that the total earned quality points by the number of credit hours attempted in a given semester. The earned grade point is computed by multiplying the value of the grade point equivalent to the corresponding Letter grade by the number of credit hours for the course (e.g., a student earning a "B" (3.0) grade in a 3 credit hours course earns 9.0 grade points).

The Cumulative GPA is calculated by the following equation:

$$CGPA = \frac{\sum C_n G_n}{\sum C_n}$$ Where, *C* is Credit Hour, *G* is Grade Point.

So, $\sum C_n G_n = C_1 G_1 + C_2 G_2 + C_3 G_3 + ... + C_n G_n$ and $\sum C_n = C_1 + C_2 + C_3 + ... + C_n$.

It is necessary to maintain a certain CGPA for a degree at UB. For example, a CGPA of C^+ (2.5) for Bachelor's degree is essential.

20. d) Withdrawal (W)

Withdrawal is assigned to a student who withdraws from the course within the deadline for withdrawal with 'W' grade. A student who withdraws after this date will earn the grade based on his performance before his withdrawal. Exception to this rule may be made on medical ground and on terms and condition of the University.

20. e) Incomplete Grade

An incomplete (I) grade is assigned only when a student has failed to complete one or more requirements of the course for unavoidable reason / accidental circumstance and has applied for (I) grade. The students who are permitted to appear in make-up examination(s) will be assigned an (I) grade for that course and this grade will stay until the student appears in the make-up examination at the first available opportunity; if she/he fails to appear in the make-up examination the (I) grade will automatically be converted to (I) grade.

20. f) Retake

If any examinee obtains grade "F" in one or more subject of any semester, he/she will have to re-register and pass those courses within the following two semesters (one year).

20. g) Grade Improvement

In the final examination, if any student obtains grade less than B, he/she will be able to appear in improvement examination. A student can sit for maximum of two improvement examinations for a particular course in the following two semester (one year). In addition, students will have the chance to participate on improvement examination for highest two (2)courses in any semester.

NB. Any examinee can apply for the improvement exam in the same course either in Midterm or Final examination, not both examinations.

20. h) Dropout

If any student doesn't complete registration in consecutive 2 (two) semesters, he/she will be counted as a dropout. If that student wants to continue his/ her study, he/ she must get re-admission.

20. i) Grades Review Procedure

The Academic Council of the University administers the Grading Regulations and Reviews course grades submitted by the Departments.

Template of Course Outline
(Outcome-Based Curriculum)

Part A

1. Course No/Course Code:
2. Course Title:
3. Course Type (Ged/Core Course/Electives/……..):
4. Year/Level/Semester/Term:
5. Academic Session:
6. Course Teacher/Instructor:
7. Pre-requisite (If any):
8. Credit Value:
9. Contact Hours:
10. Total Marks:
11. Rationale of the course:
12. Course Objectives:
13. Course Learning Outcomes (CLOs) and Mapping of CLOs with program Learning Outcomes (PLOs)

(Note: If required add/delete columns or rows)

Course Learning Outcomes (CLO) (3-5)	PLO1	PLO2	PLO3	PLO4	PLO5							

Part B

14. Course Plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs.

Week	Topic	Teaching-Learning Strategy	Assessment Strategy	Corresponding CLOs
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				
11				
12				
13				
14				
15				
16				
17				
18				

Part C

15. Assessment and Evaluation
    1) Assessment Strategy
    2) Marks distribution

Theory Course:

Class Attendance	10%
Class Test	10%
Assignment & Presentation	10%
Mid-Term Examination	30%
Final Examination	40%
Total	**100%**

Lab Course:

Class Attendance	10%
Lab Report Assessment	20%
Project/Presentation	30%
Final Examination	40%
Total	**100%**

    a) Continuous Assessment:
    b) Summative:
    3) Make-up Procedures

Part D

16. Learning Materials
    1) Recommended Readings
    2) Supplementary Readings
    3) Others….. (as applicable for the discipline/academic program)

Note: The course outline will be provided by the course teacher to the students

Template for Course File

(Outcome-based curriculum)

1. Course outline
2. Lecture materials/notes
3. Attendance sheet
4. Assignment → copies of best, mediocre and poor
5. Class test/quiz/mid-term test etc. → copies of best, mediocre and poor
6. Copies of question papers/assessment tools
7. Rubrics for assessment
8. Course evaluation by course teacher considering feedback from the students
9. Feedback from the students on the course
10. Peer review: Done/Not done
11. Evaluation of attainment of CLOs
12. Improvement plan for better attainment of CLOs